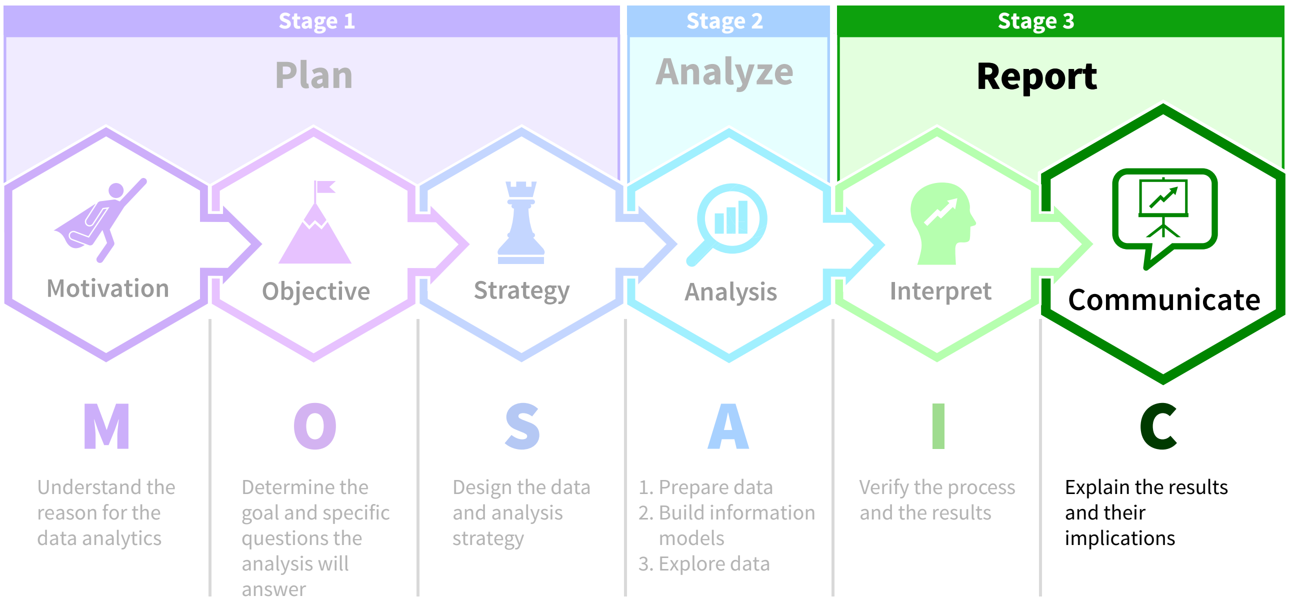
CHAPTER 9

Communicating Data Analysis Results

Chapter Preview

The last step in the data analysis process is to explain the results of our analyses and their implications. Whether the goal is to inform or persuade, effectively communicating results is critical. If the intended audience does not understand the information, then it does not matter how well the analysis was performed.

Data analysis results can be communicated through memos, reports, or oral presentations, to name a few methods. Regardless of the form of communication, analyses must be supported using data visualizations, data stories, or interactive data visualizations. This chapter focuses on best practices for the common forms of data analysis communication you will encounter in your accounting career, including preparing effective data visualizations (both static and interactive) and creating data stories.



Professional Insight: Can Data Visualization Knowledge Distinguish You From the Crowd?

Jenna is a senior accounting student who recently finished an audit internship with a large international accounting firm.

> **One of my biggest takeaways was how much high-quality communication was valued by my internship firm.** This was especially true when it came to communicating the results of data analyses.
>
> **I was the only intern on the audit client that had experience using data visualization software. I was able to create data visualizations for a presentation that the team prepared for the client.** As a result, I was invited to attend the presentation. Being involved in the client meeting allowed me to interact with the partner on the audit. **I believe it really helped me stand out from the other interns and is one of the reasons I was offered a full-time position.**

Chapter Roadmap

LEARNING OBJECTIVES	TOPICS	APPLY IT
LO 9.1 Explain how a data story communicates data analysis results.	• Develop Data Literacy • Communicate Effectively • Tell a Data Story	**Communicate Accounting Information** (Example: Financial Accounting)
LO 9.2 Summarize the steps for creating effective data visualizations.	• Verify the Data • Consider the Audience • Define the Objective	**Match Objectives to Visualization Types** (Example: Auditing)
LO 9.3 Describe the characteristics of an effective data visualization.	• Use Principles of Visual Perception • Consider Preattentive Attributes • Avoid Clutter • Use Visualization-Specific Best Practices	**Evaluate Visualizations** (Example: Managerial Accounting)
LO 9.4 Recognize misleading data visualizations.	• Omitting the Baseline • Manipulating the Y-axis • Going Against Conventions • Selectively Choosing the Data • Using the Wrong Graph	**Identify Misleading Data Visualizations** (Example: Managerial Accounting)
LO 9.5 Create an interactive data visualization presentation.	• Best Practices for Live Presentations • Creating Interactive Data Visualizations	**Create an Interactive Visualization** (Example: Financial Accounting)

Data The Data tag appears in the chapter when the data for an example, illustration, or application are available on Wiley's online learning platform.

Data analytics software is continuously changing, and there may be more recent versions of the software referenced in this chapter. For more information access the accompanying video on Wiley's online learning platform.

9.1 How Do We Tell a Data Story?

LEARNING OBJECTIVE 1

Explain how a data story communicates data analysis results.

Finding valuable insights in data and then communicating them effectively is, in the words of Google's Chief Economist Hal Varian, a "hugely important skill."[1] This is particularly important for accountants who frequently communicate data analysis findings to a variety of stakeholders.

Develop Data Literacy

Effectively communicating these findings requires **data literacy**, which is the ability to understand and communicate data. Why is data literacy so important? A study by the data visualization software company Qlik and the consulting firm Accenture found that 63% of employees use data to a make a decision at least once a week.[2] Further, another study by the consulting firm McKinsey & Company found that companies where employees consistently use data in decision-making were more likely to report revenue growth of more than 10% in the past three years.[3]

Data Let's use an example to illustrate data literacy. Imagine you are an accountant for Huskie Motor Corporation (HMC).[4] HMC is an international automobile manufacturer that produces and sells cars in 15 countries. The countries are grouped into three regions–Europe, North America, and South America (**Illustration 9.1**).

ILLUSTRATION 9.1 Huskie Motor Corporation (HMC) International Operations

Europe	North America	South America
France	Canada	Argentina
Germany	Mexico	Bolivia
Poland	USA	Brazil
Spain		Chile
Sweden		Colombia
United Kingdom		Venezuela

HMC has three brands of vehicles, and each brand has five unique models (**Illustration 9.2**).

ILLUSTRATION 9.2 HMC Brands and Models

	Brands		
	Apechete	Jackson	Tatra
Models	Chare	Brutus	Advantage
	Island	Crux	Bloom
	Pebble	Fiddle	Jespie
	Robin	Rebel	Mortimer
	Summet	Wood	Rambler

[1]McKinsey & Company, 2009. Hal Varian on how the Web challenges managers. https://www.mckinsey.com/industries/technology-media-and-telecommunications/our-insights/hal-varian-on-how-the-web-challenges-managers (accessed July 2022).
[2]Qlik and Accenture. 2020. The human impact of data literacy. Data Literacy Project. https://thedataliteracyproject.org/humanimpact (accessed July 2022).
[3]QuantamBlack AI by McKinsey, 2019. Catch them if you can: how leaders in data and analytics have pulled ahead. https://www.mckinsey.com/business-functions/quantumblack/our-insights/catch-them-if-you-can-how-leaders-in-data-and-analytics-have-pulled-ahead (accessed July 2022).
[4]Ann C. Dzuranin, Johan Perols and Dana L. Hart, "Huskie Motor Corporation: Visualizing the Present and Predicting the Future," *IMA Educational Case Journal*, Volume 11 Issue 2, 2018, bit.ly/3qHsGS2. ©2022, Institute of Management Accountants, www.imanet.org. Used with permission.

You have been asked to prepare an analysis of sales trends by month for each brand. **Illustration 9.3** provides sales information by brand for each month, but does it show what you need to understand about sales trends?

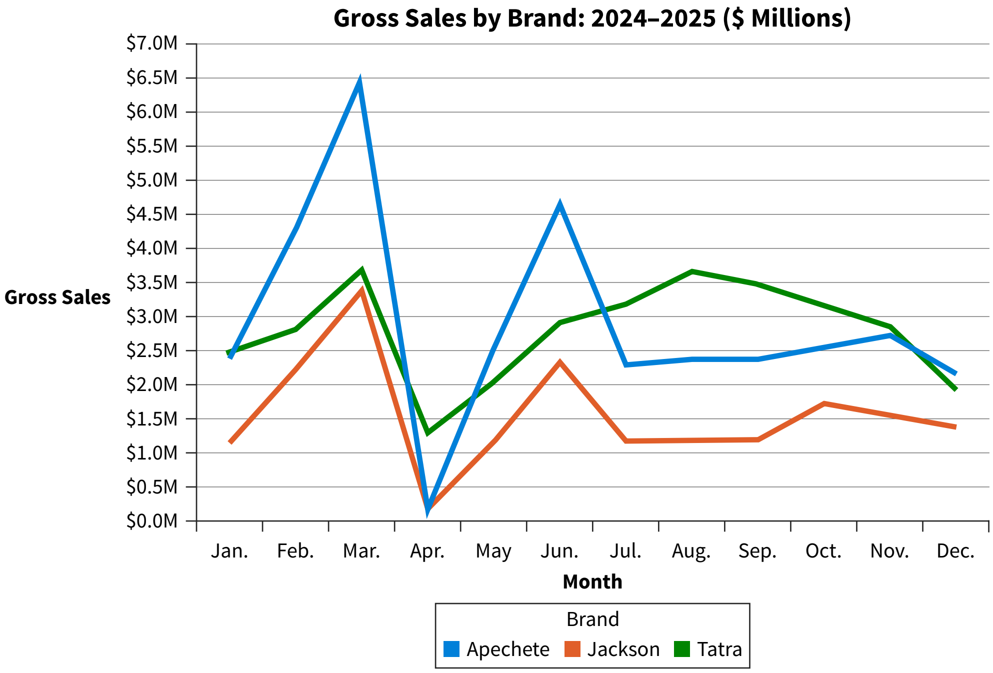
ILLUSTRATION 9.3 HMC Gross Sales by Brand

Gross Sales Trends by Brand: 2024–2025 (in $ Millions)

	Apechete	Jackson	Tatra
January	$ 2.4M	$ 1.2M	$ 2.5M
February	$ 4.3M	$ 2.2M	$ 2.8M
March	$ 6.5M	$ 3.4M	$ 3.7M
April	$ 0.2M	$ 0.2M	$ 1.3M
May	$ 2.5M	$ 1.2M	$ 2.0M
June	$ 4.6M	$ 2.3M	$ 2.9M
July	$ 2.3M	$ 1.2M	$ 3.2M
August	$ 2.4M	$ 1.2M	$ 3.7M
September	$ 2.4M	$ 1.2M	$ 3.5M
October	$ 2.6M	$ 1.7M	$ 3.2M
November	$ 2.7M	$ 1.6M	$ 2.9M
December	$ 2.2M	$ 1.4M	$ 1.9M

Analyzing this table determines which months had the highest sales and which had the lowest. Preparing a line chart would accomplish the same thing (**Illustration 9.4**).

ILLUSTRATION 9.4 HMC Gross Sales Trends Line Chart



The chart in Illustration 9.4 more clearly shows the sales trends:

- March is the highest selling month, and April is the lowest selling month.
- Sales peak again in June for the Apechete and Jackson brands, then level off for the remainder of the year.
- The Tatra brand has increasing sales from April to August.

While the figures in Illustration 9.3 can be used to determine the sales trends, it is easier to do so with Illustration 9.4. Because data literacy skills are necessary for a successful accounting career, this chapter examines the data literacy skill of communicating data analysis results.

Communicate Effectively

The final step in the data analysis process is summarizing the project's findings and communicating them to the intended audience. This requires explaining the meaning of the data by writing memos or reports or preparing presentations that explain the data analysis results clearly and concisely. This is not easy, and it takes practice. Effectively reporting the results of data analysis requires being aware of the audience, focusing on the message, putting that message in context, and making sure it is clearly presented as an engaging story.

Understand the Audience

Have you ever attended a presentation where the presenter didn't provide enough background information for you to understand the topic? It is like walking into an advanced physics class when you have never taken a physics course. Understanding to whom we are communicating is critical. The audience (or reader) must be provided with enough background and explanation to follow the presentation.

APPLYING CRITICAL THINKING 9.1: Identify the Audience

When communicating data analysis results, identifying the stakeholders of the analysis is an important first step. They are the audience to whom you are communicating **(Stakeholders)**:

- Internal stakeholders will likely have some knowledge of what you are communicating.
- When communicating to external stakeholders, consider what information they will find most relevant and focus on that.

Knowing if the audience is comprised of internal or external stakeholders makes it possible to develop communication that is both relevant and understandable. In the HMC example, if the communication is for internal stakeholders such as company management, it is safe to assume they will have knowledge of the company's background. Whereas, if the presentation was for external stakeholders such as investors, it should include more background information about the company.

Focus on the Message

Accountants are comfortable reading and interpreting numbers, so it is easy to focus solely on the numbers when communicating data analyses. However, an audience is likely more interested in the relationship between the numbers and the message. For example, if the objective of the analysis is to identify expense trends, ensure the communication explains the trends and does not only focus on the amounts.

Put it in Context

The third suggestion for communicating effectively is to put the data in **context**, or perspective. There are two aspects to context when communicating data analyses results:

1. The context of the overall purpose of the analysis. Is the purpose to inform or persuade the audience?
2. The context of the individual analyses. Are the analyses based on all the company data or just one department? Are the figures in whole dollars or in millions?

When communicating data analysis results, give the audience the information they need to understand the context of the analysis. We will discuss other aspects of context throughout this chapter.

Strive for Clarity

The fourth suggestion for effective data analysis communication is to make sure the communication is easy to understand. Do this by clearly explaining the data and the results in the narrative of the communication and including effective visualizations. The last suggestion for effective communication is to engage the audience with a memorable story.

Tell a Data Story

People have communicated knowledge and information using stories for thousands of years–they are an integral part of human communication. In fact, the human mind is wired to absorb stories. Research has shown that when we hear or read a story our brains are activated. The emotional part of the brain releases chemicals to stimulate feelings of connection, reward, and recognition.[5] In other words, a story creates both a physical and emotional response. Research has also shown that facts are up to 22% more memorable when they are part of a story.[6] So, how can storytelling help communicate the results of data analysis?

Data Story Elements

There are three elements to a data story–data, narrative, and visuals (**Illustration 9.5**).

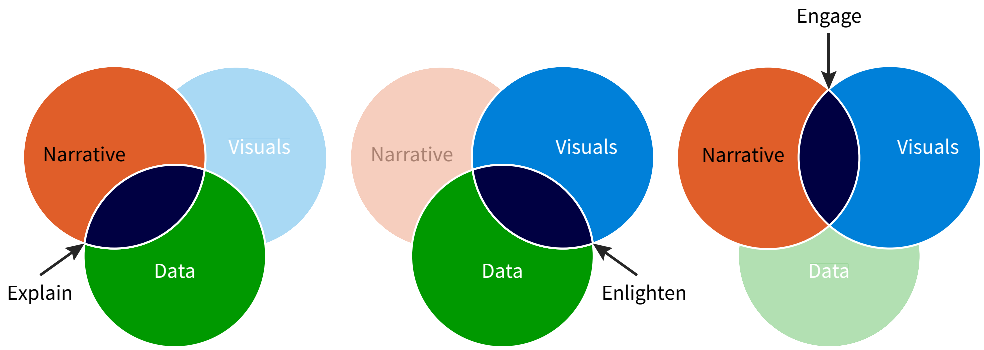


ILLUSTRATION 9.5 Elements of Effective Storytelling

The author of *Effective Data Storytelling*, Brent Dykes, describes how these elements combine to explain, enlighten, and engage the audience:[7]

- The intersection of data and narrative explains the data story. The story's narrative provides the context and commentary needed to understand the results of the analysis. It provides structure to the data and guides the reader through the meaning of the analysis.
- Data also intersects with visuals to enlighten the reader with insights. Visualizing the data reveals patterns or trends that may have gone unnoticed without the help of visualizations. In fact, humans process visual images 60,000 times faster than text.[8]
- Finally, combining narrative with visuals engages the audience in the story. A good story can hold the attention of the reader and increases the likelihood of action.

In addition to including these three elements, a good data story must be structured effectively.

Data Story Structure

Most stories have a similar structure. They introduce the characters and set the scene, then a series of events build to the most climatic or important point of the story. Following the climax of the story, the rest of the events unravel until conflicts are resolved and the story concludes.

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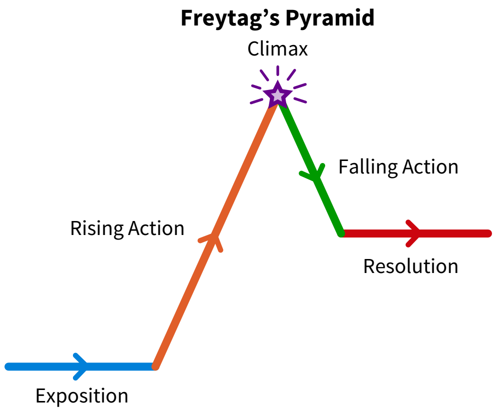
[5]Psychology Today, 2011. Rutledge, P. The psychological power of storytelling. https://www.psychologytoday.com/us/blog/positively-media/201101/the-psychological-power-storytelling (accessed July 2022).
[6]Forbes, 2015. Harrison, K. A good presentation is about data and story. https://www.forbes.com/sites/kateharrison/2015/01/20/a-good-presentation-is-about-data-and-story/#1f9a2f54450f (accessed July 2022).
[7]Dykes, B. (2020). Effective Storytelling: How to Drive Change with Data, Narrative, and Visuals. Hoboken, NJ: Wiley.
[8]Thermopylae Sciences and Technology, 2014. Humans process visual data better. https://www.t-sciences.com/news/humans-process-visual-data-better (accessed July 2022).

This structure is called Freytag's pyramid, also sometimes referred to as a storytelling arc (**Illustration 9.6**). The pyramid was developed by the German playwright and novelist Gustav Freytag to understand the structure of Greek and Shakespearian drama. It is one of the most taught dramatic structures in the world.

ILLUSTRATION 9.6 Freytag's Pyramid[9]



Freytag's pyramid can be applied to data storytelling using the HMC example:

- The audit team received an anonymous tip that one of the purchasing managers was receiving kickbacks from vendors.
- The team prepared an analysis investigating potential kickback fraud in the purchasing department. They identified the location of the possible kickback activity and the employee and vendors potentially involved.

If you were on the audit team, you would now be ready to create a story using the data analyses prepared in the audit of the purchasing department. **Illustration 9.7** provides an example of how to adapt the Freytag pyramid to a data story.

ILLUSTRATION 9.7 Anatomy of a Data Story

Freytag's Pyramid	Data Story Application	Kickback Analysis Example
Exposition	Introduce the problem or issue.	Briefly discuss the background information relevant to the analysis. Include interesting details to engage the reader's attention. Example: "Is kickback fraud occurring in the Purchasing Department? Based on an anonymous tip, we took a closer look at the purchasing department."
Rising Action	The subject of the analysis is explored at a deeper level.	In this part of the story, methodically peel back layers of the kickback analyses.
Climax	The main finding or insight is shared. This is the "aha moment" of the story.	After building the case in the previous section, announce the suspected employee(s) and vendor(s).
Falling Action	Share the solution.	Identifying a suspected employee and vendor does not prove kickback fraud has occurred. Next, provide more details and recommendations as to specific transactions that should be investigated further.
Resolution	Conclude the story and provide next steps.	Make suggestions for additional internal controls to avoid future kickback fraud.

Following the storytelling arc described in Illustration 9.7 is a simple and effective way to structure a data story. Once the structure is set, the elements of a story (data, narrative, visuals) can be applied to bring the data story to life.

[9]Writers.com, 2020. Glatch, S. The 5 elements of dramatic structure: understanding Freytag's pyramid. https://writers.com/freytags-pyramid (accessed July 2022).

Apply It 9.1
Communicate Accounting Information

Data **Financial Accounting** U.S. Outdoor Adventures is a retail company that sells camping travel supplies in the United States. They specialize in building camping packages with all the supplies customers will need for their outdoor adventures. The company has three categories of products: Camping Gear, Paddle, and Tents. Customers of U.S. Outdoor Adventures are categorized into segments: Consumer, Corporate, and Travel Agency. The list of products, sales by segment, and financial information for 2025 are provided. (Note: the numbers in the 2025 financial statement may not add due to rounding.)

U.S. Outdoor Adventures 2025 Sales by Segment

Segment	Total Sales
Consumer	$ 512,711
Corporate	$ 359,264
Travel Agency	$ 232,095
Grand Total	**$ 1,104,070**

U.S. Outdoor Adventures Products Offered

Camping Gear	Paddle	Tents
Camp stove	Kayak	Backpacking Tent
Chairs	Life Vest	Base Camp Model
Cook set	Paddle Board	Mountain Tent - 4 Person
Fasteners	Paddles	Northface subzero
Firestarter Kit		
First Aid Kit		
Micro Cooking unit		
Propane Portable Grill		
Sleeping Bag		

U.S. Outdoor Adventures 2025 Financial Information

	Camping Gear	Paddle	Tents	Total
Sales	$ 321,964	$ 243,580	$ 538,526	$ 1,104,070
Discount	$ 34,170	$ 40,030	$ 96,864	$ 171,063
Net Sales	$ 287,794	$ 203,550	$ 441,663	$ 933,006
Cost of Goods Sold	$ 114,903	$ 87,682	$ 193,681	$ 396,265
Shipping Cost	$ 36,900	$ 31,066	$ 64,646	$ 132,611
Profit	$ 135,992	$ 84,802	$ 183,336	$ 404,130

The company would like to expand their business internationally, but they must raise additional capital to do so. If you were tasked with presenting this information to a group of potential investors, describe why each of the following are important to consider and give an example of each related to U.S. Outdoor Adventures.

1. Identify the audience.
2. Focus on the message, not the numbers.
3. Put the data in context.
4. Make it easy to understand.
5. Create a memorable story.

SOLUTION

1. Provide enough background and explanation so the audience can follow the presentation. In this case, potential investors are the audience. They will likely understand the numbers but may need additional background about the business.
2. This audience probably wants to know how the numbers relate to the message being communicated, which is how likely the business is to be successful internationally.
3. First, there is the context of the overall objective of the analysis, which is to attract investment into the company. There is also is the context of the individual analyses. These include the company's customers, the products they sell, and their current profits. Finally, there is the context of how the company compares to its competitors.
4. Clearly explain the data and the results. Make communication easy to understand by using effective visualizations. In this example, it may make sense to create visuals other than the tables.
5. Engaging the audience with a data story makes it easier for the reader to remember the results. The story in this example could focus on U.S. Outdoor Adventures' prior success and how investing in an international expansion will build on that success.

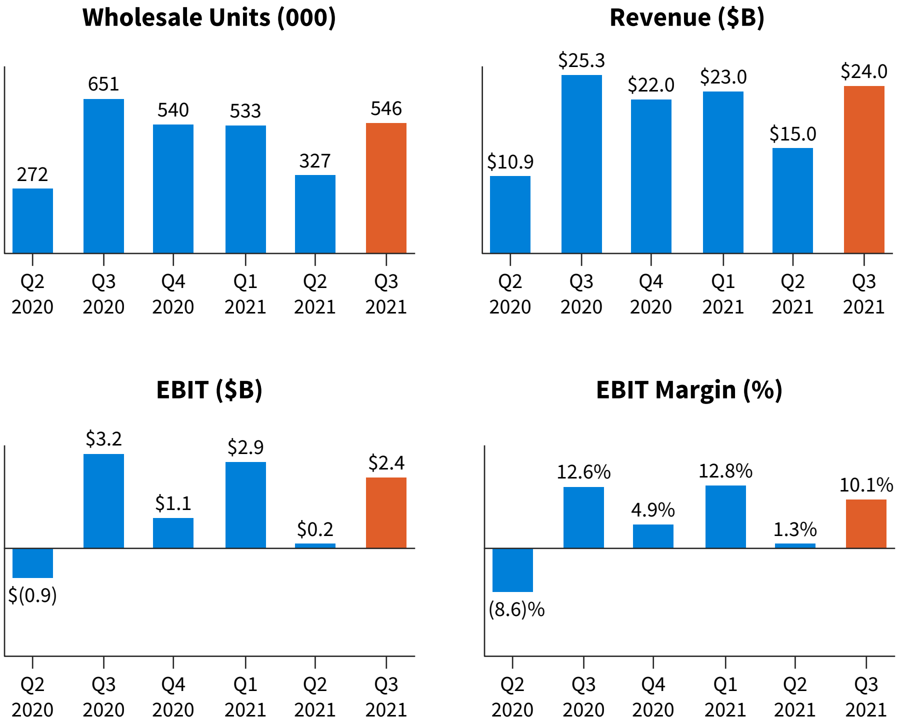
9.2 What are the Steps for Creating Effective Data Visualizations?

LEARNING OBJECTIVE 2

Summarize the steps for creating effective data visualizations.

Data visualization is the process of displaying data to provide meaning and insights for the audience. A well-designed visualization conveys the results of an analysis clearly and concisely. **Illustration 9.8** is a visualization Ford Motor Company used in their presentation of third quarter 2021 results to investors.

ILLUSTRATION 9.8 Ford Motor Company Earnings Presentation



Source: Ford, Q3 2021 Earnings Review, October 27, 2021.

As an overview of Ford's North America performance, this illustration is an example of a well-designed visualization. It combines information about the number of vehicles sold (wholesale units), revenue generated from those sales (revenue), earnings before interest and taxes on those sales (EBIT), and the percent profit margin of earnings before interest and taxes (EBIT Margin %). Note the color differentiation of the third quarter data. By using a different color for Q3 2021, the objective of the visualization (to convey third quarter 2021 results) is immediately clear. Creating well-designed visualizations like this starts with verifying the data, considering the audience, and defining the objective of the analysis.

Verify the Data

The saying "garbage in, garbage out" is as relevant to data analysis communication as it is to performing data analyses. Incorrect data leads to incorrect visualizations. To avoid this, data should have the attributes of accuracy, completeness, consistency, freshness, and timeliness.

Accurate Data

Accurate data are free from errors. They are reliable and representative of the problem or issue being visualized. Imagine preparing information for the shareholders of the fictitious HMC company. The goal is to provide information about monthly sales in 2025 for each brand. The data provided in **Illustration 9.9** are representative of 2025 sales performance of the Apechete brand. If the data are confirmed to be free of error, then they are also reliable.

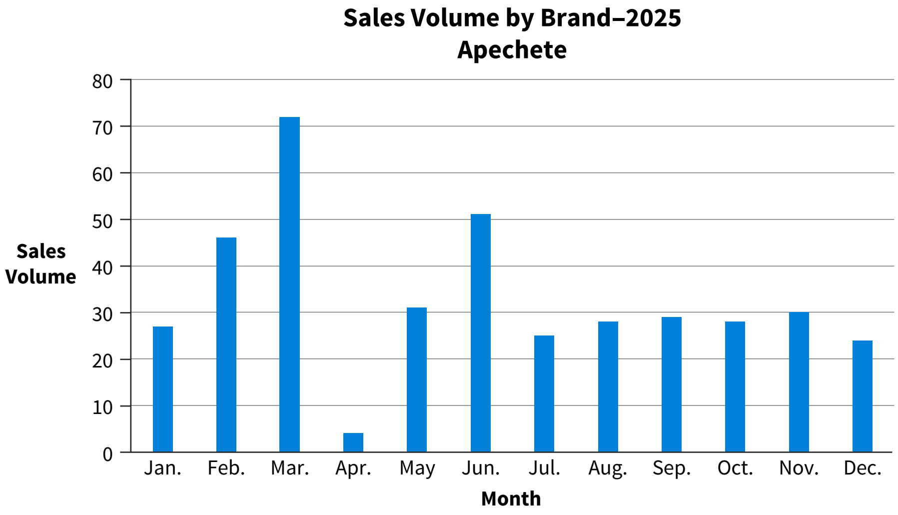


ILLUSTRATION 9.9 HMC 2025 Sales Performance

Complete and Consistent Data

Data are complete when there are no missing data. Illustration 9.9 shows monthly sales volume, but it is missing data for the month of April. It would be unusual to have only four sales for the entire month, so the next step is to confirm if the data are complete.

Are the data consistent across all periods? For example, data should be formatted identically across the periods that are displayed. Consistency also relates to the attributes of the data. In other words, is there the same level of detail for each period being displayed? **Illustration 9.10** is an example of inconsistent data in a visualization. The visualization for gross sales in 2024 shows the totals above the bars in thousands, and the visualization for 2025 shows the gross sales totals in millions. This could be confusing for viewers who do not notice the notations.

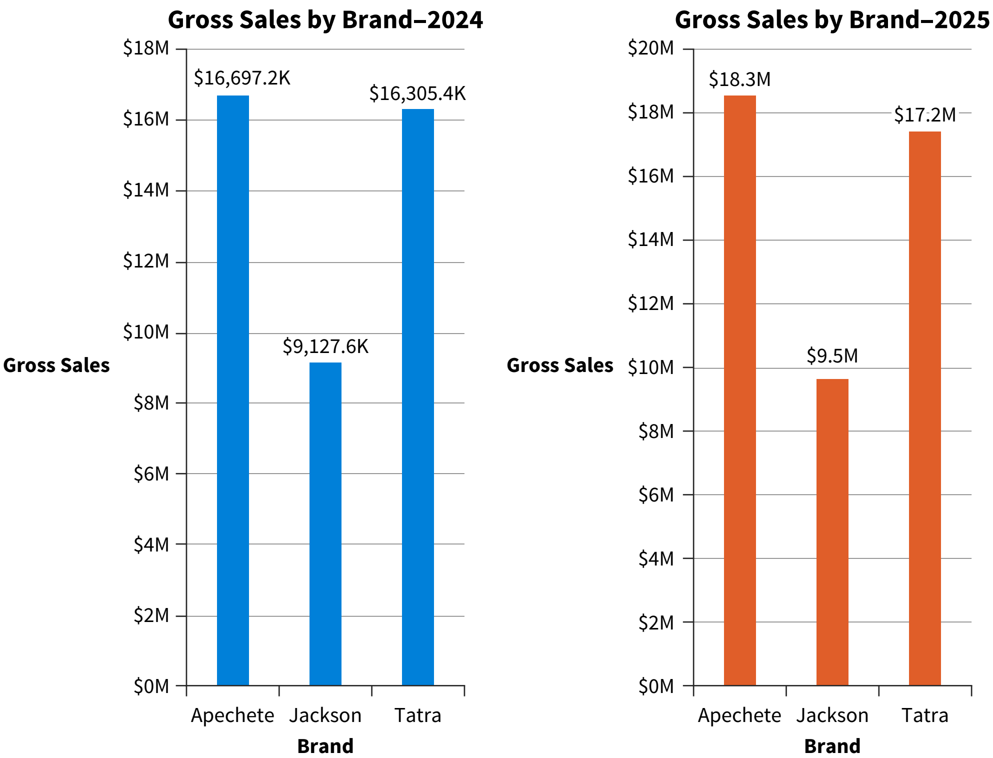


ILLUSTRATION 9.10 HMC Gross Sales by Brand: 2024 and 2025

Fresh and Timely Data

Data is considered fresh if they are the most recent data available. Avoid using outdated data in visualizations. Consider the earnings presentation in Illustration 9.8. If Ford had used data from the second quarter in their third quarter presentation, investors would not have useful information about third quarter results.

Data are considered timely when they are available in time to be used for a visualization. When designing a visualization, ensure that the targeted data are available so that the visualization is fresh.

Consider the Audience

After verifying the data, consider who will be viewing the visualization. Audiences can be separated into four categories (**Illustration 9.11**).[10]

ILLUSTRATION 9.11 Types of Audiences

Category	Description	What They Want
Novice	Has never encountered the information.	Enough detail to gain understanding.
Managerial	Has some knowledge of the topic.	Actionable results.
Expert	Has deep knowledge of the topic.	Investigation and discovery.
Executive	Has a broad, high-level knowledge of the topic.	Only the most important insights.

Novice Audiences

A novice audience, which can be internal or external to the organization, needs enough background information to understand the results. An analysis for an external client or one presented to an internal department unfamiliar with the topic would both have novice audiences. **Illustration 9.12** is an example of the type of visualization that might be used to explain sales trends to a novice audience.

[10]Harvard Business Review, April 2013. Stikeleather, J. How to tell a story with data. https://hbr.org/2013/04/how-to-tell-a-story-with-data (accessed July 2022).

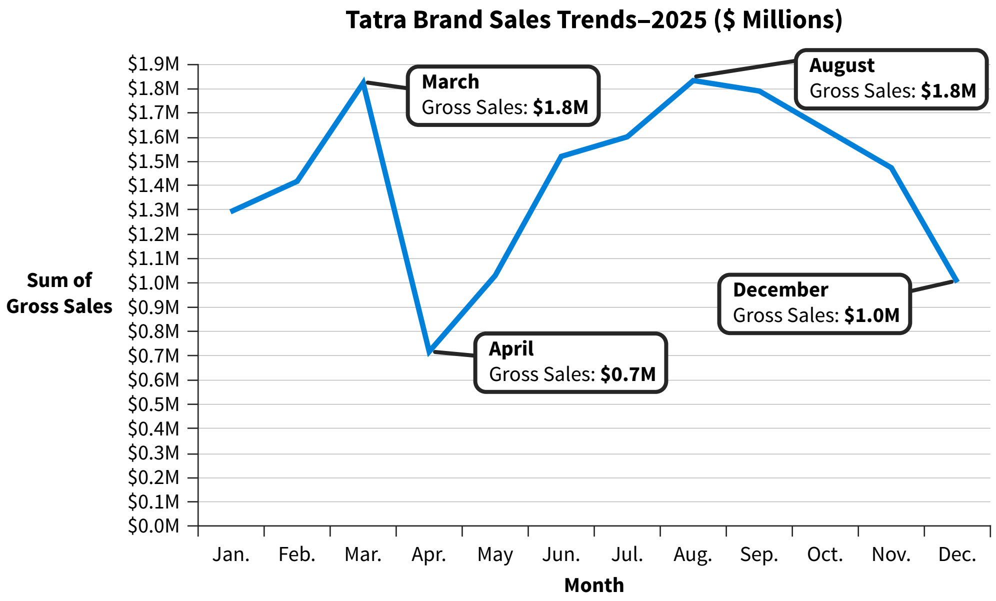


ILLUSTRATION 9.12
Visualization Sample for a Novice Audience

The visualization is clearly labeled to show the viewer the highest and lowest months of sales.

Managerial Audiences

A managerial audience generally has some knowledge of the topic, so detailed background information may be unnecessary. However, this audience is looking for actionable results, so the visualization should include recommendations for actions based on the results.

The dashboard in **Illustration 9.13** communicates to the managers profit margin as well as gross sales and cost analysis. (Data How To 9.1 at the end of the chapter explains how to create this dashboard in Tableau.) The filters for year sold and brand let the managers customize the visual by year and brand. It provides quick, actionable information.

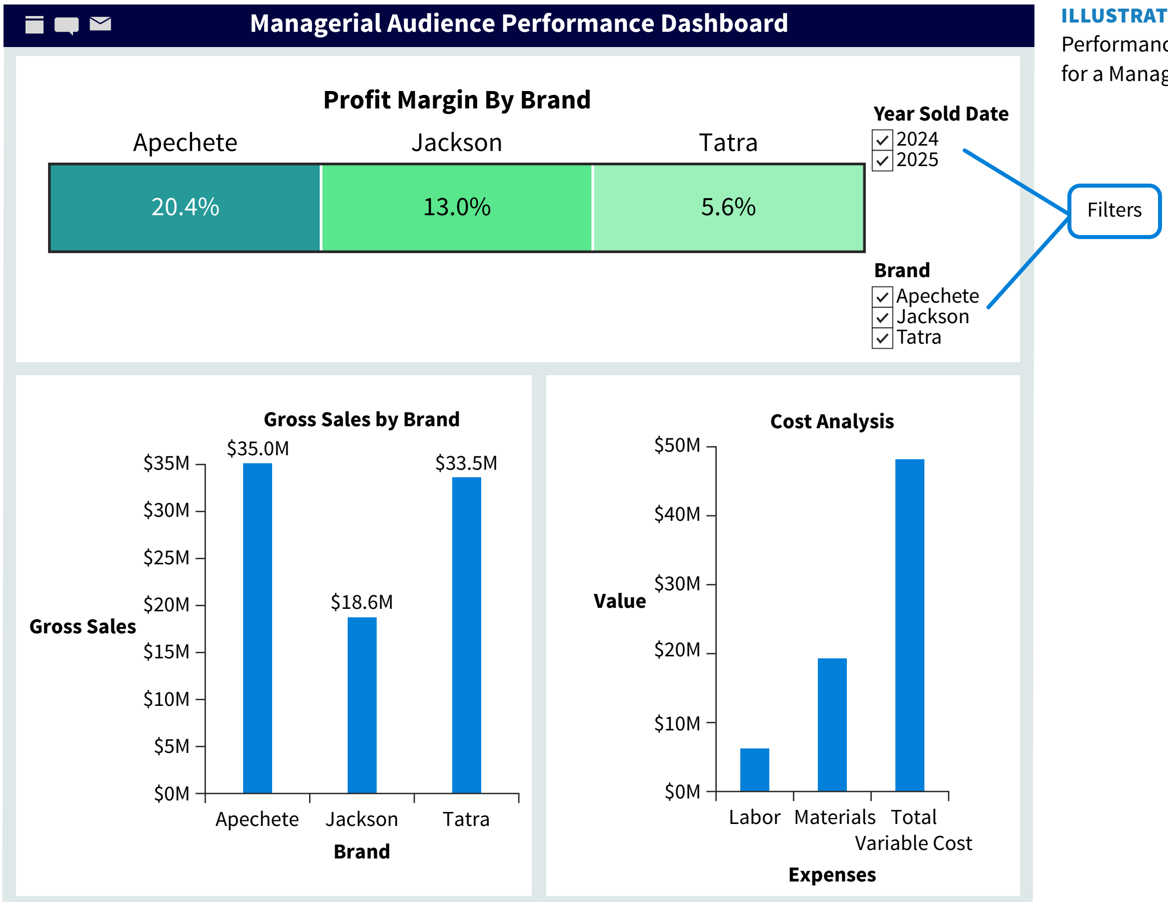
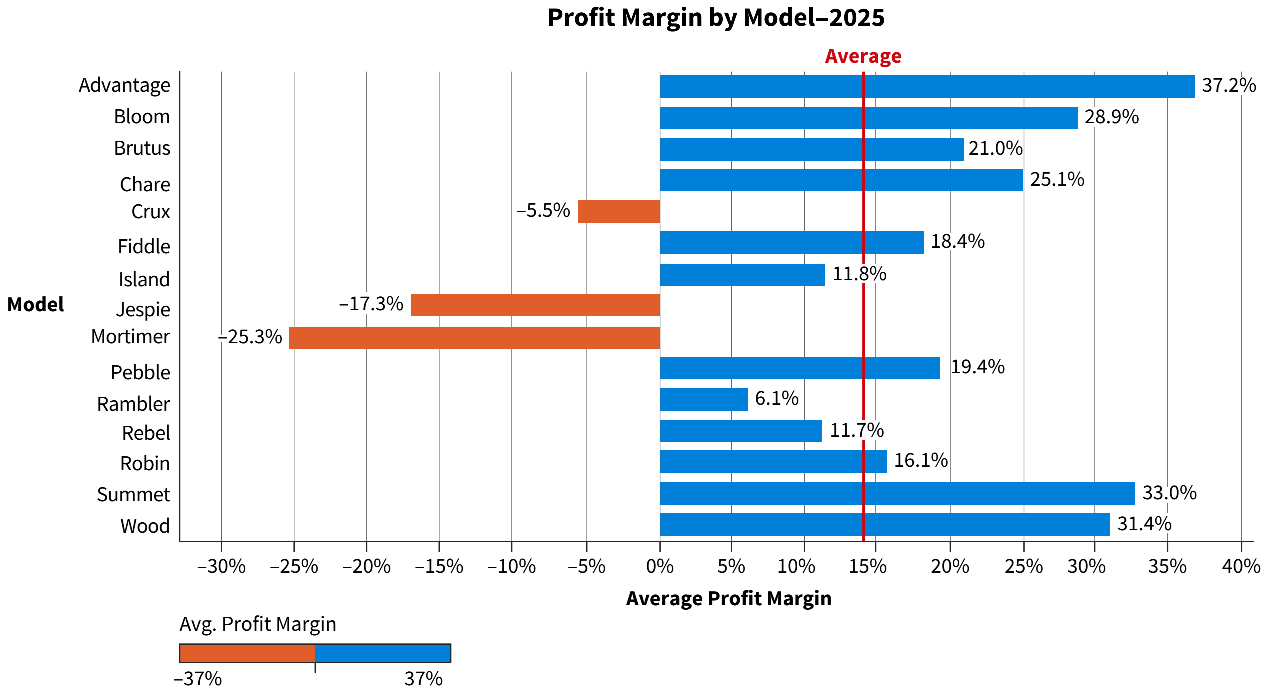


ILLUSTRATION 9.13
Performance Dashboard Sample for a Managerial Audience

Expert Audiences

Because experts already have deep knowledge about the subject of the analysis, this type of audience does not need basic background information. Instead, they are interested in the investigative aspect of the story. For example, experts would be interested in digging deeper into profit margin for HMC to see if there are specific models that should be investigated further. **Illustration 9.14** is a visualization used to communicate an analysis of profit margin by model.

ILLUSTRATION 9.14 Sample Visualization for an Expert Audience



This illustration provides detailed information about each model's profit margin as compared to the average profit margin for all HMC models. The models with negative profit margin are clearly identified by the orange bars. From here the expert can determine which model to investigate further.

APPLYING CRITICAL THINKING 9.2: Give the Audience the Information They Need

It is easy to lose the audience's attention if they lack background knowledge or information necessary to interpret and understand data analysis results **(Knowledge)**:

- Consider what information is necessary to understand the message being communicated.
- Include background necessary for the audience to understand the analysis.

For the HMC analysis prepared in Illustration 9.15. the audience are executives, so they understand how to use contribution margin, total fixed costs, and income before taxes to decide whether to discontinue a product. Therefore, they do not need detailed information about how to interpret those figures.

Executive Audiences

An audience composed of executives will be interested in only the most important insights. Discuss the important insights first, then discuss the support for those insights. **Illustration 9.15** is a visualization that might be used when communicating to an audience of executives who want details about the two least profitable models, which are Mortimer and Jespie.

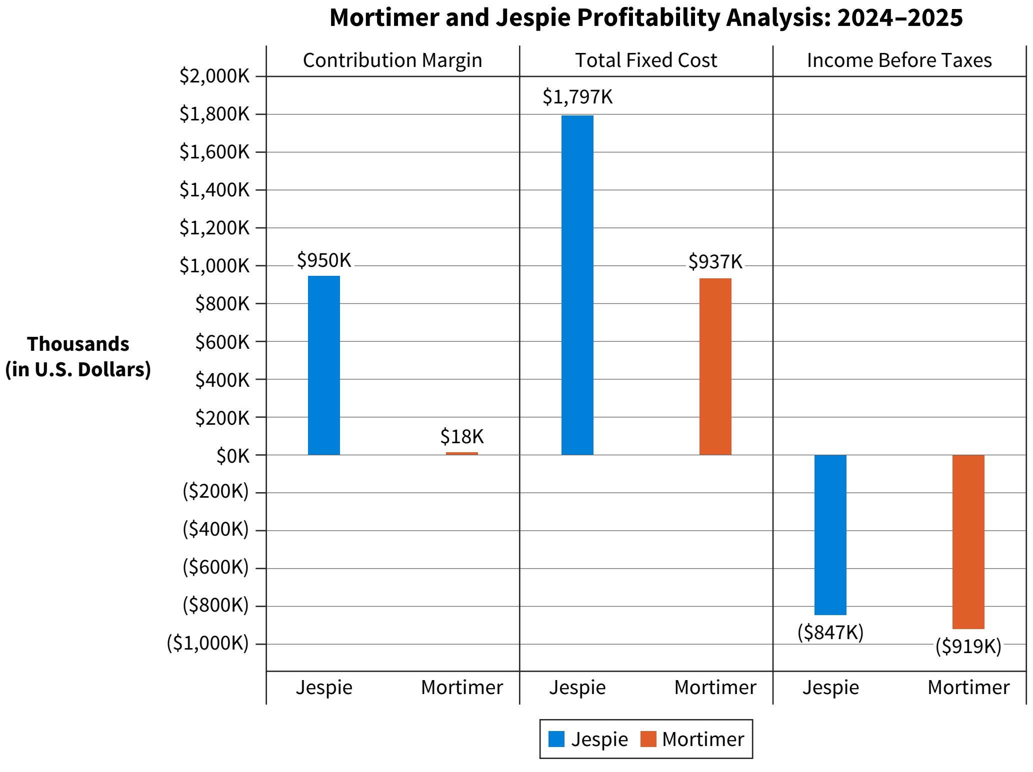


ILLUSTRATION 9.15
Sample Visualization for an Executive Audience

Illustration 9.15 shows the contribution margin, total fixed costs, and income before taxes for each model. If the decision being made is whether to discontinue the models, this visual communicates information relevant to that decision. Specifically, it shows how much contribution margin would be lost and the impact of fixed costs on the decision. Executives would need to determine if the fixed costs are avoidable if the models are discontinued.

APPLYING CRITICAL THINKING 9.3: Address the Objective

It is common to prepare many visualizations in the process of performing your analysis. When you decide which visuals to use in presentations, stay focused on the purpose of the analysis **(Purpose)**. This helps exclude visuals that are not related to the original objective of the analysis.

The purpose of the HMC analysis is to evaluate how the various brands are performing. Remain focused on that objective while preparing the presentation. For example, if the purpose is to compare the performance of five categories, choose a column chart rather than an area chart.

Define the Objective

Once data are verified and the audience has been considered, the next step is to understand the objective of the analysis, or the question/issue to be addressed in the visualization. The objective of the analysis helps determine the types of visualizations that are appropriate. It might be to show composition, relationships, distributions, trends, or comparisons of the data (**Illustration 9.16**).

ILLUSTRATION 9.16
Visualization Objectives

Objective	Explanation	Example
Composition	Show how part of the data compares to the whole.	How much revenue has each region contributed to total revenue?
Relationships	Show how the data are related.	Is there a relationship between machine hours and maintenance expense?
Distributions	Reveal how data are spread out or grouped.	Are there transactions that might be considered outliers?
Trends	Display patterns in the data.	Is there a seasonal pattern in revenue?
Comparisons	Compare values between groups of data.	How does the revenue for 2024 compare to 2025 by product?

Once the objective of the data analysis communication is identified, choose the visualization that will best convey the message. **Illustration 9.17** is a decision tree for selecting a visualization if the objective is to show composition, relationships, or distributions.

ILLUSTRATION 9.17
Visualization Decision Tree for Showing Composition, Relationships, and Distributions

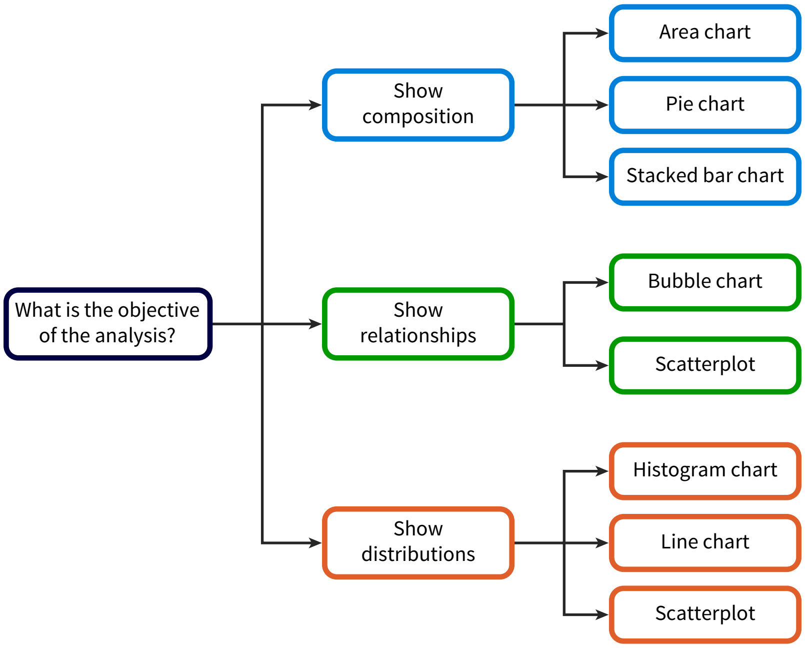
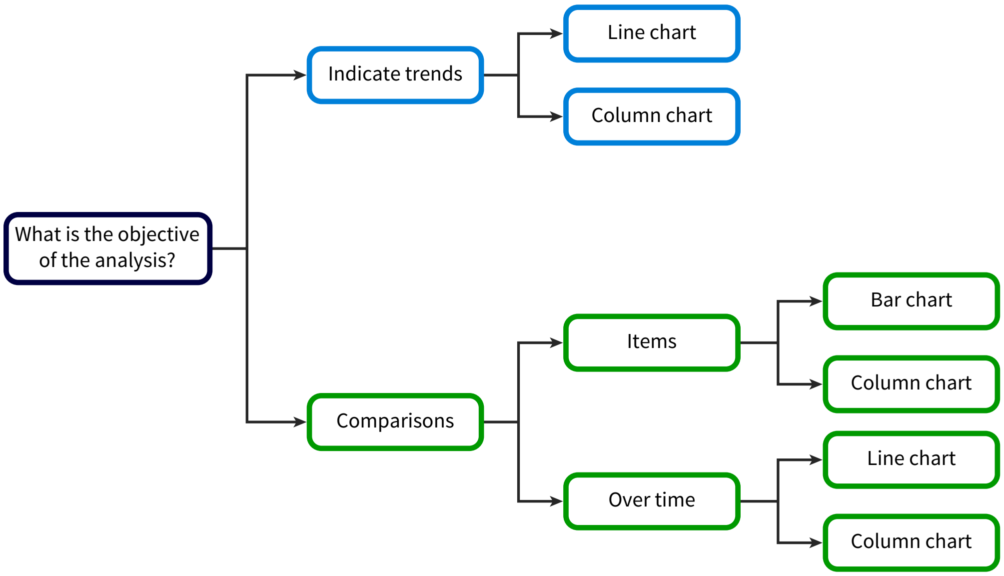


Illustration 9.18 provides guidance when the objective is to show trends or comparisons.

ILLUSTRATION 9.18
Visualization Decision Tree for Showing Trends or Comparisons



Next, we discuss best practices for creating visualizations.

Apply It 9.2
Match Objectives to Visualization Types

Auditing You have been assigned to the audit team for U.S. Outdoor Adventures. You are responsible for preparing the data visualizations that will communicate the results of some of the audit tests. The following is a list of analyses that have been prepared for the audit. Match the appropriate visualization(s) that could be used for each analysis.

Matching Choices: Scatterplot, column chart, stacked column chart, line chart, bar chart

Audit Test	Possible Visualization
Comparison of units sold and amounts billed for every sales transaction.	
Sales revenue by quarter and year.	
Number of tents sold by tent type for current year and three prior years.	
Discount offered per transaction to identify unusual discounts.	
Actual versus expected revenue.	
Changes in general ledger account balances.	

SOLUTION

Audit Test	Possible Visualization
Comparison of units sold and amounts billed for every sales transaction.	Scatterplot
Sales revenue by quarter and year.	Column chart or stacked column chart
Number of tents sold by tent type for current year and three prior years.	Stacked column chart or separate line charts for each product
Discount offered per transaction to identify unusual discounts.	Scatterplot
Actual versus expected revenue.	Line chart
Changes in general ledger account balances.	Bar chart

9.3 What Are the Characteristics of Effective Visualizations?

LEARNING OBJECTIVE 3

Describe the characteristics of an effective data visualization.

Once a visualization type is chosen, the next step is to apply best practices to ensure its effectiveness.

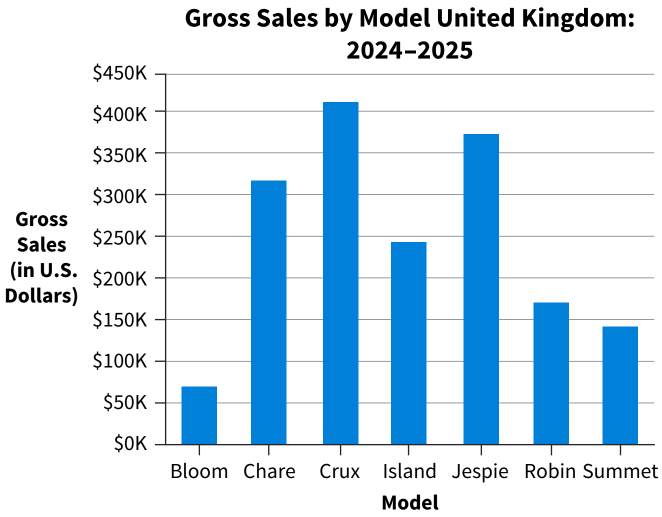
Use Principles of Visual Perception

The human brain prefers simplicity and order in visual images because it prevents us from becoming overwhelmed with information. We can process simple patterns faster than complex patterns. Gestalt is an area of psychology that was the foundation for the modern study of perception. **Gestalt principles of visual perception** describe how humans gain meaning from the stimuli around them. These principles address the natural need for humans to find order. We can create effective visualizations by considering relevant principles such as continuity, similarity, proximity, and focal point.

Continuity

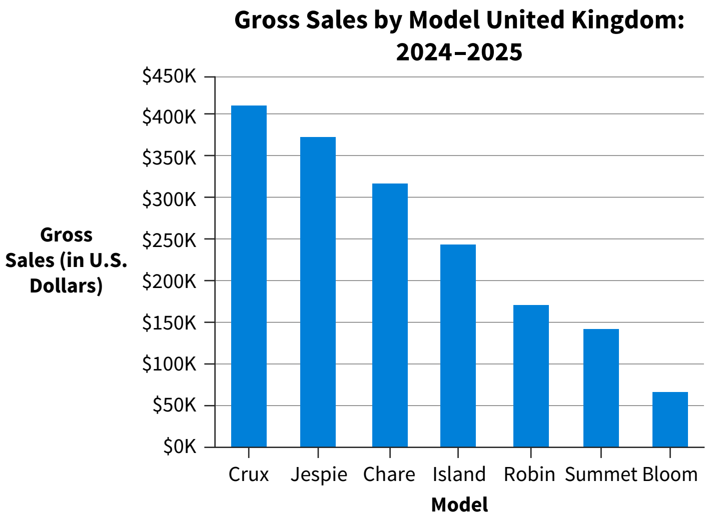
The **law of continuity** refers to how people tend to perceive any line as continuing in its established direction and how objects aligned with each other are perceived as a single path or shape. We follow lines, curves, or a sequence of shapes to determine if there is a relationship between the elements. Effective data visualizations will arrange visual objects in a line to simplify grouping and comparison. **Illustration 9.19** shows gross sales by model for HMC sales in the United Kingdom.

ILLUSTRATION 9.19 Gross Sales by Model-United Kingdom



The bars in this chart are in alphabetical order, an organization that makes it difficult for the viewer to compare the models. In contrast, **Illustration 9.20** shows the bars in descending order by gross sales. Illustration 9.20 is a more readable chart because the viewers' eyes follow a continuous path. The viewer can easily see which are the highest selling and lowest selling models.

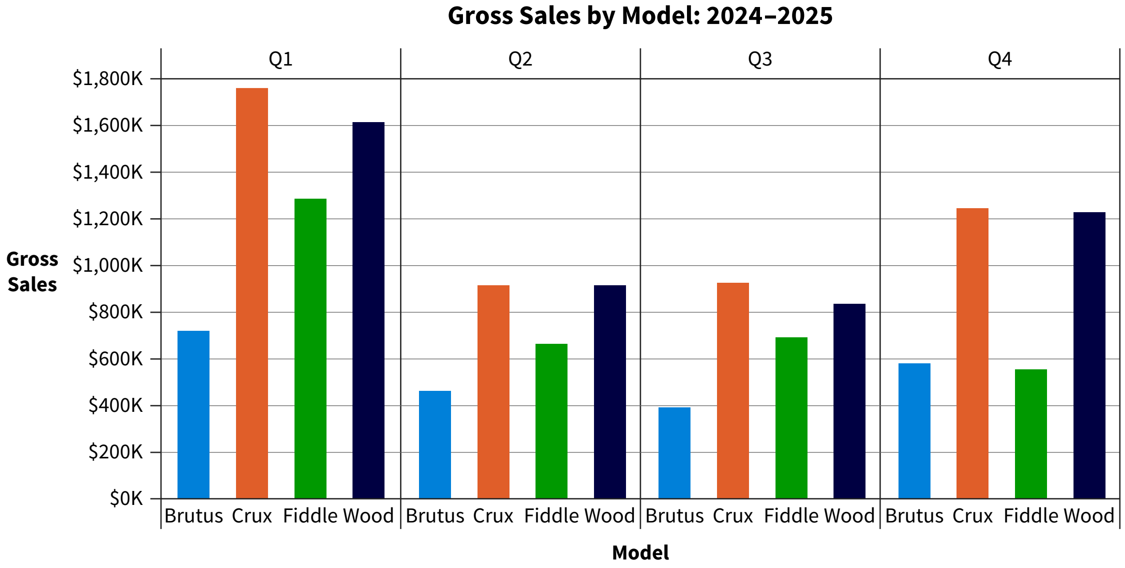
ILLUSTRATION 9.20 Gross Sales by Model Sorted in Descending Order



Similarity

The **law of similarity** states that similar elements tend to be perceived as a unified group. Items similar in color, shape, size, or location evoke the perception that they belong to the same group. **Illustration 9.21** is a bar chart showing sales by quarter for four HMC car models. This chart makes it easy to compare the brands within each quarter, but it is difficult to compare the sales of the individual brands over time.

ILLUSTRATION 9.21 Quarterly Sales by Model



If the visualization's purpose is to provide a comparison of brand performance over time, then the brands should be grouped together as shown in **Illustration 9.22**.

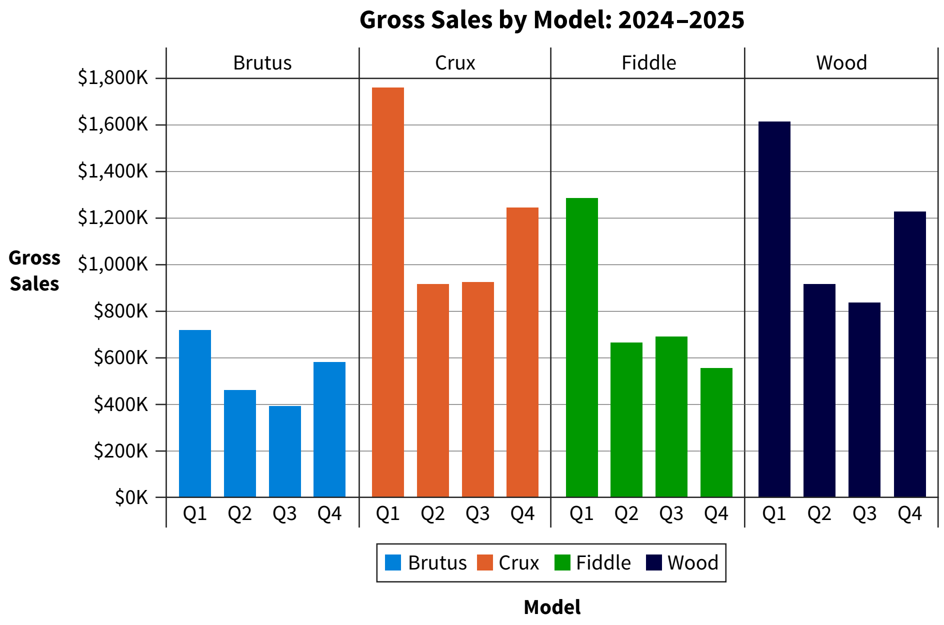


ILLUSTRATION 9.22 Quarterly Sales by Model (Revised)

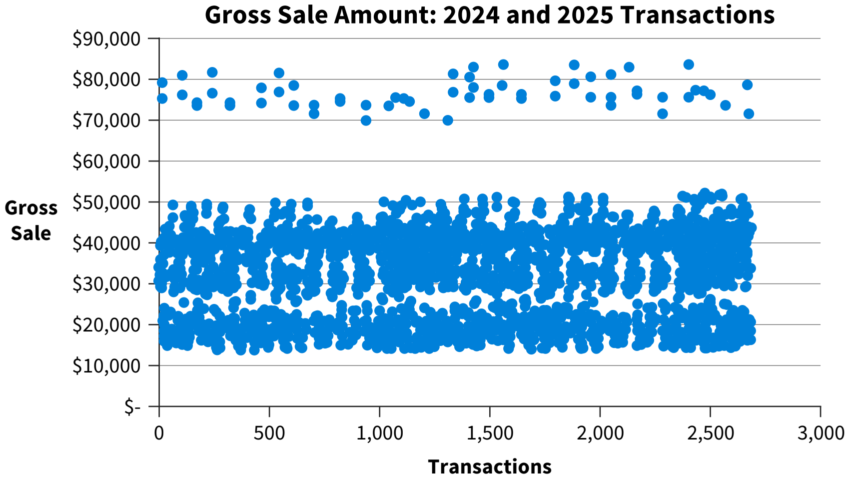
Now we can compare quarterly brand performance, as it is easy to perceive the grouping by brand. Using the law of similarity can help viewers identify the groups to which the displayed data belong.

Proximity

The **law of proximity** states that people will perceive visual elements based on how closely they are positioned to one another. Each point in the scatterplot in **Illustration 9.23** is an individual sale. It is obvious that there are two groupings of sales:

1. Sales between $70,000 and $85,000.
2. Sales between $11,000 and $55,000 thousand.

ILLUSTRATION 9.23 Gross Sale Amount by Transaction

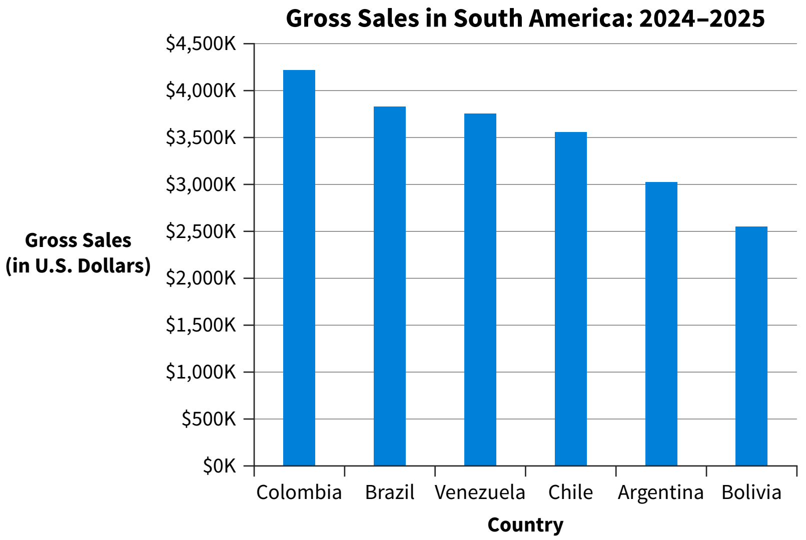


The law of proximity helps a viewer make sense of a large set of data very quickly.

Focal Point

The **law of focal point** refers to how we are more attentive to whatever stands out visually. The focal point tends to be the starting point for the viewer. Illustration 9.21 and 9.22 are examples of the focal point principle. Imagine you have prepared an analysis of HMC sales in Venezuela. You begin your presentation with an overview of sales for all the South America countries shown in **Illustration 9.24**.

ILLUSTRATION 9.24 Sales Performance in Venezuela



You have followed the law of continuity by sorting the bars from highest to lowest. However, you can improve this visual by highlighting the country you want your viewers to focus on (**Illustration 9.25**).

ILLUSTRATION 9.25 Sales Performance Graph with Focal Point

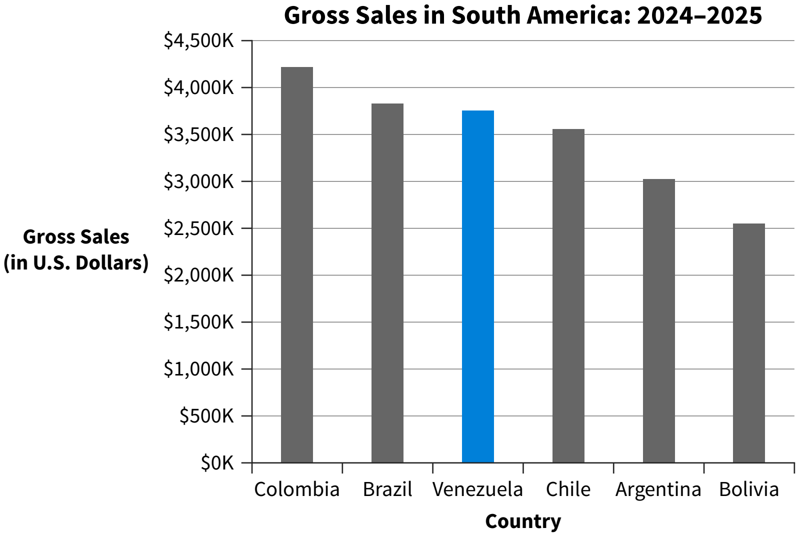


Illustration 9.25 still provides the information about the other South American countries, but using a different color for Venezuela focuses the viewers' attention on that country.

Consider Preattentive Attributes

Gestalt principles help create visualizations that are easily consumed by viewers. But how do we capture their attention? Studies have shown that someone will decide within 3 to 8 seconds whether to continue looking at a visualization or turning their attention to something else.[11] **Preattentive attributes** are visual properties we notice without realizing it. Size, color, and position are preattentive attributes in visualizations that can direct the audience's attention.

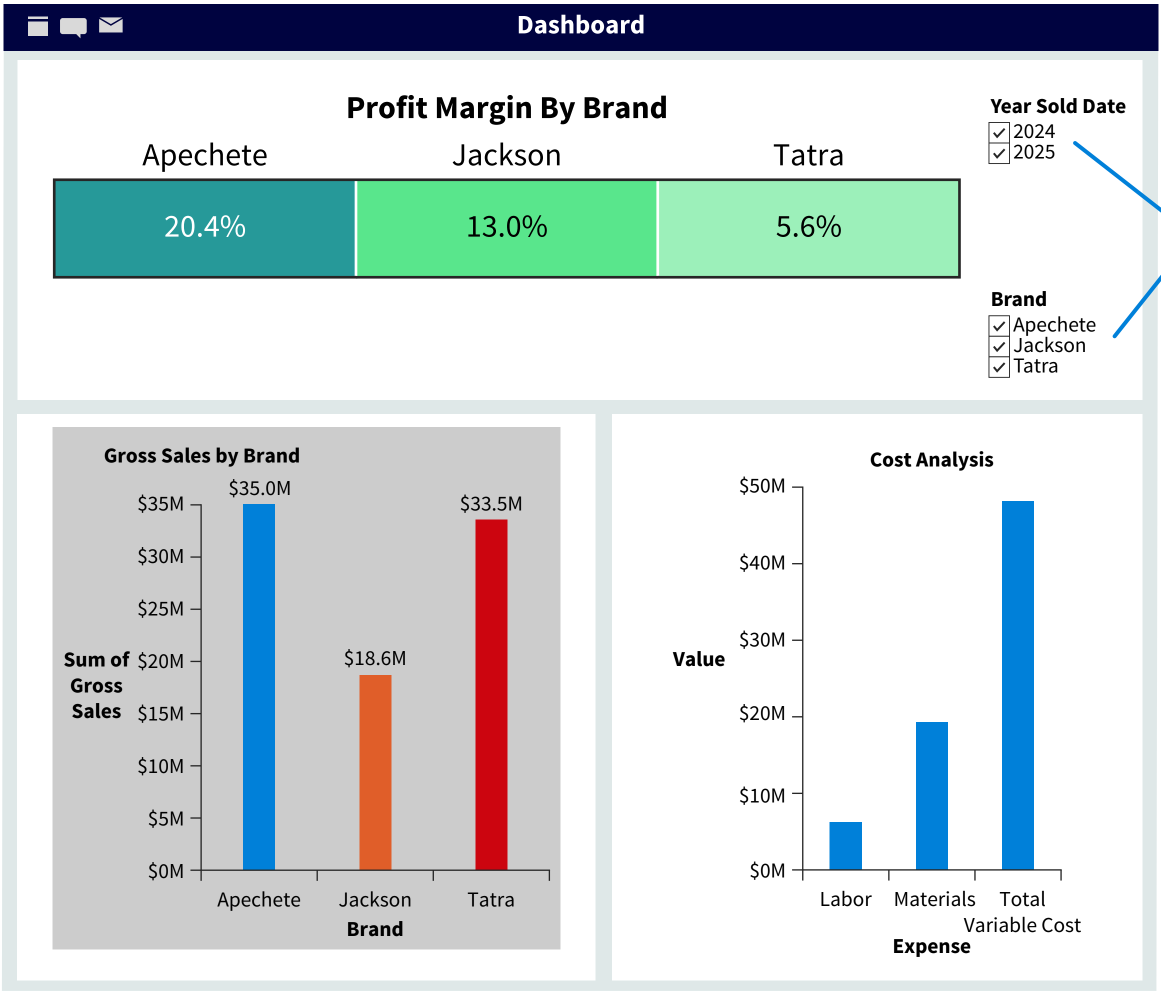
Size

If elements of a visualization vary in size, the audience assumes those size differences matter. In other words, relative size will be interpreted as relative importance. This is true in individual visualizations as well as in dashboards.

Illustration 9.26 is a dashboard designed to help management monitor the sales and revenue of their brands. The visualization with the largest font size in the dashboard is the profit margin by brand. Because this visual is larger than the others, the viewer focuses on it before viewing gross sales and the cost analysis.

[11]Knaflic Nussbaumer, C. (2015). Storytelling with Data. Hoboken, NJ: Wiley.

ILLUSTRATION 9.26 Profit Margin by Brand Dashboard



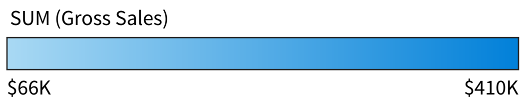
While a visualization's relative size indicates its importance, the size of the text within the illustration, relative to other text, will also indicate importance.

Color

As you learned in the discussion of Gestalt principles, color can make visualizations more effective. There are three ways to use color in visualizations:

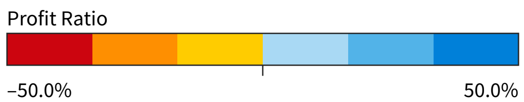
1. Sequential: Color is ordered from low to high (**Illustration 9.27**).

ILLUSTRATION 9.27 Sequential Color Scale



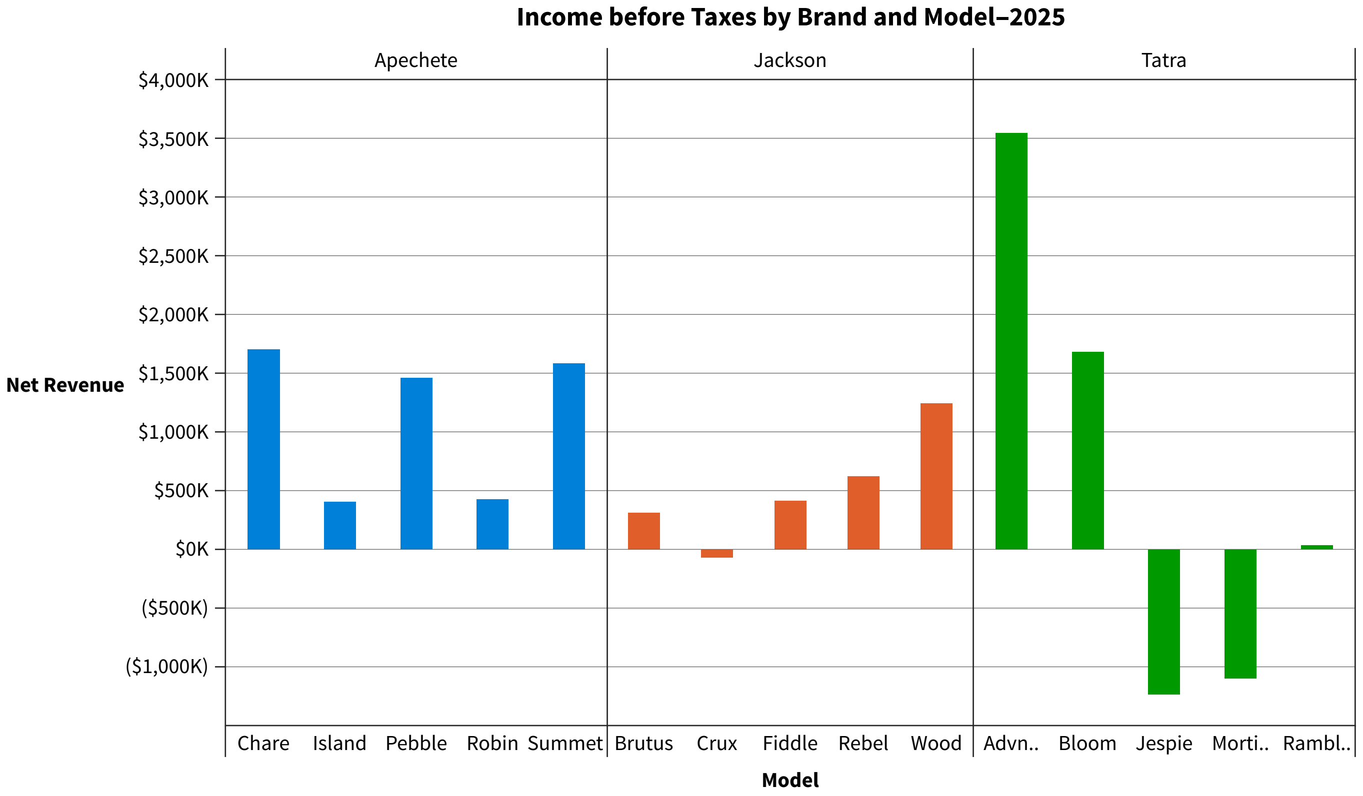
2. Diverging: There are two sequential colors with a neutral midpoint (**Illustration 9.28**). This type of scale is useful for showing gains and losses.

ILLUSTRATION 9.28 Diverging Color Scale



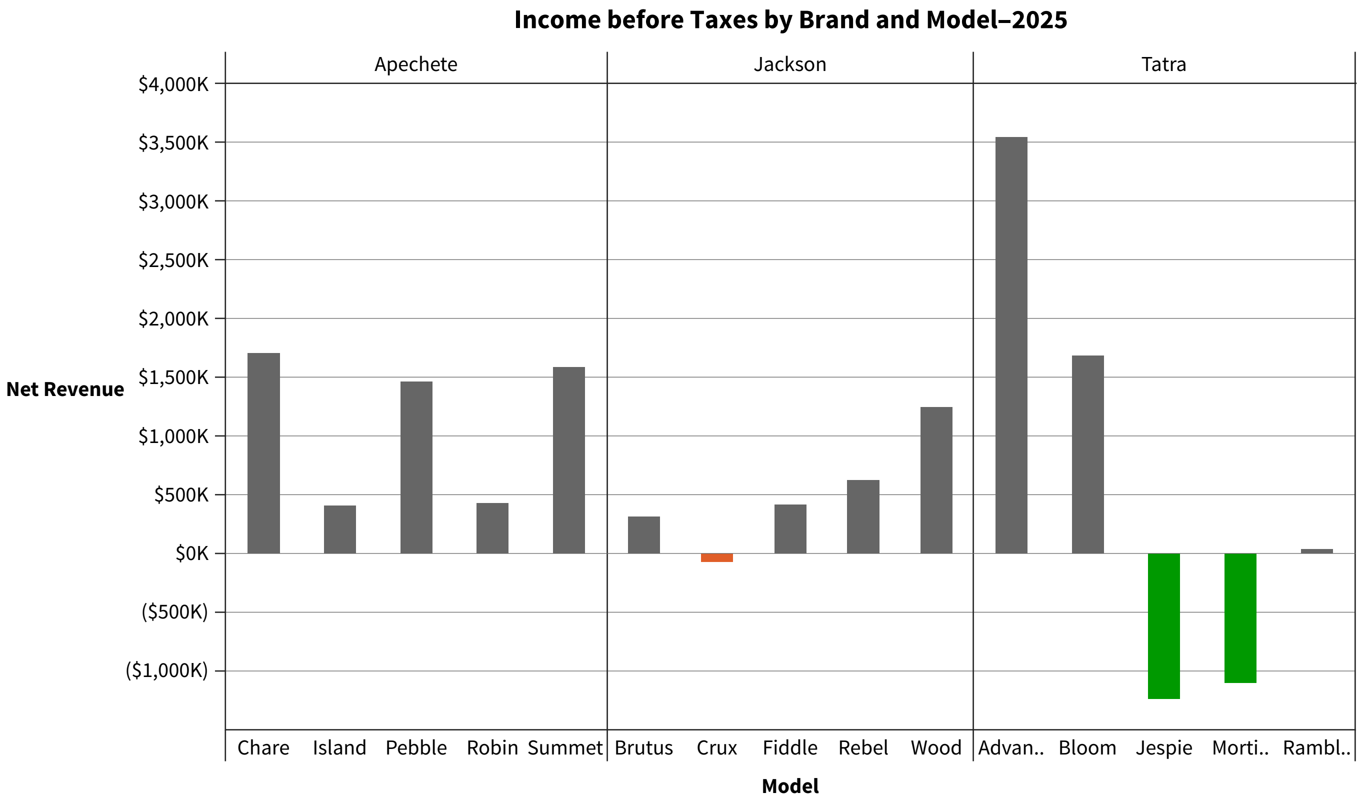
3. Categorical: There are contrasting colors for individual comparison. This is a common use for color when comparing categories. **Illustration 9.29** shows how contrasting colors can be used in a column chart.

ILLUSTRATION 9.29 Categorical Color Use in a Column Chart



Color can also highlight or alert the audience to focus on something specific in the visualization. **Illustration 9.30** shows the use of highlighting. The visualization is identical to Illustration 9.29, but this version only highlights the brands with net losses.

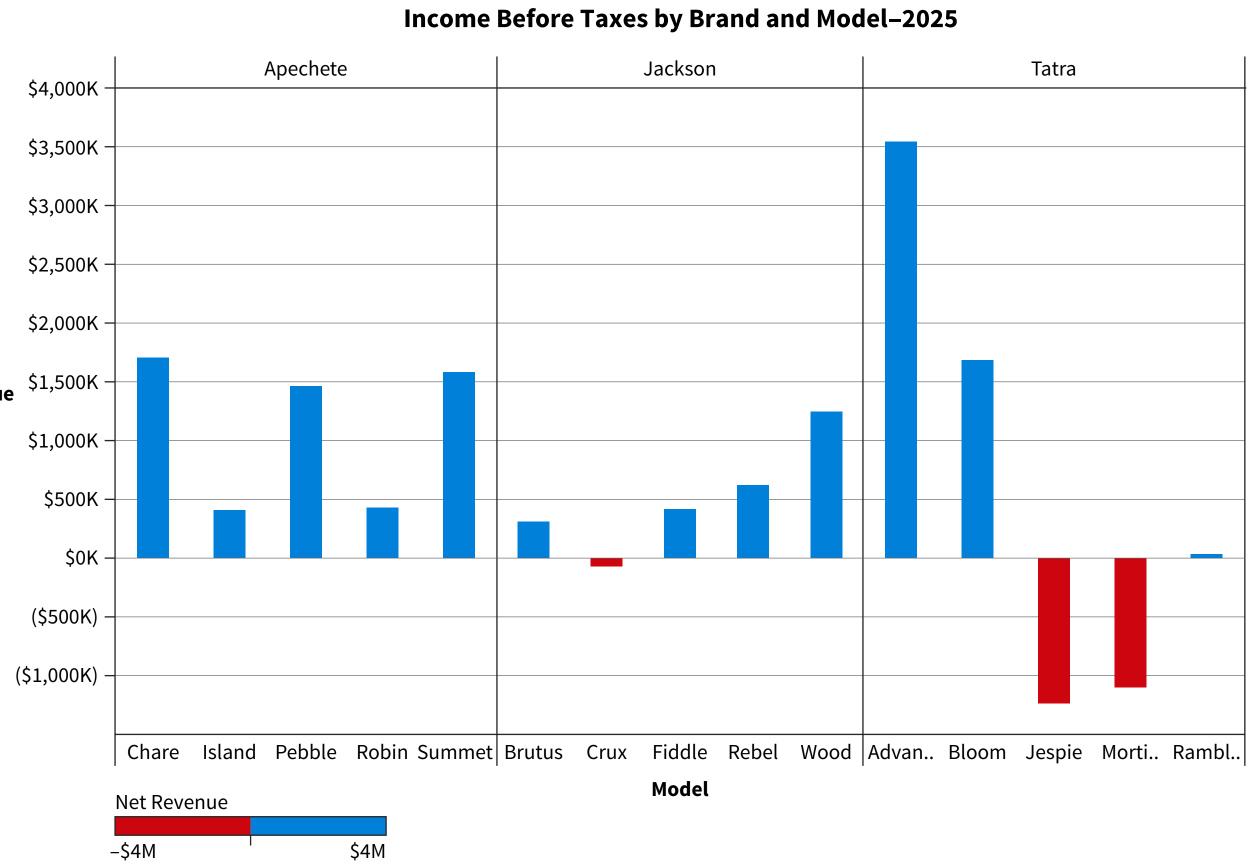
ILLUSTRATION 9.30 Using Color to Highlight



Highlighting the brands with net losses and making all the other bars fade out encourages the audience to immediately focus on the darker bars. Using color this way is also another example of applying Focal Point Law.

What if the intention is to grab the audience's attention and alert them to an issue or problem? **Illustration 9.31** changes the color of the losses in Illustration 9.30 to convey a sense of urgency.

ILLUSTRATION 9.31 Using Color to Alert the Viewer



Color evokes emotion, so be thoughtful of the tone conveyed by the color. For example, red evokes a sense of urgency. In visuals with financial information, red is also an indication of poor performance. By highlighting the models with negative net revenue (Illustration 9.31) the viewer can quickly see that the Crux, Jespie, and Mortimer are losing money. Regardless of its tone, color should be used both consistently and sparingly:

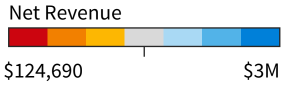
- If the visualization includes color, variables should be indicated by the same color to avoid confusion.
- Interpreting too many colors can overwhelm the audience, so only add color that makes it easier to interpret the visualization.

Illustrations 9.32 visualizes net revenue by region using multiple colors in a diverging color scale.

ILLUSTRATION 9.32
Visualization with Too Much Color

Income before Taxes by Region

Region	Apechete	Jackson	Tatra
Europe	$ 1,246,353	$ 1,010,446	$ 136,006
North America	$ 2,013,345	$ 963,145	$ 2,719,779
South America	$ 2,372,282	$ 625,278	$ 124,690

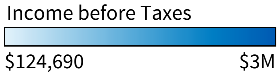


Due to the additional cognitive load created by the colors, interpreting the table is more difficult. The viewer must interpret the numbers, regions, brands, and seven different colors to understand the table.

On the other hand, the single-color gradient scale in **Illustration 9.33** makes it easier for the audience to quickly see that the Tatra in North America has the highest net revenue.

ILLUSTRATION 9.33 Revised Visualization Using a Single-Color Gradient Scale

Income before Taxes by Region			
Region	**Apechete**	**Jackson**	**Tatra**
Europe	$ 1,246,353	$ 1,010,446	$ 136,006
North America	$ 2,013,345	$ 963,145	$ 2,719,779
South America	$ 2,372,282	$ 625,278	$ 124,690

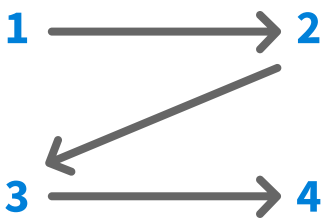


Finally, consider users with colorblindness when creating visualizations. Color blindness affects 8% of men and 0.5% of women. The most common color blindness is the inability to distinguish between red and green shades, so avoid using red and green in the same visualization.

Position

Item placement in visualizations and dashboards matters. Most people begin viewing a visualization from the top left corner and then scan in a zig-zag motion through the entire visualization (**Illustration 9.34**). Think about viewing the dashboard visualization in Illustration 9.26. You may have started by reading the title "Profit Margin by Brand." Therefore, be sure to put the most important information at the top left of the visualization.

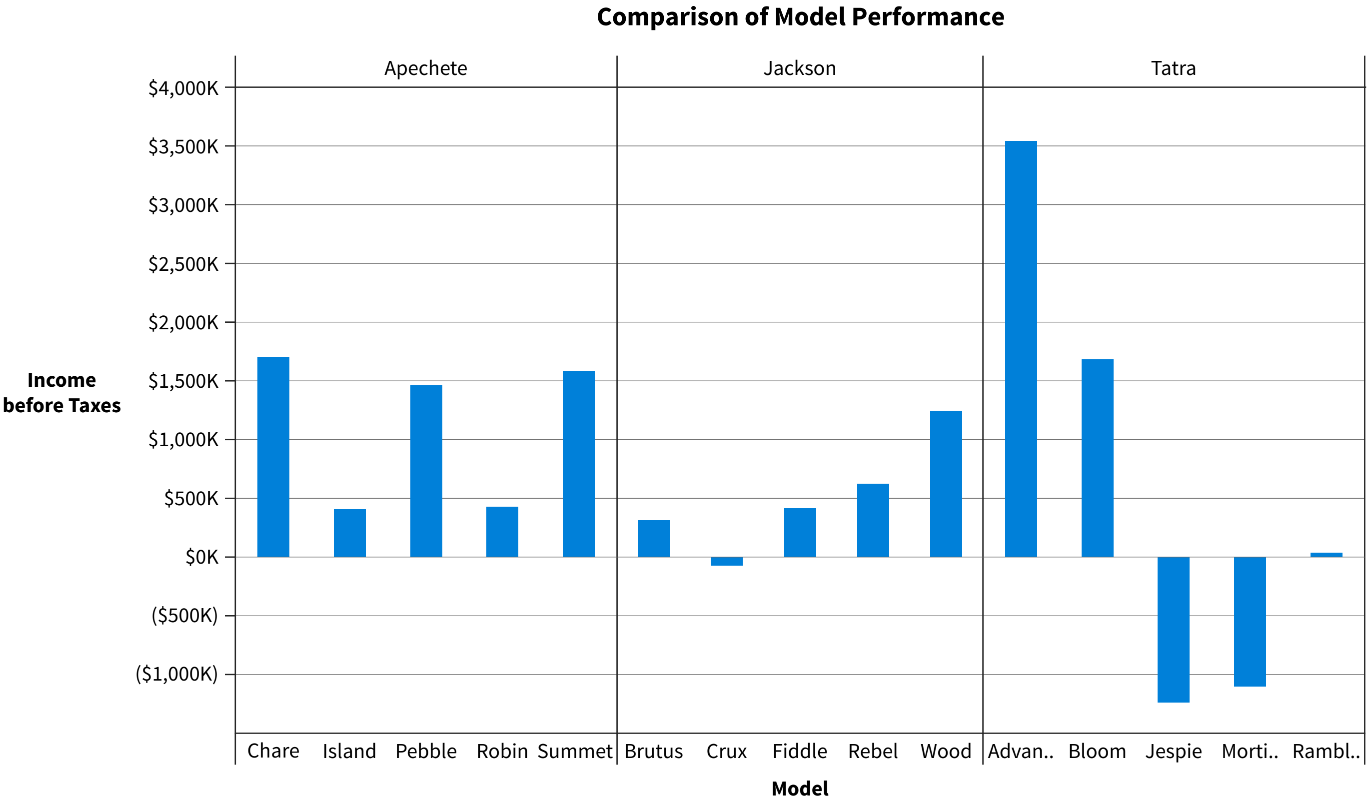
ILLUSTRATION 9.34 How Information is Viewed on a Screen or Page



Titles

Finally, include a factual and neutral title. Avoid using unnecessary descriptive words. The title should be a noun that represents what was measured and when. In Illustration 9.31 the title is simply: Income before Taxes by Model–2025. Income before taxes is being measured, so the reader knows what the measurement is and the time period. **Illustration 9.35** is an example of what not to do.

ILLUSTRATION 9.35 Incorrect Way to Title a Visualization

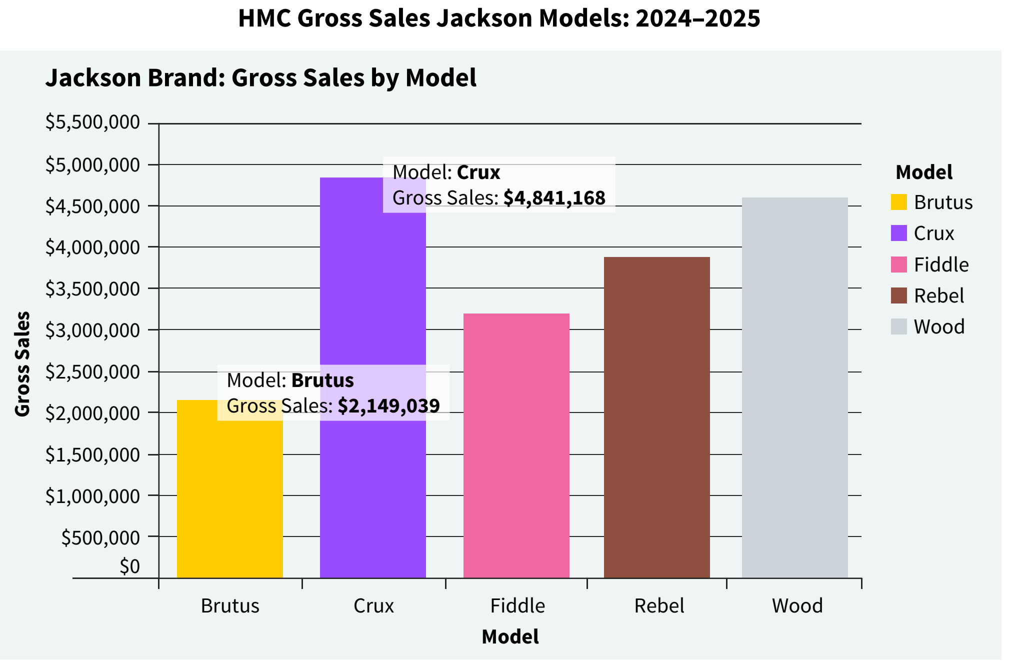


The title of the visual does not give the reader a clear description of what is being measured (income before taxes by model) or the time period represented (2025). A better title would be: "Income before Taxes by Model–2025."

Avoid Clutter

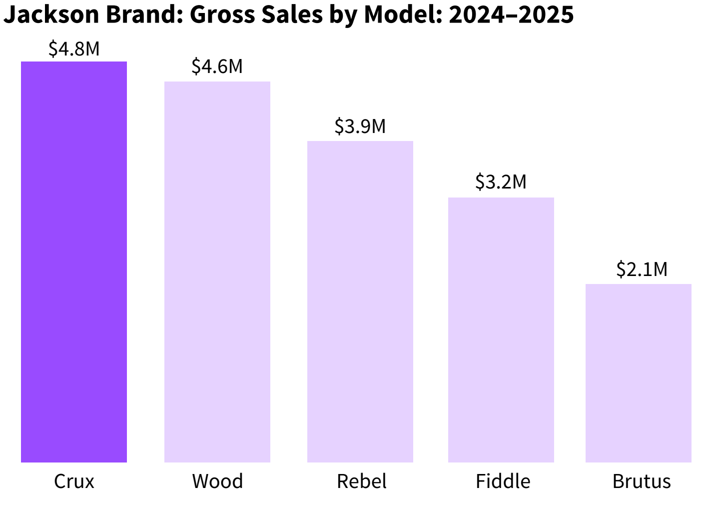
After considering principles of perception and preattentive attributes, consider visual clutter in visualizations. Clutter is the enemy of a good visualization. The more cluttered the visualization, the harder it is for the viewer to understand the results. Remove any non-data related details from the visualization and check for redundant information. The purpose of **Illustration 9.36** is to show gross sales for each model produced under the Jackson Brand and identify the bestselling model.

ILLUSTRATION 9.36
Visualization Containing Clutter



The visual does show sales for the Jackson brand models. However, there are so many elements that the viewer could become overwhelmed. **Illustration 9.37** uses the same data but removes the unnecessary aspects of the visualization.

ILLUSTRATION 9.37 Revised Visualization with Clutter Removed



Recall that the purpose of the visualization was to show how the various models are performing for the Jackson brand. The changes are summarized in **Illustration 9.38**.

ILLUSTRATION 9.38 Summary: Before and After Reducing Visualization Clutter

Cluttered Visualization	Revised Visualization
Too many colors are not adding to the meaning of the graph because the models are also directly labeled.	The visual is a single color.
Axis labels are provided, but the audience must guess the exact amount.	Label each bar directly with the sales volume amount and remove the axis. Sales volume is in the title of the graph, so remove the axis header.
Annotation text boxes identify the bestselling model.	Sort the data from highest to lowest.
Background color of the graph is unnecessary.	Remove background color. Remove gridlines since we are removing the axis and labeling the columns directly.
Models are listed in alphabetical order. There is no indication which model the audience should focus on.	The bestselling model is highlighted and the others are grayed out to draw attention to it.

The background in Illustration 9.36 was shaded a gray color, but it is not necessary to shade the background of visualizations. It is especially important to avoid using a black background because although it might look striking, a black background can make lighter colors and words appear blurry to anyone with an astigmatism. According to the World Health Organization (WHO), 43 % of the population have astigmatism, a visual condition where there is a refractive error that impedes the eye from focusing light evenly on the retina.

Illustration 9.39 is a checklist, based on the best practices discussed here, to use when evaluating your data visualizations.

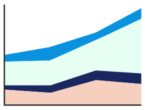
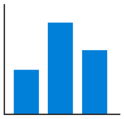
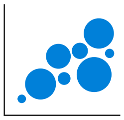
ILLUSTRATION 9.39 Data Visualization Checklist

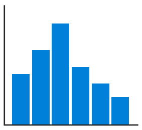
1.	Data have been verified.	11.	Graph does not have border line (reducing clutter).
2.	Visualization addresses the objective or question being addressed in the analysis.	12.	Data are labeled directly rather than in a separate legend.
3.	It is appropriate for the intended audience.	13.	Redundant labels are removed.
4.	The type of visual is appropriate for the data and the level of precision needed.	14.	Y-axis scales start at zero.
5.	A descriptive title is left-justified in the upper left corner.	15.	Proportions are accurate.
6.	Subtitle and/or annotations provide additional information.	16.	Data are intentionally ordered.
7.	Significant findings or conclusions are highlighted.	17.	Display is free from distractions.
8.	Text size is hierarchical and readable.	18.	Color is used sparingly and consistently.
9.	Text is horizontal when possible.	19.	Color is legible for people with colorblindness.
10.	Gridlines, if used, are muted.	20.	Spatial flow is intuitive for the viewer.

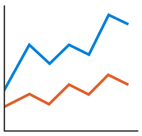
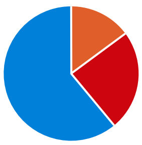
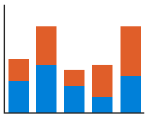
Use Visualization-Specific Best Practices

The best practices previously discussed and those listed in Illustration 9.39 are an essential starting point, but there are also best practices based on the type of visualization to consider. **Illustration 9.40** is a summary of best practices for common data visualizations.

ILLUSTRATION 9.40 Best Practices by Visualization Type

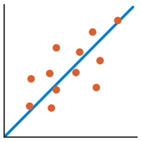
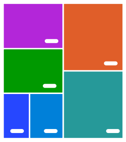
Visualization	Best Practices
Area Chart 	• Do not use with more than four categories to avoid confusion and clutter. • Start the y-axis at zero. • Put highly variable data on the top and data with low variability on the bottom.
Bar Chart  Column Chart 	• Use horizontal bars if there are more than 7 categories or long category labels. • Use horizontal labels for better readability. • Space bars appropriately and consistently. • Use color sparingly, or as an accent. • Always have a zero baseline (the y-axis begins at zero). • Compare 2–7 categories with vertical columns.
Bubble Chart 	• Label bubbles and make sure they are visible. • Scale bubble size by area and not diameter. • Do not use bubbles if they are all similar in size.

Histogram Chart 	• Use a zero baseline.
	• Choose an appropriate number of bins:
	  • Bins are numbers that represent the intervals into which the data will be grouped.
	  • Bins define the groups used for the frequency distribution.
	  • Generally, include between 5–15 bins.

Line Graph 	• Time runs from left to right. • Be consistent when plotting time points. • Use solid lines, not dotted. • Use a zero baseline. • Do not plot more than four lines. Use multiple charts, instead.
Pie Chart 	• Most impactful with small data sets. • Best to use when showing differences within groups based on one variable. • Ensure the data adds to 100%. • Limit the chart to a maximum of five segments. • Start the first segment at 12 o'clock position.
Stacked Bar Chart 	• Can be vertical or horizontal. • Follow same best practices as bar charts. • Used to show comparisons of subcomponents between categories. • Best used when there are not too many subcomponents. • Consider using a 100% stacked bar to make comparisons between the bars and subcomponents easier.

(Continued)

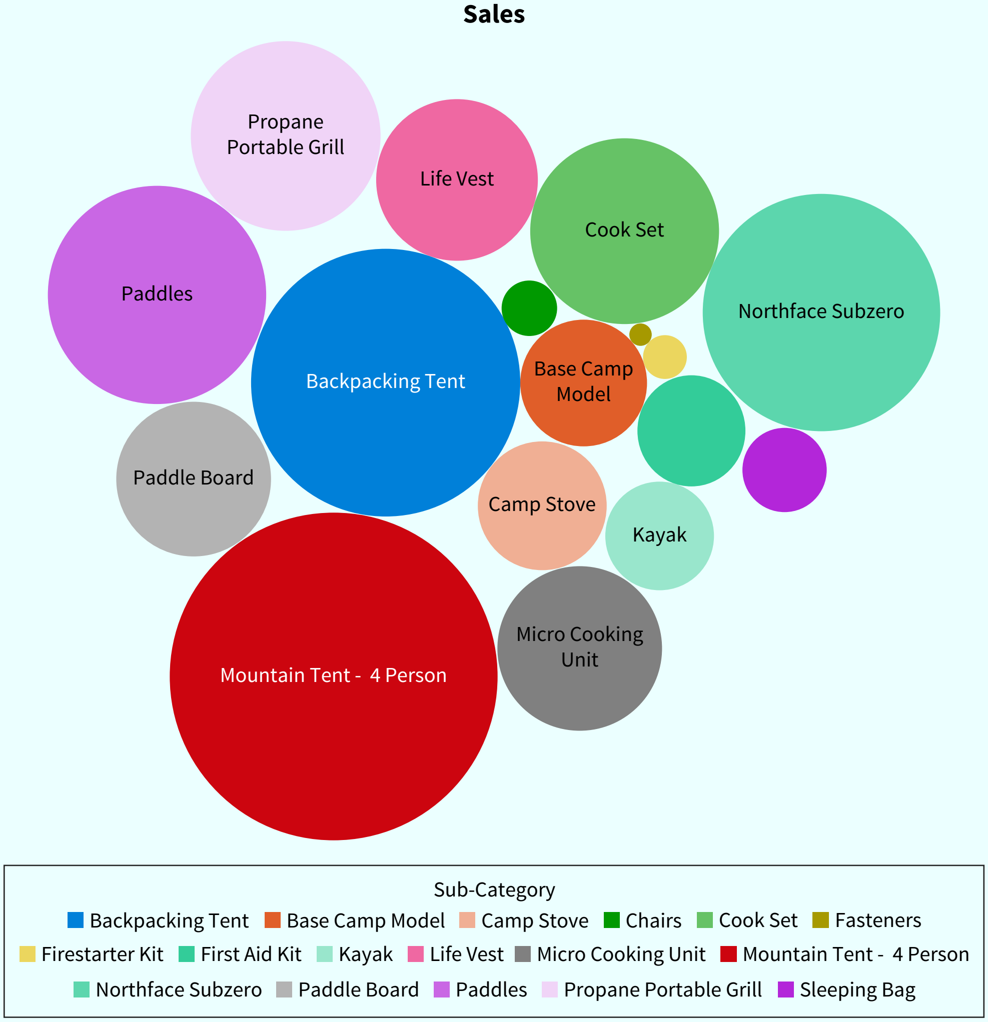
ILLUSTRATION 9.40 *(Continued)*

Visualization	Best Practices
Scatter Chart 	• Data set should be in pairs with an independent variable (x-axis) and a dependent variable (y-axis). • Use if order is not relevant–otherwise use a line graph. • Do not use if there are only a few pieces of data or if there is no correlation.
Tree Map 	• Appropriate when precise comparisons are not important. • Use bright, contrasting colors so that each box is easily defined. • Label boxes with text or numbers.

Applying best practices for each specific visualization, the principles of visual perception, and appropriately using preattentive attributes will help ensure you are creating a good visualization and communicating results effectively. Failing to do so can lead to visualizations that are misleading, which is the topic of the next section.

Apply It 9.3
Evaluate Visualizations

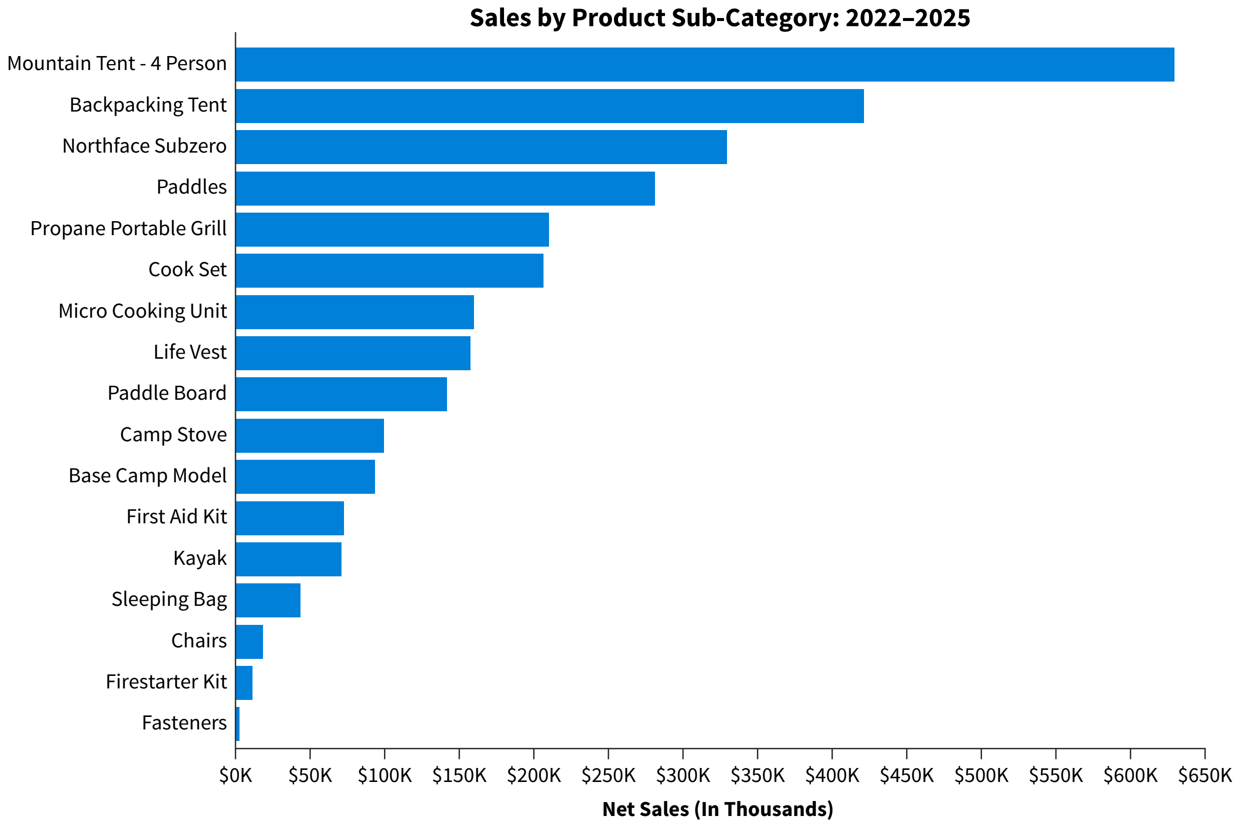
Data **Managerial Accounting** U.S. Outdoor Adventures has prepared the following visualization to help better understand which products are their bestselling products.



1. Discuss the effectiveness of this visualization using the best practices for data visualizations.
2. Use the U.S. Outdoor Adventures data set to create an improved visualization.

SOLUTION

1. Some issues with the visualization include:
   - The title is not clear. It should be more specific and include dates from the data.
   - A bubble chart is not the correct choice. There are too many bubbles, and it is difficult to interpret the extent of the difference between the sub-categories.
   - The visualization uses too many colors.
   - There are no numbers or scale, so the sales in each sub-category are unclear.
   - It is unclear if the visualization is showing sales dollars or number of sales.
2. The suggested visualization has a clear title and sorts the products from highest selling to lowest. The scale is clearly indicated as net sales in thousands of dollars.



9.4 What Makes Data Visualizations Misleading?

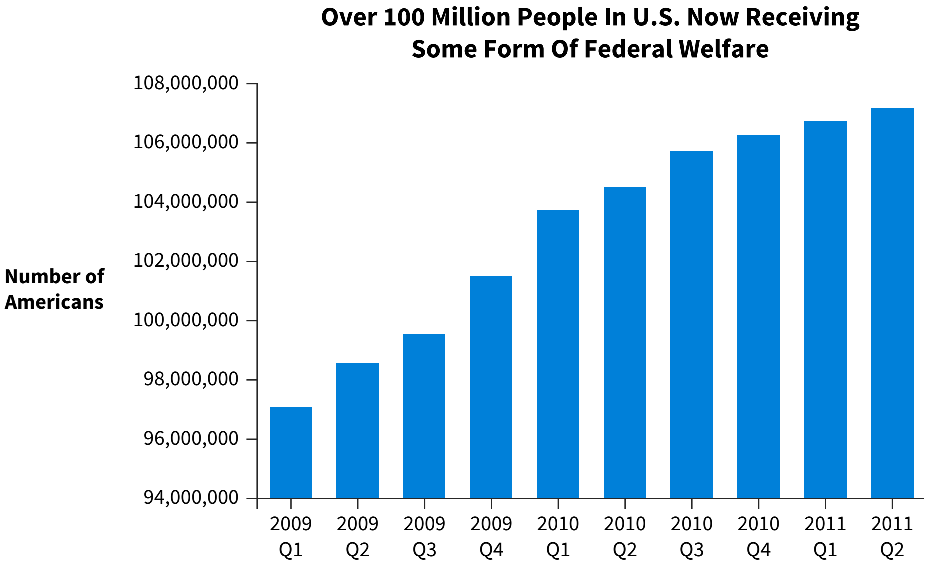
LEARNING OBJECTIVE 4

Recognize misleading data visualizations.

Ethics and data visualization best practices are intertwined. Because visualizations can significantly influence how data are used to make decisions, there is an ethical obligation to not mislead the viewer. Using best practices is the first step for mitigating the risk of creating a misleading data visualization. The second is developing an awareness of how visualizations can mislead to avoid making those mistakes. Visualizations may mislead by omitting the baseline, manipulating the y-axis, selectively picking the data, using the wrong type of graph, and going against conventions.

Omitting the Baseline

Illustration 9.41 is a visualization that was posted in a USA Today blog.



Source: US census survey / U.S Department of Commerce / Public Domain.

ILLUSTRATION 9.41 Federal Welfare Spending: Omitted Baseline

The data shows that there was an alarming increase in federal welfare spending. However, notice the starting point is 94 million, not zero. The same data is presented in **Illustration 9.42** with a zero baseline.

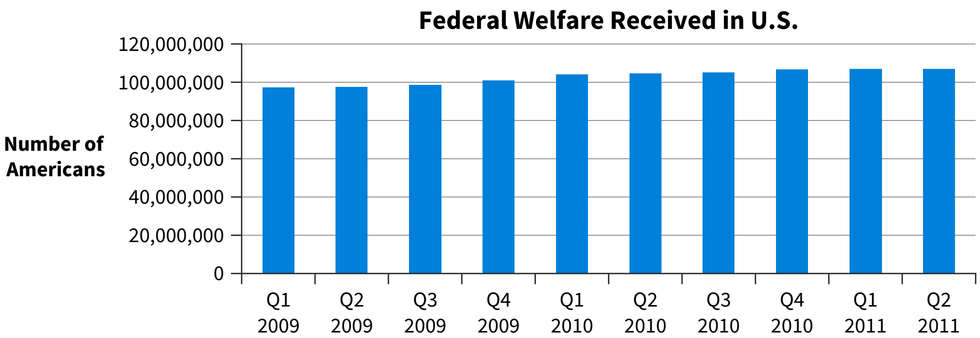


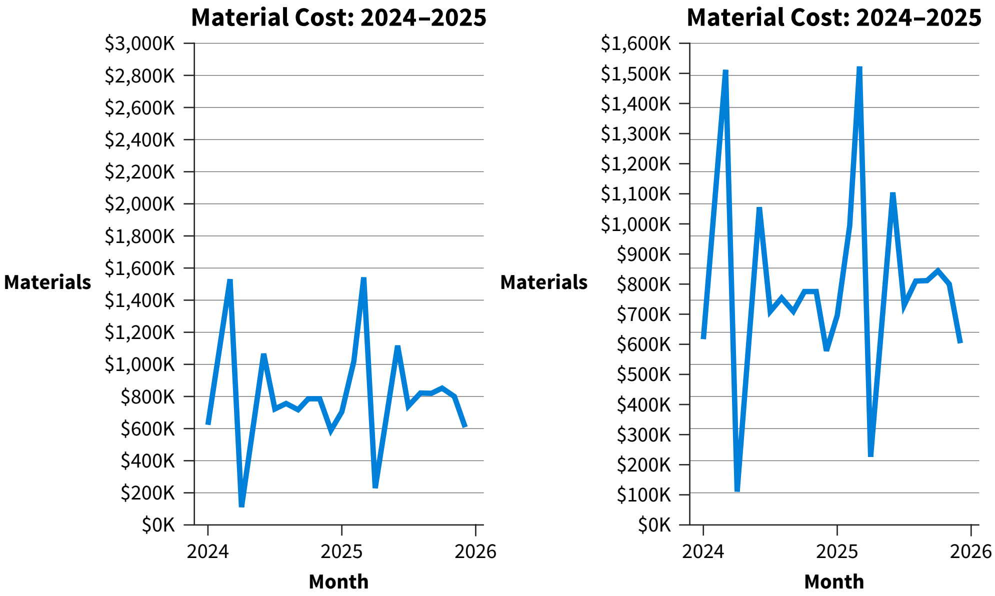
ILLUSTRATION 9.42 Federal Welfare Spending: Zero Baseline

Notice the difference between the two visualizations? The increase in welfare spending does not seem nearly as dramatic. Always graph data with a zero baseline to avoid misleading the audience.

Manipulating the Y-Axis

Like omitting the baseline, manipulating the scale on the y-axis can affect how the data are interpreted. Expanding or compressing the scale on the y-axis can make changes in the data seem more or less significant. **Illustration 9.43** is a comparison of two visualizations using the same data.

ILLUSTRATION 9.43
Manipulating the Y-Axis Examples

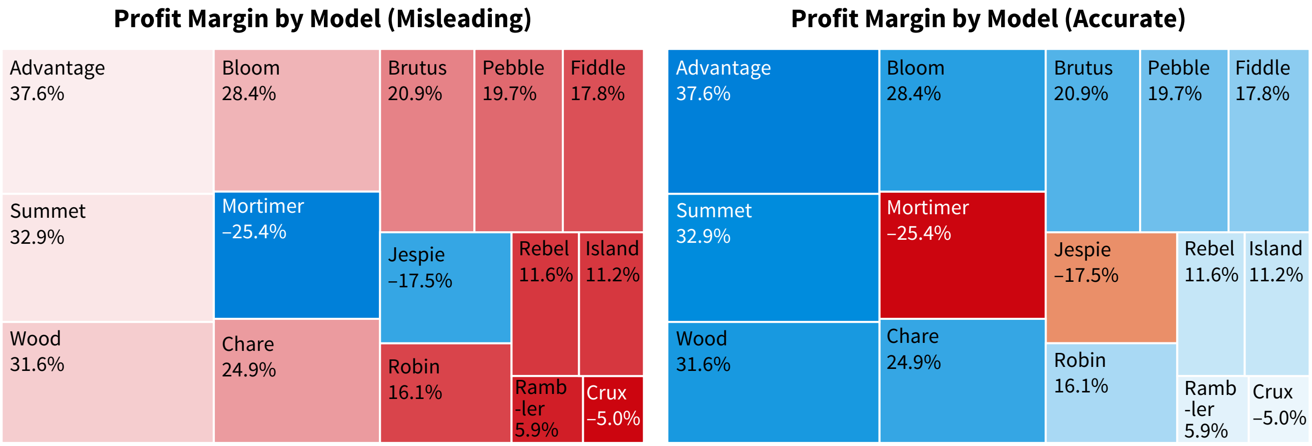


Material costs look much more volatile in the graph on the right. If the preparer wanted to make costs look less volatile, the y-axis can be manipulated like the graph on the left by increasing the scale maximum.

Going Against Conventions

A general standard in data visualization is that darker colors indicate higher numbers in a color scale. **Illustration 9.44** provides an example of what happens when that standard is violated.

ILLUSTRATION 9.44 Going Against Conventions Example



If the viewers did not concentrate on the numbers in the visualization, they would assume the Mortimer has the highest profit margin according to the tree map on the left. The tree map on the right follows general color conventions, and it is immediately clear that the Mortimer has the lowest profit margin.

APPLYING CRITICAL THINKING 9.4: Consider Visualization Risks

When creating data visualizations, think about the risks involved **(Risk)**:

- The viewer may not understand the visual.
- The visual is unclear.
- The visual is misleading.

For the HMC visualization in Illustration 9.44, if you did not consider the risk of creating a misleading visualization, you may not realize that the color convention is misleading.

Selectively Choosing the Data

Including only some data points in a visualization may also create a false impression of the data. **Illustration 9.45** is an example of what happens when data are selectively used.

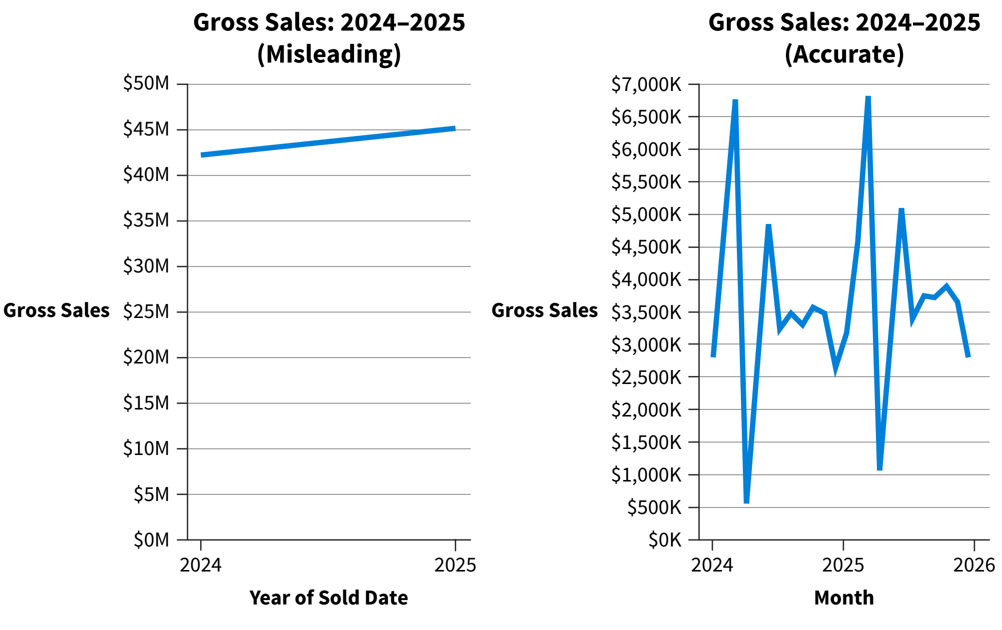


ILLUSTRATION 9.45 Selectively Choosing the Data Example

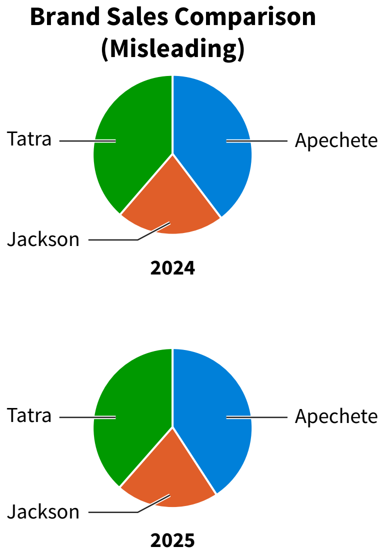
The graph on the left plots the sum of sales for the year 2024 and the year 2025, whereas the graph on the right plots sales for each month. Notice that the graph on the left creates the impression that sales have gradually increased. However, as the graph on the right shows, there is a great deal of variability in monthly sales.

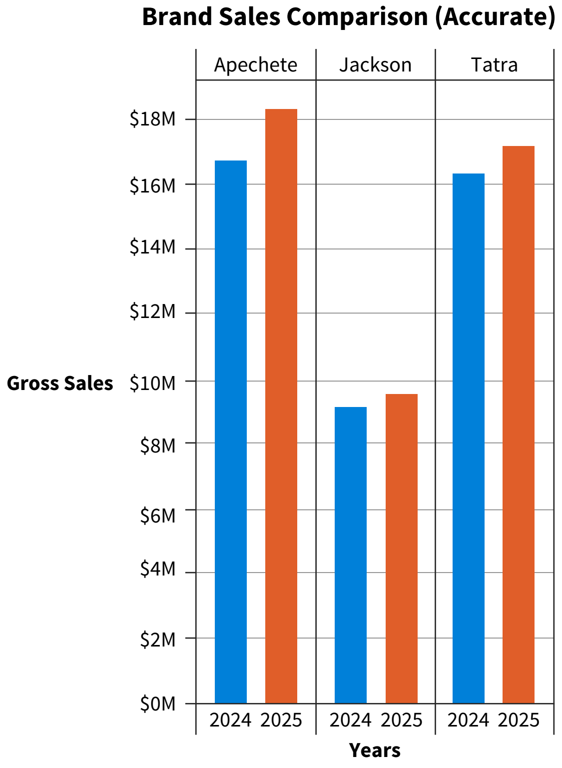
Using the Wrong Type of Graph

Sometimes choosing the wrong type of graph can make it difficult to interpret the data and results in misleading the viewer. A visualization that is not appropriate for the type of data or analysis results being reported makes it difficult for the audience to interpret the message.

Consider the two visualizations in **Illustration 9.46**.

ILLUSTRATION 9.46 Using the Wrong Type of Graph Example



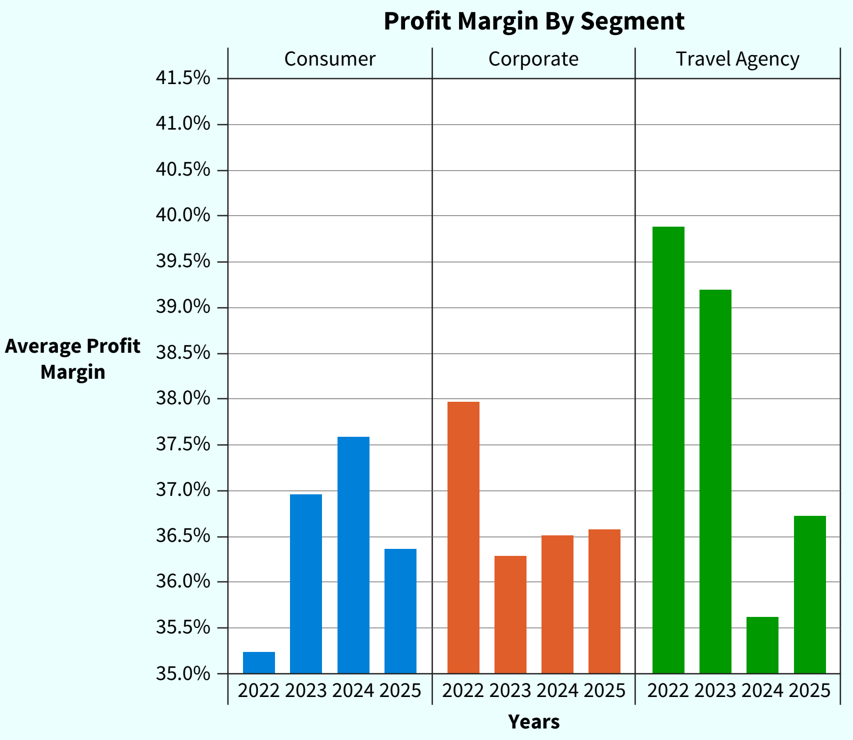


If the objective is to show a comparison of how each brand performed by year, a bar chart is a better choice. Unlike a pie chart, bar charts make it easy to determine if amounts have increased or decreased. This is because it is difficult for the human eye to interpret proportions and changes in proportions in a pie chart.

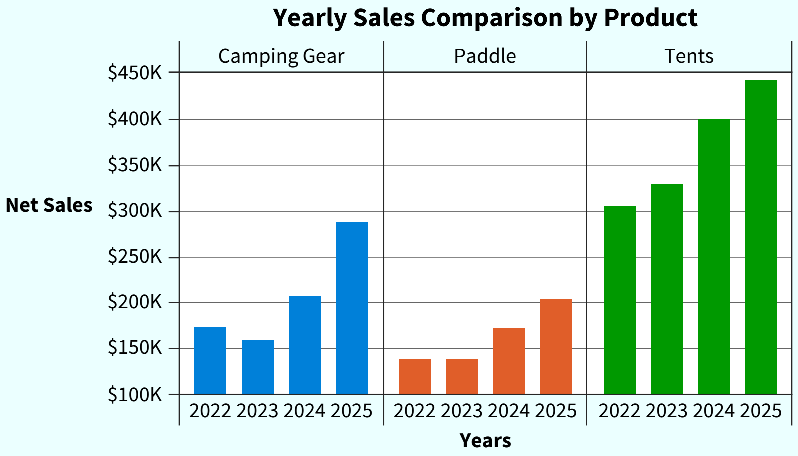
Apply It 9.4
Identify Misleading Data Visualizations

Data **Managerial Accounting** You are a managerial accountant for U.S. Outdoor Adventures reviewing data analyses prepared by someone in your department. For each visualization, identify if it is or is not misleading, and correct those that are misleading.

1.



2.



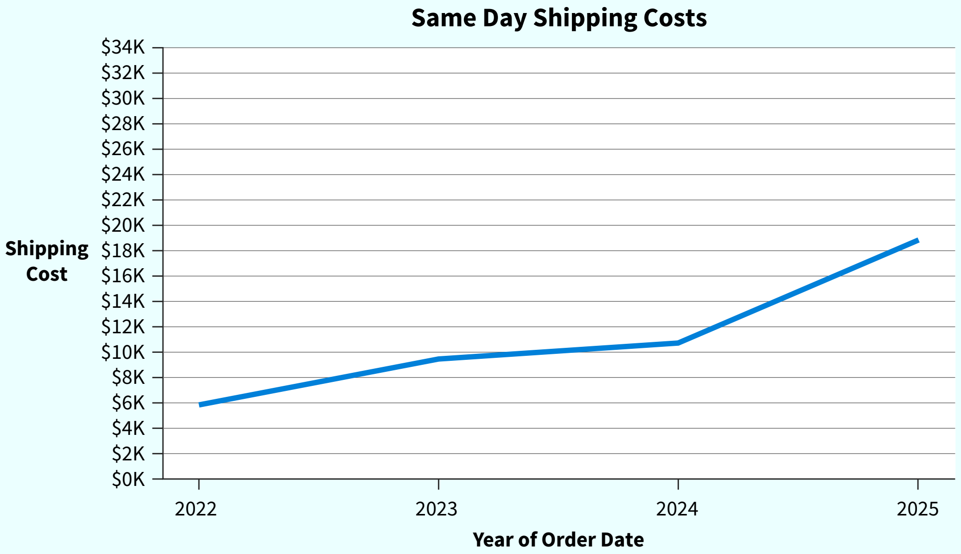
3.

Shipping Costs by Order Priority and Shipping Mode				
Order Priority	**First Class**	**Same Day**	**Second Class**	**Standard Class**
Critical	$ 5,505	$ 2,664	$ 1,654	
High	$ 16,132	$ 13,516	$ 7,687	$ 13,881
Medium	$ 13,215	$ 2,611	$ 8,898	$ 41,628
Low				$ 5,220

Shipping Cost

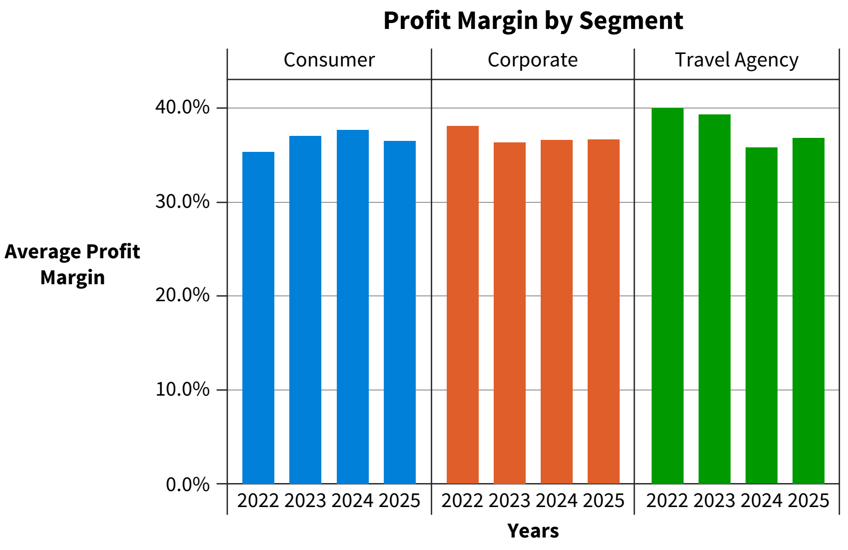
$1,654 $41,628

4.

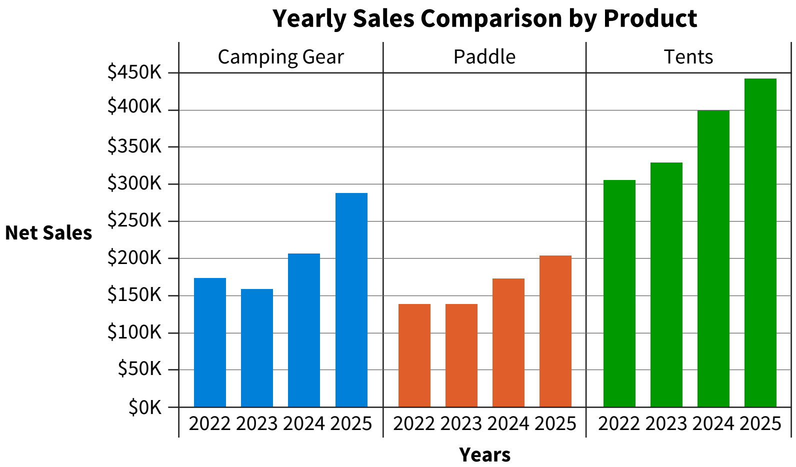


SOLUTION

1. This is misleading because the baseline is not zero. Corrected visualization:



2. This is misleading because the baseline is not zero. Corrected visualization:



3. This visual is misleading because it goes against conventions by reversing the color scale. Corrected visualization:

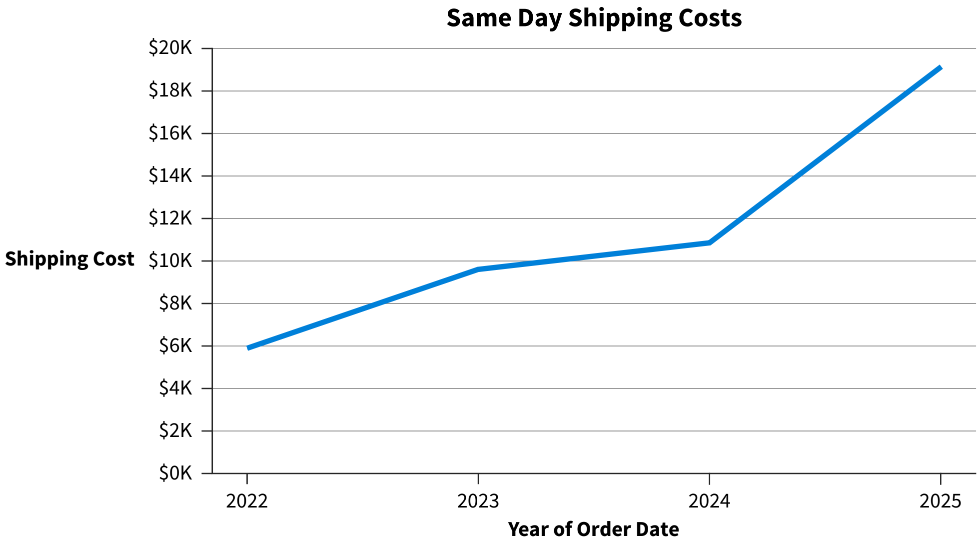
Shipping Cost by Mode and Order Priority: 2024–2025

Order Priority	First Class	Same Day	Second Class	Standard Class
Critical	$ 5,505	$ 2,664	$ 1,654	
High	$ 16,132	$ 13,516	$ 7,687	$ 13,881
Medium	$ 13,215	$ 2,611	$ 8,898	$ 41,628
Low				$ 5,220

Shipping Cost

$1,654 $41,628

4. This is misleading because the y-axis was manipulated. Corrected visualization:



9.5 How are Data Used in Live Presentations?

LEARNING OBJECTIVE 5

Create an interactive data visualization presentation.

Because accountants are often asked to present data stories, oral communication skills are essential for a successful career. Results of data analyses are commonly communicated via a live presentation to the intended audience, either in-person or in a virtual meeting. These presentations often include interactive data visualizations.

Best Practices for Live Presentations

Whether presenting to a live audience or virtually, a presentation should be clear and engaging. There is nothing worse than preparing a great data analysis and then presenting it to an uninterested audience. Business writer Joel Schwartzberg provides seven best practices to follow when presenting data analyses:[12]

1. Ensure the audience can see the data. A visualization that looks fine on a screen may be too small for someone in the back of the room to see.
2. Focus on the points the data illustrates by explaining the meaning of the data analysis. Stating facts without showing how they tell a story will leave the audience confused.

[12]Harvard Business Review, 2020. Schwartzberg, J. Present your data like a pro. https://hbr.org/2020/02/present-your-data-like-a-pro (accessed July 2022).

3. Share one major point on each chart to avoid overwhelming with details. Rather than showing several visualizations, only share those that support the data story.
4. Label chart components clearly. Review them and ask: "If I was seeing this for the first time, would I understand it?"
5. Visually highlight the "a-ha" point, or the insight or discovery, in the story. Smart presenters explain the relevance of the "a-ha" moment both orally and with a visual highlight in the chart or graph.
6. Slide titles should reinforce the data's point. Avoid generic titles and choose those the audience will notice and remember.
7. Present to the audience by looking at them and not reading from a slide presentation. Engaging the audience requires connecting with them, and the best way to do that is by focusing on them rather than the slides.

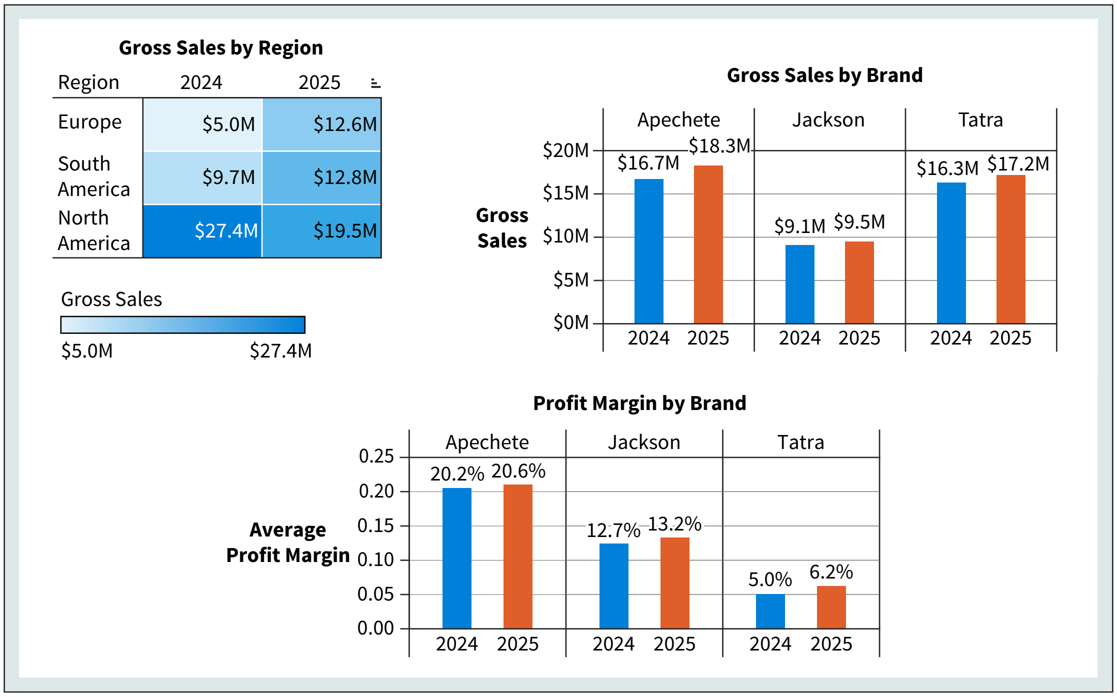
Following these seven best practices will help create a powerful presentation. However, the work does not stop there, as an effective oral presentation requires planning and practice.

Creating Interactive Data Visualizations

A powerful way to communicate data analysis results in a live presentation is by inviting the audience to explore the representation of the data. An **interactive data visualization** is a visualization that allows users to explore, manipulate, and interact with graphical representations of data. They help connect the presenter with the audience by drilling down into the data based on the audience's needs. This allows the presenter to quickly answer any questions from the audience.

Besides following best practices for creating data visualizations and telling a story with data, interactive data visualizations should provide an easy and intuitive way for the user to interact with the data. Consider the visualization in **Illustration 9.47**, which is a dashboard that reflects gross sales and profit margin by brand of vehicle sold by Huskie Motor Corporation (HMC).

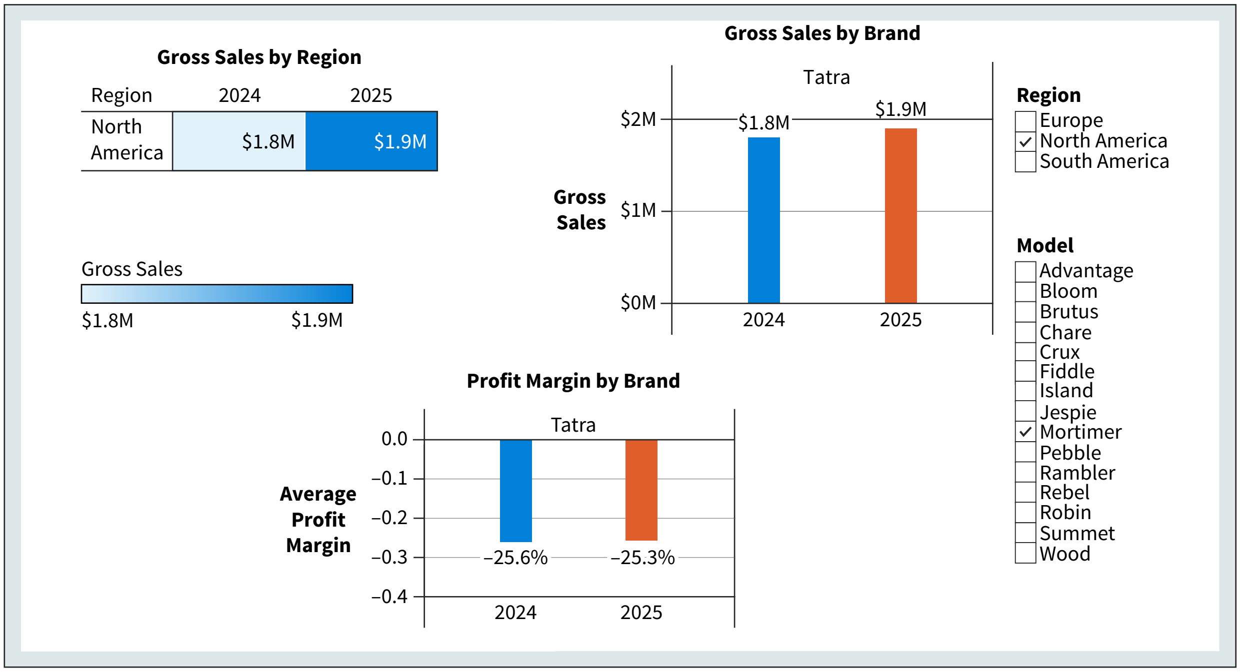
ILLUSTRATION 9.47 Static Dashboard of Gross Sales and Profit Margin



This static visualization does not have the functionality to dive deeper into the data. What if, during the presentation for this analysis, the CEO of HMC asked how the model Advantage performed? An interactive visualization would make it possible to answer the CEO's question quickly. How do we make visualizations interactive? The most common method is to add a filter that allows filtering for the specific information. The best interactive visualizations allow the user to view detailed information.

Adding filters to the dashboard in Illustration 9.47 makes it interactive, letting the user see the same information at a more detailed level–by region and model (**Illustration 9.48**).

ILLUSTRATION 9.48 Interactive Dashboard of Gross Sales and Profit Margin



How To

Users of Illustration 9.48 can choose the region and the car model and see either all the data or any combination of the data. In this illustration the user has chosen the North American region and the Mortimer model. They can quickly see how that model performed in 2024 and 2025. (**Data** How To 9.2 illustrates how to create an interactive dashboard in Tableau.)

The benefit of interactive data visualization is that it allows engagement with the data. An interactive visualization helps users:

- Quickly identify trends.
- Efficiently identify relationships.
- Provide useful data storytelling.
- Simplify complex data.

Keep in mind that interactive data visualization is not limited to live presentations. In general, the more interactive the visual, the more engaged the user will be regardless of the communication medium.

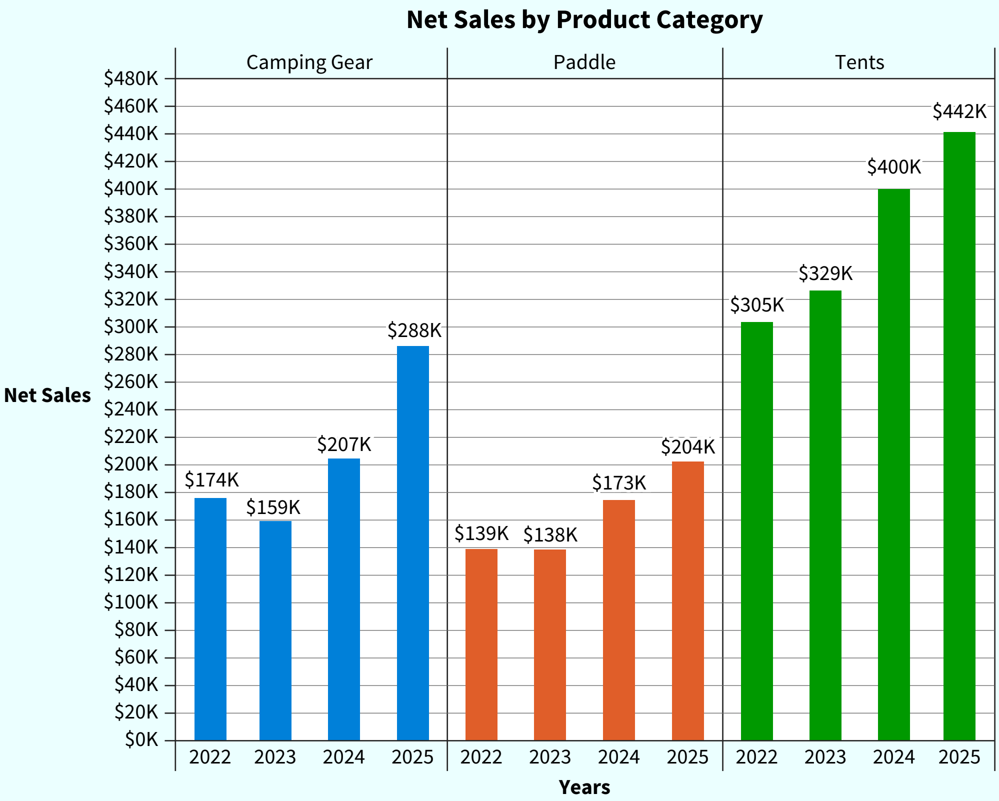
APPLYING CRITICAL THINKING 9.5: Plan the Best Presentation

Creating data visualizations for a presentation takes planning. Part of that planning should include considering different ways to present your findings as well as possible alternative visualizations **(Alternatives)**:

- Be opened-minded about visualizations so that you consider all possible options and create the best presentation.
- For example, when preparing the presentation of the HMC dashboard, considering an interactive versus static visualization opens the opportunity for a more powerful presentation.

Apply It 9.5
Create an Interactive Visualization

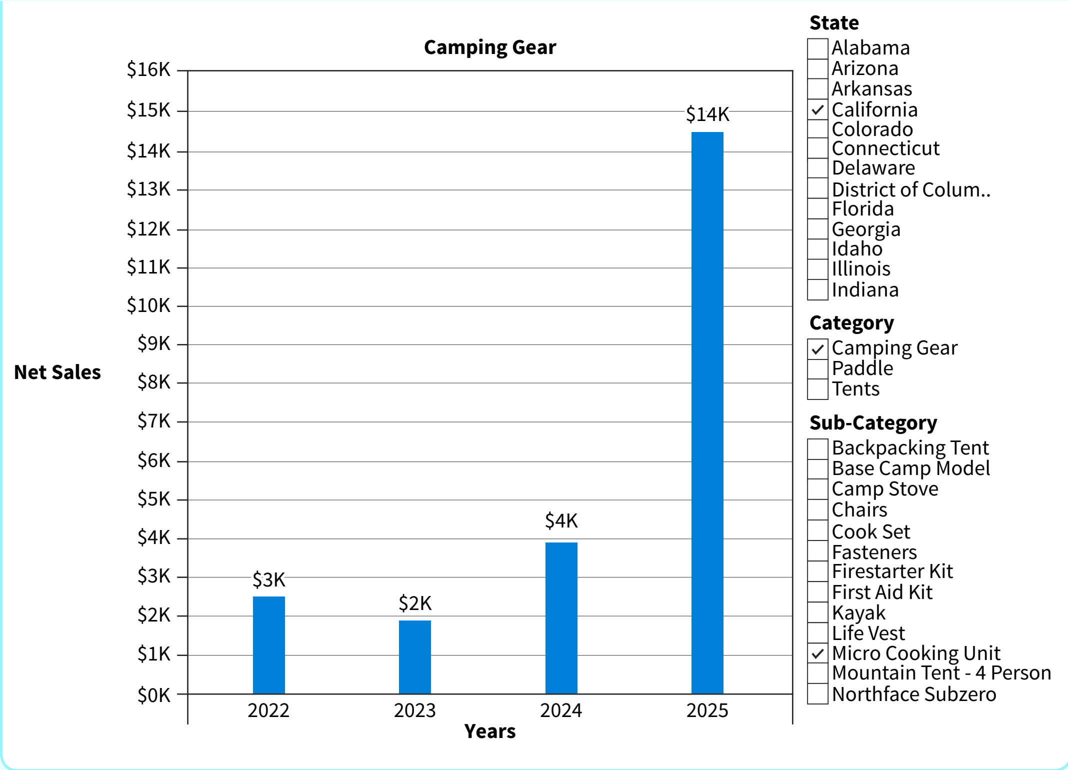
Financial Accounting As a financial accountant at U.S. Outdoor Adventures, you prepared an analysis of net sales by product category. As part of the presentation of its results, you prepared the following visualization:



During the presentation of the analysis to the accounting team, you were asked about the sales for a specific product (Micro Cooking Set) in the Camping Gear product category for the year 2024 in California. List what should be included in this visual to make it interactive so you can answer the question posed during the presentation.

SOLUTION

To create an interactive visualization, identify which aspects of the visual a user would be interested in changing. In this example a filter is necessary to see how a specific sub-category of the product is performing. To quickly answer questions about sales in a specific state, a filter for state is also necessary. It is also a good idea to have a filter for product category to answer questions about the entire category, as well as specific sub-categories.



Chapter Review and Practice

Learning Objectives Review

1 Explain how a data story communicates data analysis results.

The final step in data analysis is to summarize the findings and communicate them to the intended audience:

- Data literacy is the ability to understand data and communicate their meaning clearly and concisely.
- Communicate data analysis results effectively by understanding the audience, focusing on the message rather than the numbers, providing context for the data, making it easy to understand the results, and telling a data story.
- A data story includes three elements: data, narrative, and visuals. Effective data stories have a specific structure. First, the problem or issue is introduced and explored. Next, the main insight is identified and a solution is shared. The final part of the story is a conclusion that includes next steps.

2 Summarize the steps for creating effective data visualizations.

There are three steps to follow before creating effective data visualizations:

- Verify the data: Make sure they are accurate, complete, consistent, fresh, and timely.
- Consider the audience: Who will be viewing the visualizations? Consider what the audience needs so that they are engaged and understand the visualization.

- Define the objective: Is the objective to show composition, relationships, distributions, trends, or comparisons? Choose the visual that matches the project's goal.

3 Describe the characteristics of an effective data visualization.

Once the visualization is chosen, follow best practices to be sure it effectively communicates the results.

Consider the principles of visual perception:

- Continuity: Effective visualizations arrange visual objects in a line to simplify grouping and comparison.
- Similarity: Items similar in color, shape, size, or locations evoke the perception that they belong to the same group.
- Proximity: People perceive visual elements are related by how closely they are positioned to one another.
- Focal point: The viewer of a visualization will pay more attention to whatever stands out visually.

Preattentive attributes are visual properties we notice without realizing it:

- Size: Relative size is interpreted as relative importance.
- Color: Use color sparingly and consistently.
- Position: People typically start at the top left corner of a visualization and then scan it in a zig-zag motion. Put important information at the top left of a visualization.
- Titles: They should be factual and neutral. Avoid using unnecessary descriptive words. Titles should include a noun that represents what was measured and when.

Avoid cluttering visualizations:

- Remove any non-data related details from the visualization.
- Eliminate redundant information.

4 Recognize misleading data visualizations.

Ethics and data visualization are intertwined. It is important to be able to identify misleading visualizations and to not create them. The common ways visualizations are misleading include:

- Omitting the baseline: This occurs when the baseline does not start at zero and therefore gives the impression that changes are more dramatic.
- Manipulating the y-axis: Increasing or decreasing the axis range can alter perception of the data.
- Going against conventions: Unusual color conventions can confuse and mislead an audience.
- Selectively choosing the data: Omitting data points to alter the perception of trends in the data can be misleading.
- Using the wrong type of graph: Choosing a visualization that is not easily interpretable based on the data.

5 Create an interactive data visualization presentation.

A common way to communicate the results of data analyses is an oral presentation to the intended audience:

- There are certain practices for oral presentations:
  - Make sure the audience can see the data.
  - Explain the meaning of the data.
  - Share one point per chart.
  - Label visualizations clearly.
  - Highlight the important insights or discoveries.
  - Slides should reinforce the data's point.
  - Look at the audience and not at your slides.
- Create an interactive data visualization that allows the viewer to explore, manipulate, and interact with graphical representations of data.

Key Terms Review

How To Walk-Throughs

How To

HOW TO 9.1 Create a Dashboard in Tableau

The dashboard in Illustration 9.13 was created using the visualization software Tableau. Follow these steps to create it yourself.

What You Need: **Data** The How To 9.1 data file.

STEP 1: Open the workbook to find the three visualizations that have already been created: profit margin by brand, gross sales by brand, and cost analysis.

STEP 2: Open a dashboard worksheet by selecting **Dashboard** in the main toolbar or by clicking the dashboard icon next to the worksheets (**Illustration 9.49**).

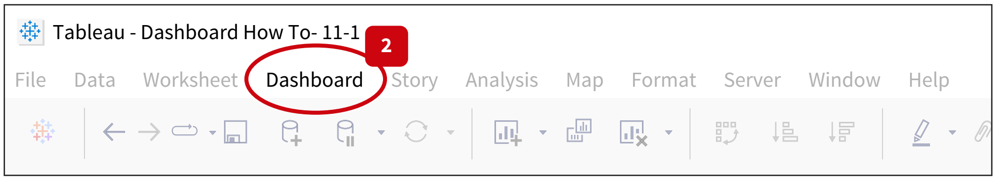


ILLUSTRATION 9.49 Tableau Dashboard Option in the Toolbar

STEP 3: In the dashboard worksheet, click and drag any of the visualizations listed under **Sheets** to the box **Drop Sheets Here**.

STEP 4: In the **Objects** section on the left, choose **Floating** (See **Illustratiuon 9.50**). Manually locate and size the visualizations.

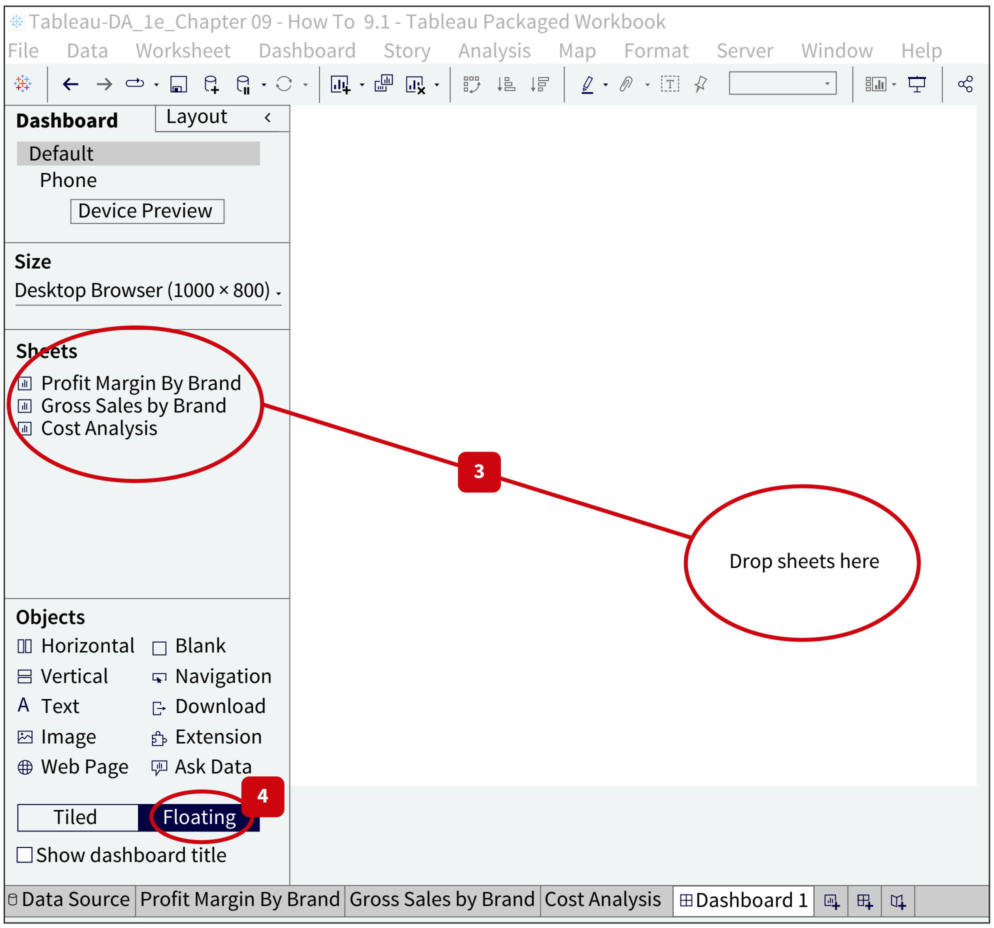
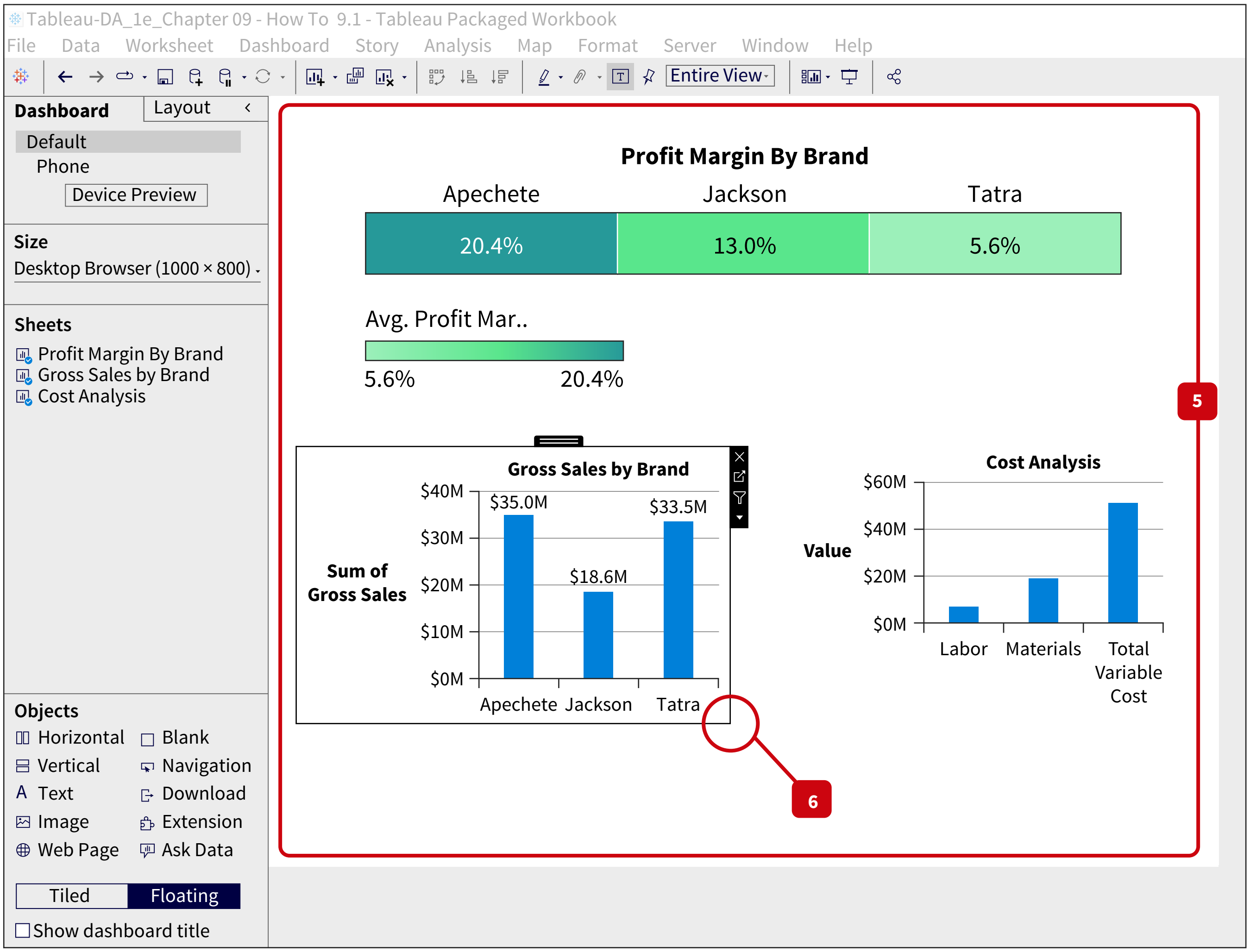


ILLUSTRATION 9.50 Tableau Dashboard Canvas

STEP 5: Recreate the dashboard shown in Illustration 9.13 by dropping the visualizations into the dashboard. (**Illustration 9.51**)

STEP 6: Move the visualizations around the dashboard by clicking on the visual and dragging it. It can be re-sized by clicking on the visual and then hovering at the corner. (Illustration 9.51)

ILLUSTRATION 9.51 Moving Visualizations and Sizing Dashboard Visualizations



HOW TO 9.2 Create an Interactive Dashboard in Tableau

How To

The interactive dashboard in Illustration 9.48 was created using the visualization software Tableau. Follow these steps to create it.

What You Need: **Data** The How To 9.2 data file.

STEP 1: Open the workbook. There are three visualizations that have already been created: gross sales by region, gross sales by brand, and profit margin by brand.

STEP 2: In each visualization, confirm there is a filter for model and region. This will allow you to filter to the lowest level of detail within the dashboard (**Illustration 9.52**).

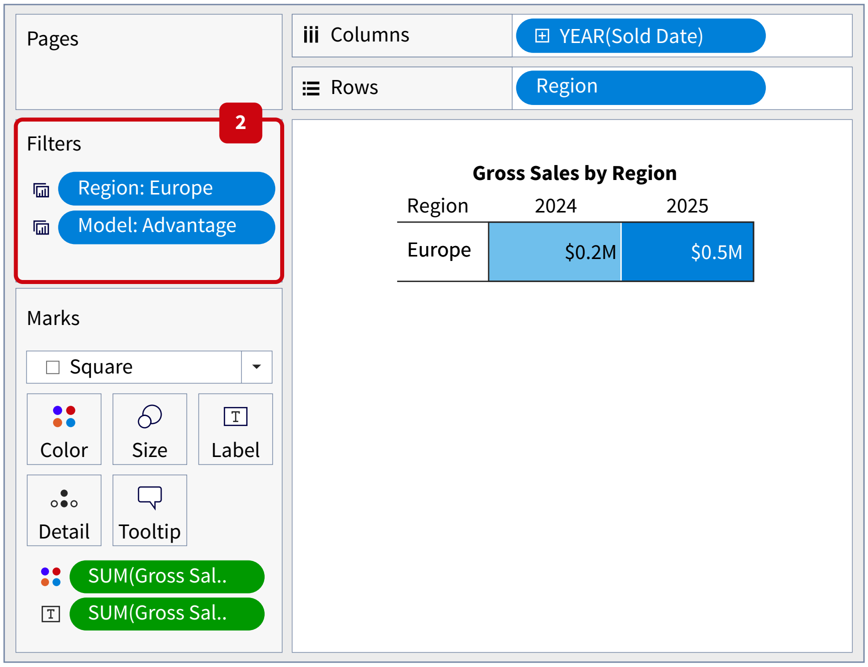
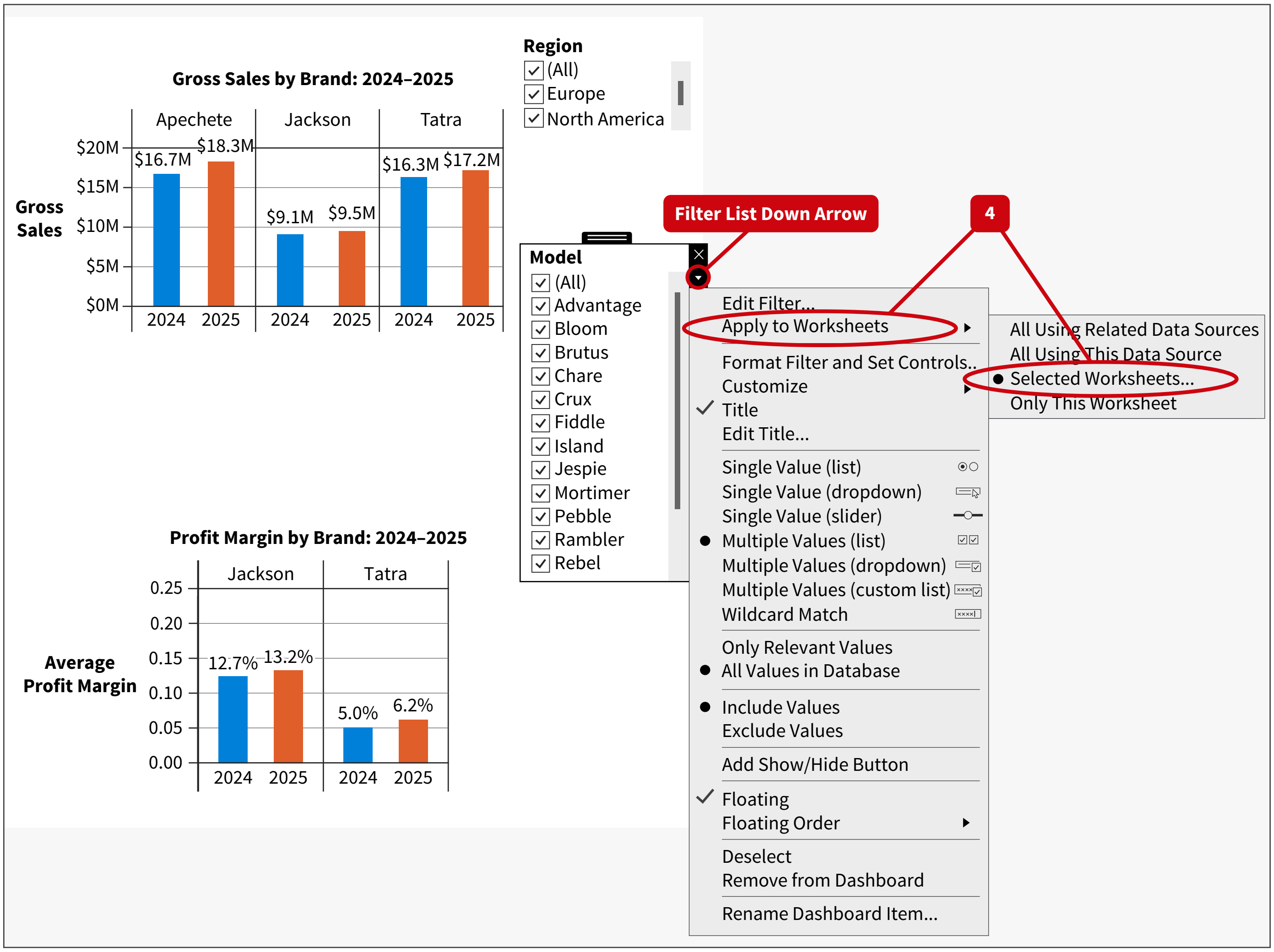


ILLUSTRATION 9.52 Confirm Filters for Region and Model

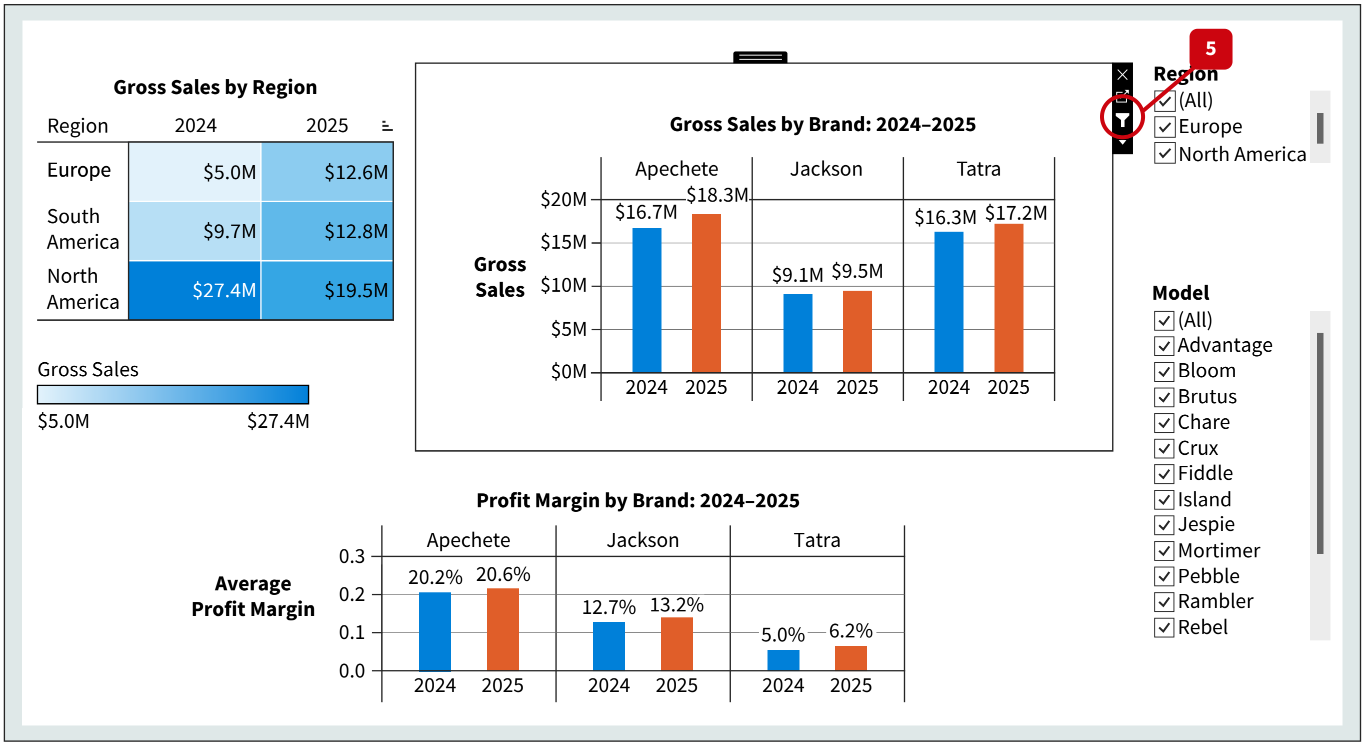
STEP 3: Follow the instructions in How To 9.1 to drag all three visualizations onto the dashboard canvas and format to mirror the dashboard in Illustration 9.48.

STEP 4: To link the worksheets in the dashboard so the filters will apply to each, click on the **Filter List**. Choose the option **Apply to Worksheets** and then **Selected Worksheets** (**Illustration 9.53**).

ILLUSTRATION 9.53 View of Dashboard after the Filter List Down Arrow is Selected

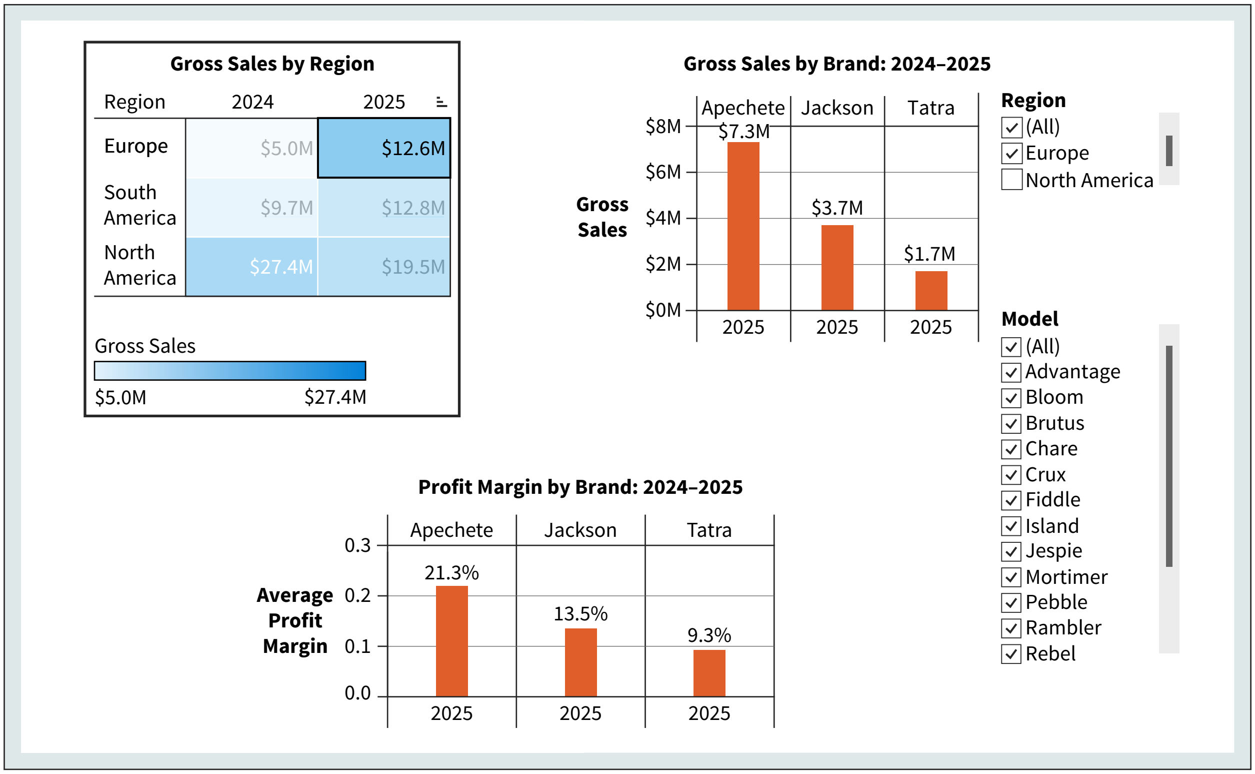


STEP 5: Click each visual in the dashboard and create a filter by selecting the filter icon.



That will allow clicking on one visual and applying the selection to all of them. For example, clicking the $12.6M in Europe 2025 means that all the visuals will show only Europe 2025 results. (**Illustration 9.54**)

ILLUSTRATION 9.54 Interactive Dashboard



Data The Data tag appears when the data required to answer a question or complete an exercise are available on Wiley's online learning platform.

Multiple Choice Questions

1. **(LO 1)** The ability to read, write, and communicate data in context is/are referred to as
   - **a.** data communication skills.
   - **b.** effective data communication.
   - **c.** data literacy.
   - **d.** data storytelling skills.

2. **(LO 1)** Dara prepared a data analysis of profitability by product for the current year for her company. She felt it was also important to show how this year's profitability compared to prior years. This is an example of
   - **a.** putting the analysis in context.
   - **b.** understanding the audience.
   - **c.** creating a memorable story.
   - **d.** a composition analysis.

3. **(LO 1)** Sara is a financial accountant for a retail company who was asked to do an analysis of sales over the past five years to prepare a sales forecast. The company is seeking additional investors so they can expand their operations. Sara's supervisor has asked her to prepare a report that will be given to the potential investors. The investors would be considered the ____________ of the report.
   - **a.** stakeholders
   - **b.** preparers
   - **c.** purchasers
   - **d.** regulators

4. **(LO 1)** Marcus is a financial accountant for a computer software company. He was asked to do an analysis of cashflows over the past five years and to prepare a revenue forecast. The company is seeking additional investors so that they can expand their operations. Marcus' supervisor has asked him to prepare a report that will be given to the potential investors. If the investors have a bias against the type of software the company sells, Marcus should consider this bias as a(n)
   - **a.** key alternative.
   - **b.** uncontrollable event.
   - **c.** risk.
   - **d.** negative outcome.

5. **(LO 1)** The combination of narrative and data __________ the story.
   - **a.** explain
   - **b.** enlighten
   - **c.** engage
   - **d.** evolve

6. **(LO 1)** Methodically peeling back layers of the analysis in a data story is the __________ section of the story.
   - **a.** exposition
   - **b.** rising action
   - **c.** climax
   - **d.** falling action

7. **(LO 2)** A well-designed __________ conveys the results of an analysis in a clear and concise manner.
   - **a.** analysis
   - **b.** data visualization
   - **c.** statistic
   - **d.** report

8. **(LO 2)** A(n) __________ audience is primarily concerned with actionable results.
   - **a.** novice
   - **b.** managerial
   - **c.** expert
   - **d.** executive

9. **(LO 2)** An area chart is best used for which of the following objectives?
   - **a.** Showing composition.
   - **b.** Showing relationships.
   - **c.** Showing distributions.
   - **d.** Showing comparisons.

10. **(LO 2)** A scatterplot is best used for which of the following objectives?
    - **a.** Showing composition.
    - **b.** Showing relationships.
    - **c.** Indicating trends.
    - **d.** Showing comparisons.

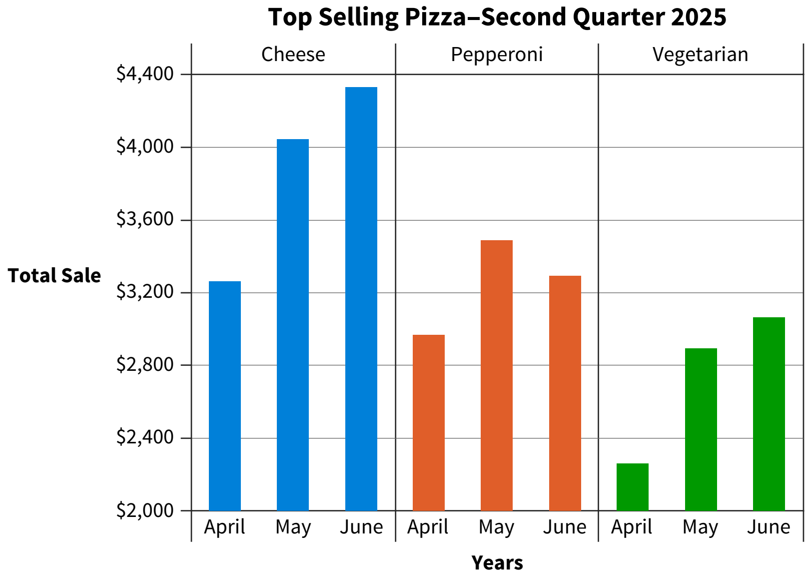
11. **(LO 2)** Jamie was preparing a visualization to show the results of an analysis of sales trends. Jamie has data for each month's sales from all the company's regions. All the sales data is reported in whole dollars except for the east region. The east region reports sales in thousands. This is an example of what type of data verification attribute?
    - **a.** Accuracy
    - **b.** Completeness
    - **c.** Consistency
    - **d.** Freshness

12. **(LO 3)** An example of a preattentive attribute is
    - **a.** clutter.
    - **b.** accuracy.
    - **c.** timeliness.
    - **d.** color.

13. **(LO 3)** If there are 9 categories in a column chart, which of the following would improve the visualization and make it easier for the reader to interpret?
    - **a.** Using vertical bars.
    - **b.** Using horizontal bars.
    - **c.** Using 9 different colors to differentiate the categories.
    - **d.** Using vertical labels to fit all the category descriptions in the graph.

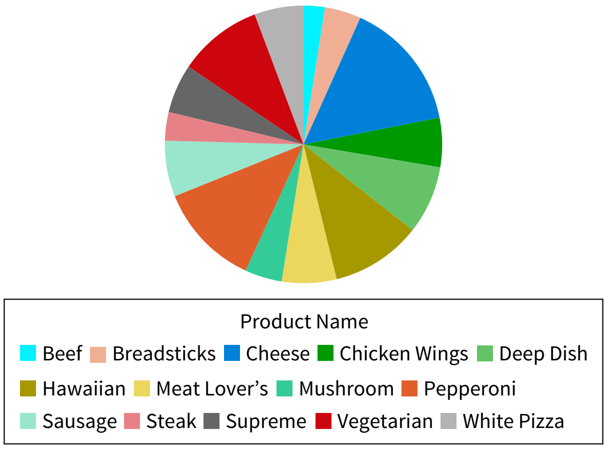
14. **(LO 3)** When showing correlation between two variables, which is the most appropriate chart to use?
    - **a.** Tree map
    - **b.** Stacked bar chart
    - **c.** Bubble chart
    - **d.** Scatterplot

15. **(LO 4)** Identify how the following visualization may be misleading.



- **a.** There are too many colors.
- **b.** The data has been manipulated.
- **c.** The axis does not start at zero.
- **d.** The wrong type of visualization was used.

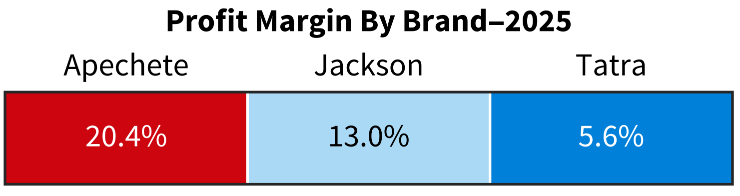
16. **(LO 4)** The following visualization was prepared to allow the viewer to compare sales quantity by type of product for Pizza My Heart Food truck.



How might this visualization be misleading?

- **a.** There is no baseline.
- **b.** The data has been selectively chosen.
- **c.** The wrong type of visualization was used.
- **d.** The color scale goes against conventions.

17. **(LO 4)** The following visualization was prepared for a management dashboard to monitor profit margin.



How might this visualization be misleading?

- **a.** There is no baseline.
- **b.** The data has been selectively chosen.
- **c.** The wrong type of visualization was used.
- **d.** The color scale goes against conventions.

18. **(LO 5)** Which of the following is *not* a best practice for preparing a presentation of data analysis results?

a. Slide titles should be generic so that all can understand the title.

b. There should only be one major point per chart to avoid overwhelming the audience.

c. Visually highlight the "a-ha" point or insight.

d. Ensure the audience can see the data.

19. **(LO 5)** A static visualization

a. allows the user to dig deeper into the analysis.

b. is best suited for live presentations.

c. does not allow the user to dig deeper into the analysis.

d. cannot be used in a data story.

20. **(LO 5)** A visualization that allows the user to explore and manipulate the data is

a. an interactive data visualization.

b. required for all live presentations.

c. not recommended for a live presentation.

d. only used in dashboards.

Review Questions

1. **(LO 1)** List and discuss each suggestion for effectively communicating to your audience.

2. **(LO 1)** What is meant by "putting the data in context?"

3. **(LO 1)** Imagine you are presenting a data analysis that shows expense trends for the last three years. You have found that one region had a significant jump in labor expenses in the second year of the analysis. Upon further analysis, you identified a specific location that is paying much higher wage rates than other locations in the same region. Discuss how you would build a data story for these analyses. Include the elements of an effective data story.

4. **(LO 1)** Discuss why data stories are an effective way to communicate data analysis results.

5. **(LO 2)** Discuss why it is important to verify your data prior to preparing your data analysis communication.

6. **(LO 2)** Compare and contrast communicating to a novice audience versus an executive audience.

7. **(LO 2)** Compare and contrast communication to a managerial audience versus an expert audience.

8. **(LO 2)** Discuss how a dashboard can help communicate data analysis results to managers.

9. **(LO 2)** Discuss how the objective of an analysis can help determine the type of visualization that should be used.

10. **(LO 3)** Discuss how the law of continuity applies to data visualization.

11. **(LO 3)** Discuss how the law of similarity applies to data visualization.

12. **(LO 3)** Discuss how the law of proximity applies to data visualization.

13. **(LO 3)** Discuss how the law of focal point applies to data visualization.

14. **(LO 3)** What are preattentive attributes, and why do they matter when preparing a visualization?

15. **(LO 4)** Give an example of how manipulating the y-axis could cause a visualization to be misleading.

16. **(LO 4)** In terms of creating a visualization, discuss what going against conventions means and give an example.

17. **(LO 4)** Discuss why omitting the baseline in a visualization may be misleading.

18. **(LO 5)** Compare and contrast static visualizations with interactive visualizations.

19. **(LO 5)** Discuss the benefits of using interactive data visualizations and give an example of how an interactive data visualization could be used in a live presentation.

20. **(LO 5)** Discuss best practices for presenting data analysis results to a live audience.

Brief Exercises

BE 9.1 **(LO 1)** **Financial Accounting** Your company is considering several investment opportunities. You have collected data on the investments and have prepared a risk analysis. Describe why each of the following are important to consider as you begin to prepare your communication of the risk analysis:

1. Understand the audience.
2. Focus on the message.
3. Put the data in context.
4. Make it easy to understand.
5. Create a memorable story.

BE 9.2 (LO 1) **Tax Accounting** U.S. Outdoor Adventures is concerned about sales tax compliance. The tax department director has asked you to analyze company sales this year and the sales tax that should have been collected. You have performed the analysis and are preparing to communicate your results to the director. Describe how each of the following apply to the presentation of your results to the tax director.

1. Understand the audience.
2. Focus on the message.
3. Put the data in context.
4. Make it easy to understand.
5. Create a memorable story.

BE 9.3 (LO 1) **Auditing** You are an auditor who has performed an analysis of your client's purchase card (P-card) transactions. The objective of the analysis was to test the controls the client has over the use of employee purchase cards. You used all the P-card transactions data for the year under audit to evaluate the following controls:

- Employees have not exceeded their spending limit per transaction.
- Employees have not exceeded their monthly spending limit.
- All purchases are recorded with descriptions.
- All purchases have supervisor approval.

During your analysis you discovered there were several violations of policy, but they were generally small and involved various employees/supervisors. However, one employee and supervisor had very suspicious violations. Upon deeper analysis, you were able to identify a pattern of suspicious spending. Use Freytag's pyramid to draft how you will tell this story.

BE 9.4 (LO 2) **Financial Accounting** Your company is considering several investment opportunities. You have collected data on the investments and have prepared a risk analysis. What knowledge will your audience need to understand the communication of your results?

BE 9.5 (LO2) **Auditing** **Financial Accounting** **Managerial Accounting** Match each of the following to the best visualization. Visulization choices can be used once, more than once, or not at all.

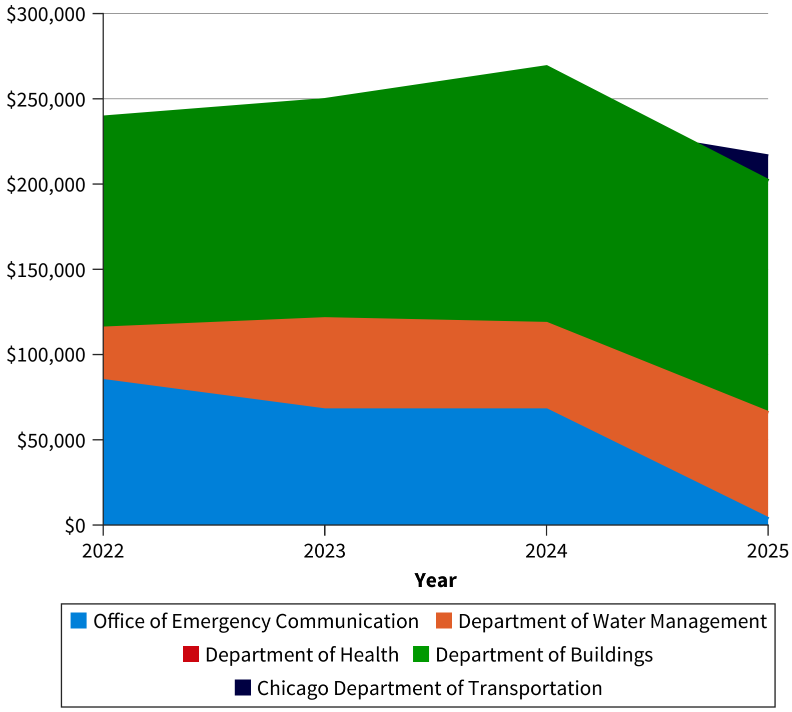
a. Bar chart
b. Line chart
c. Stacked bar chart
d. Scatterplot
e. Bubble chart

Purpose	Visualization
1. Show the distribution of prices for a specific product.	
2. Show the composition of total expenses.	
3. Show the relationship between temperatures and soup sales at a restaurant.	
4. Show sales trends over time.	
5. Show the distribution of sales by country.	
6. Show sales comparisons by year.	

BE 9.6 (LO 3) **Data** **Financial Accounting** **Managerial Accounting** Using the available data, create a line chart that shows the trend in overtime pay over the year. Be sure to follow best practices for creating line charts.

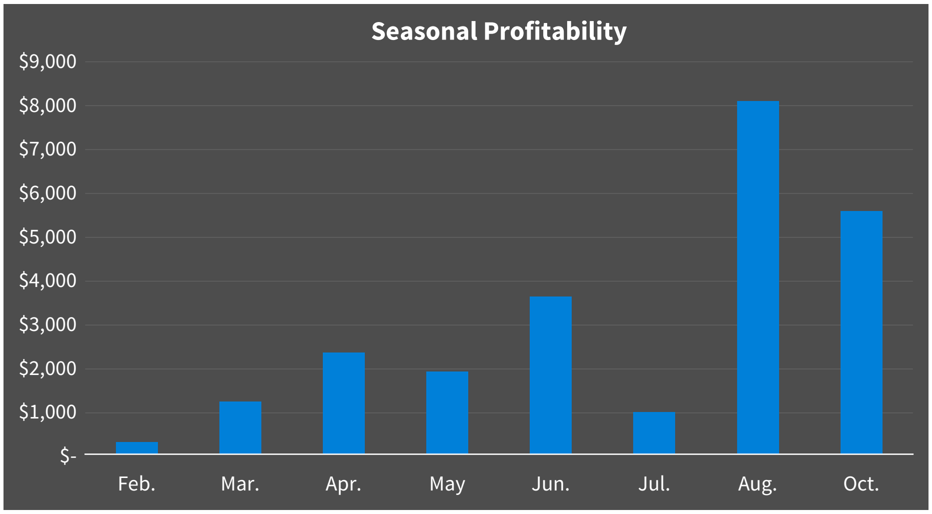
BE 9.7 (LO 3) **Data** **Financial Accounting** Create a column chart comparing product sales in 2024 and 2025 for the following cities: Denver, Loveland, Lafayette, and Brookfield. Be sure to follow best practices for creating charts.

BE 9.8 **(LO 2, 3)** Data Financial Accounting Tax Accounting The following visualization was prepared to communicate the analysis of employee reimbursement to city employees. The audience for the visualization is the internal audit team. The objective is to identify the departments with the highest amount of employee reimbursement over the years 2022 to 2025.



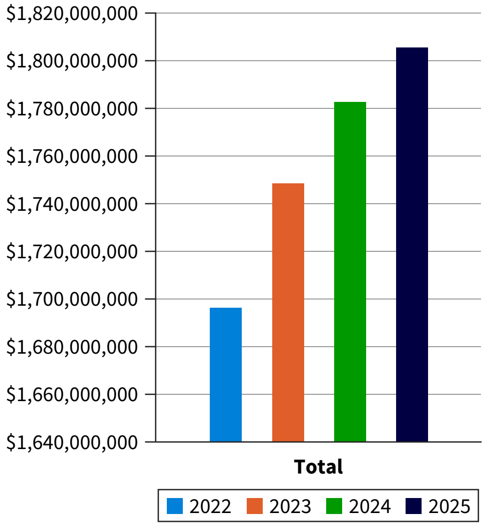
1. Identify possible corrections/improvements for this visualization.
2. Prepare a corrected visualization.

BE 9.9 **(LO 3)** Auditing Financial Accounting Managerial Accounting The following visualization was prepared to show seasonal profitability of a bakery for the prior year.



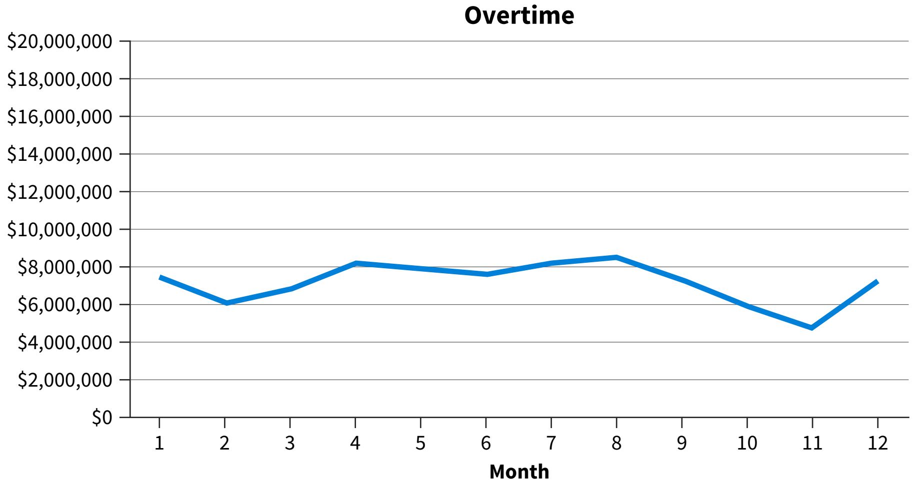
1. Identify possible corrections/improvements that should be considered to improve the visualization.
2. Discuss why your suggested improvements are necessary.

BE 9.10 (LO 3, 4) **Data** **Financial Accounting** A hotel chain prepared a visualization showing the growth in revenue from 2022 to 2025. The audience for the visualization is potential investors.



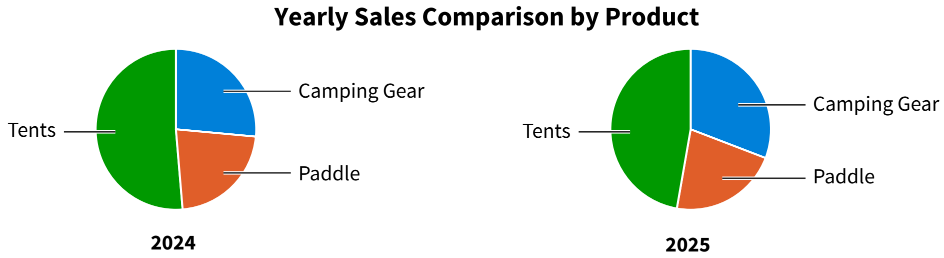
1. Identify and discuss any potential misleading elements to the visualization.
2. Identify if there are any violations of best practices.
3. Prepare a corrected visualization.

BE 9.11 (LO 4) **Data** **Financial Accounting** You are a city accountant analyzing overtime pay for the fire department. Review the visualization prepared by someone else in your department.



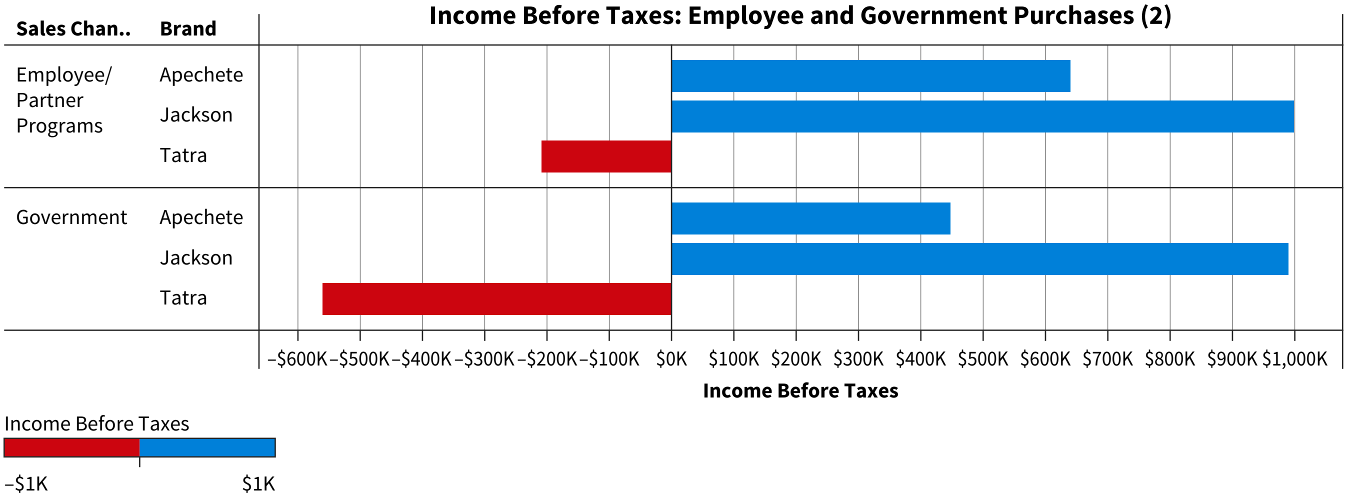
1. Discuss whether the visualization is misleading, and if so, how?
2. Prepare a corrected visualization.

BE 9.12 **(LO 4)** **Data** **Financial Accounting** **Managerial Accounting** U.S. Outdoor Adventures has asked for an analysis of sales for 2024 and 2025. The objective of the analysis is to compare sales from 2024 to 2025 and determine if any product categories are increasing or decreasing. The following analysis was prepared for the management team.



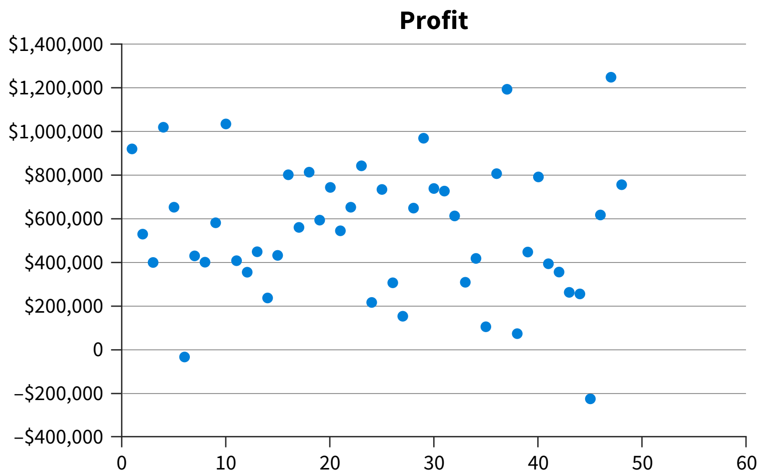
1. Is this data visualization appropriate for this analysis? Discuss why it is or is not appropriate.
2. Prepare another visualization. Explain why your visualization is more appropriate.

BE 9.13 **(LO 5)** **Data** **Managerial Accounting** The following visualization was prepared for Huskie Motor Corporation to analyze sales channels. HMC would like to better understand the profitability of employee, government, and purchase payment options (cash, financing, or leasing).



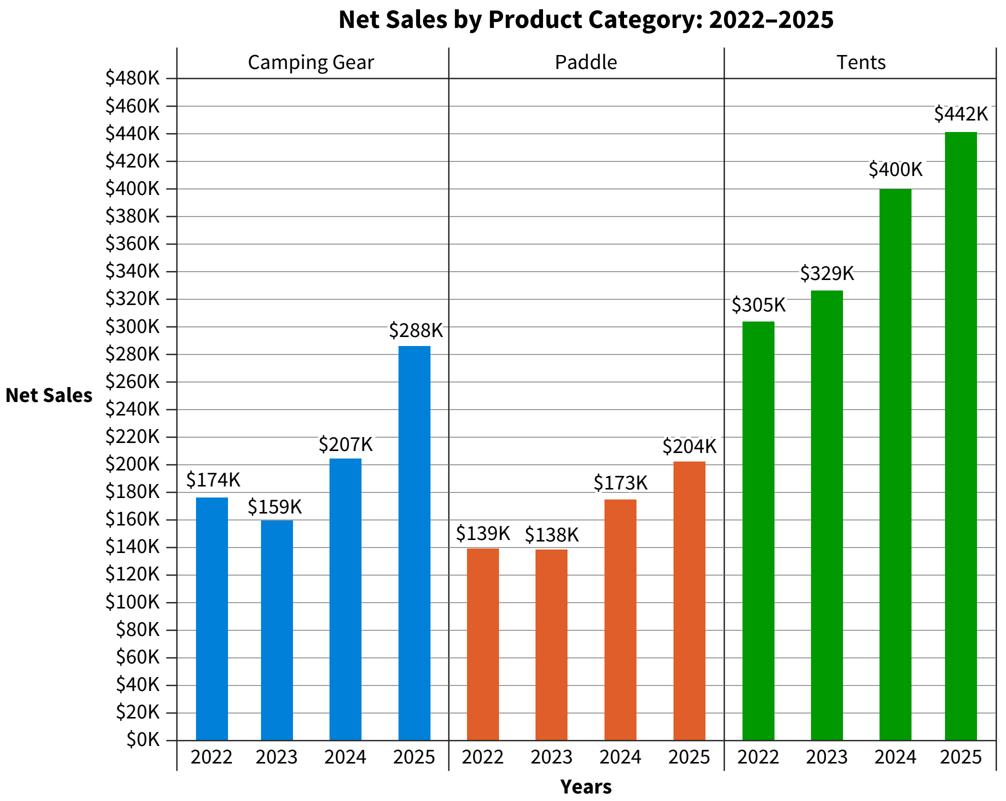
1. Recommend how this visualization could be made interactive.
2. Create the interactive visualization you recommended.

BE 9.14 **(LO 2, 5)** **Data** **Auditing** **Financial Accounting** **Managerial Accounting** The following visualization was prepared to help Denton Hospitality analyze profit by hotel location. Each dot on the scatterplot represents an individual hotel location. The audience for the presentation will be experts.



1. Based on the audience being experts, what would you change or enhance for a live presentation?
2. Prepare a corrected scatterplot based on your recommendations.

BE 9.15 **(LO 5)** **Data** **Financial Accounting** **Managerial Accounting** The following visualization was prepared for U.S. Outdoor Adventures to help them analyze sales.



Identify ways to make this visualization interactive so the user can drill down and see how the various products are performing. Then, create the visualization.

Exercises

EX 9.1 **(LO 1, 2, 3, 5)** **Data** **Managerial Accounting** **Create Visualizations to Analyze Sales and Profitability** Super Scooters is a company that manufactures and sells four different scooters: Captain, Celeritas, Kicks, and Lazer. You are an accountant at Super Scooters, and the CEO has asked for an analysis of sales and profitability of the models for the last three years.

1. Prepare an analysis of sales and profitability. Follow best practices outlined in the chapter.
2. Discuss how you can make the analyses engaging for the audience to which you are presenting.

EX 9.2 **(LO 2, 3, 5)** **Data** **Tax Accounting** **Use Visualizations to Analyze Deductible Expenses** You are a tax accountant for Ace Software, a computer software company, tasked with performing an analysis of business entertainment expenses. Under the current tax law, business meals are 50% deductible. Entertainment costs (golf, sport event tickets, etc.) are not deductible.

1. Prepare a visualization that summarizes spending on entertainment. Follow best practices outlined in the chapter.
2. Design an interactive visualization that will allow analysis down to the employee level.

EX 9.3 **(LO 2, 3)** **Data** **Managerial Accounting** **Create Visualizations to Analyze Variable Costs** Super Scooters manufactures and sells four different scooters: Captain, Celeritas, Kicks, and Lazer. You have been asked to do an analysis of variable costs by model and year.

1. Prepare an explanatory data visualization of variable costs. You may use Excel, PowerBI, or Tableau to prepare the visualizations. Be sure to follow all best practices.

2. Discuss how you would communicate your analysis to each of the following audiences:
   - Novice
   - Managerial
   - Expert
   - Executive

EX 9.4 (LO 1, 2, 3) Data Auditing Use Visualizations to Analyze Revenue Trends You are an auditor for the accounting firm Banes, Kent, and Williams. As part of the audit of Super Scooters you have been asked to create a visualization to show trend analysis of sales revenue. The objective of the analysis is to determine if there have been any unusual changes in sales from prior years or other trends that might affect risks of material misstatement.

1. Prepare a visualization that can analyze changes in sales and sales trends.
2. Discuss why the visualizations you chose are appropriate.
3. Discuss how you would communicate your analysis to each of the following audiences:
   - Novice
   - Managerial
   - Expert
   - Executive

EX 9.5 (LO 3) Data Financial Accounting Create Visualizations to Analyze Payroll Data Perform an analysis of overtime paid by department and month for the city of Chicago. Use visualization software to create visualizations showing the following:

1. Departments with the five highest total overtime.
2. The monthly trend of overtime for the department with the highest total overtime.
3. The employees with the five highest overtime totals for the year in the department with the highest amount of overtime.

EX 9.6 (LO 5) Data Financial Accounting Create Interactive Visualizations U.S. Outdoor Adventures would like to use an interactive dashboard to evaluate product sales and profits. Review the static visualizations in one of the Excel, PowerBI, or Tableau files prepared by the accounting department at U.S. Outdoor Adventures. Convert the static visualizations into interactive visualizations that management can use to monitor the profitability of products by sub-category and location.

EX 9.7 (LO 1, 2, 3) Data Managerial Accounting Create Visualizations to Inform Decisions You are a financial analyst working in the operations group at Super Scooters. The executive team is interested in the sales trends by scooter models. Specifically, they want to understand gross sales by model and sales volume by model to identify those models that are increasing in these measures. Those high performing models should receive more marketing dollar allocation.

1. Analyze sales dollars, sales volume, and sales volume mix by model for the last three years. Present visualizations to communicate the results to the executive team. You will need more than one visualization to create an adequate story for the executive team.
2. Use Freytag's pyramid to tell the story. Provide recommendations to the executive team regarding the allocation of marketing dollars to high performing models.

EX 9.8 (LO 2, 3) Data Financial Accounting Create Visualizations to Describe Warranty Expenses High-End Hubs (HEH) is a private entity that manufactures and sells bicycle wheel parts. Their primary customers are high-end mountain bike manufactures. The controller has asked you to understand warranty returns this year compared to last year. Your objective is to identify models and part numbers that have warranty issues in the current year and to identify assumptions to use in the warranty accrual calculation for year-end. Prepare descriptive visualizations of warranty returns compared to sales by model for this year and last year.

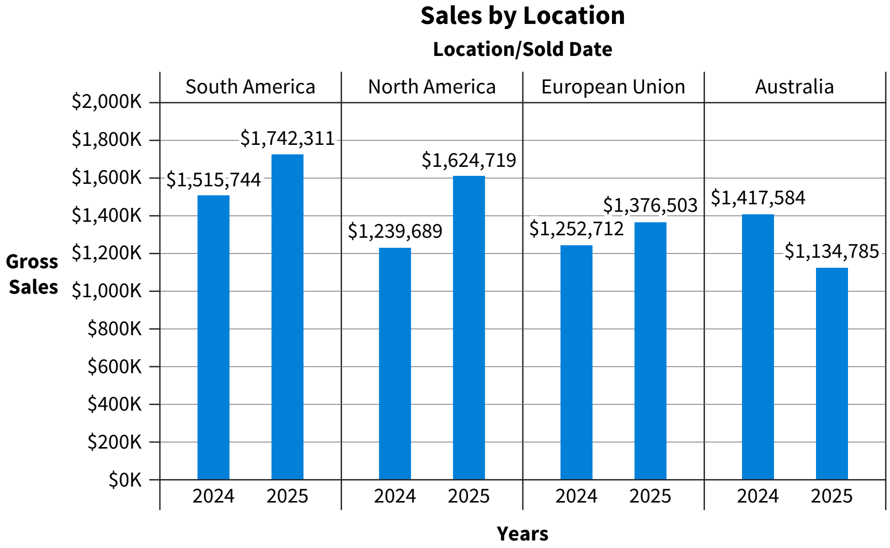
EX 9.9 (LO 1, 2, 3) Data Audit Communicate Risk of Material Misstatement Using Visualizations You are an auditor for the accounting firm Banes, Kent, and Williams. As part of the audit of Super Scooters, you must create a visualization to communicate the trend of sales revenue by model. The objective of the analysis is to communicate changes in sales trends from prior year that inform your consideration of the risk of material misstatement related to revenue recognition for the current year audit.

1. Create a visualization to show yearly gross sales by location.
2. Create a visualization to show yearly gross sales by model and location.
3. Use Freytag's pyramid to tell the story of each visualization and how it might inform the audit of Super Scooters. Remember that your audience for these visualizations and story are the audit team and the audit files.

EX 9.10 (LO 1, 2) **Financial Accounting** **Managerial Accounting** **Create a Story Based on Audience** You are a financial analyst for SWI, Inc. SWI is a manufacturer and distributor of microchips and microprocessors. The company sells its products to customers in several countries and regions including Australia, European Union, North America, and South America. You have prepared the following table and visualization that show sales in each of the various regions in 2024 compared to 2025.

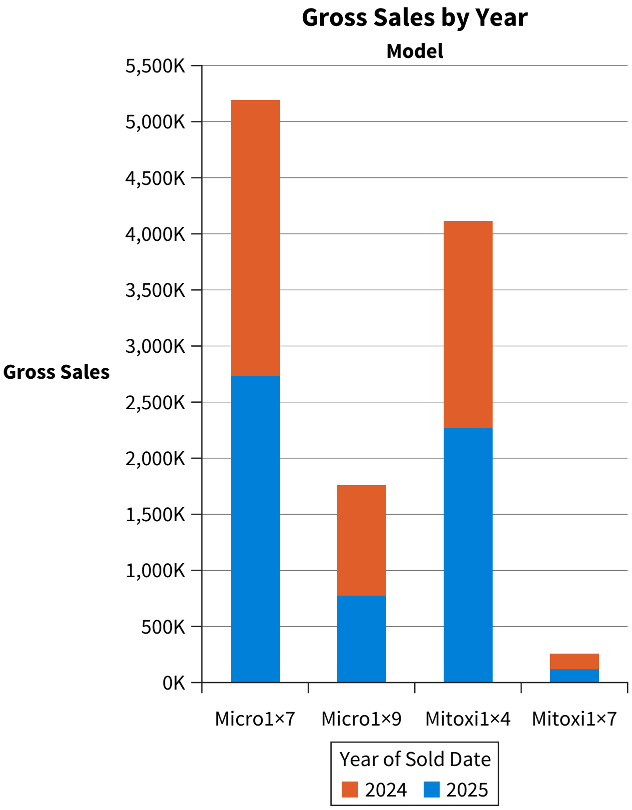
Sales by Location

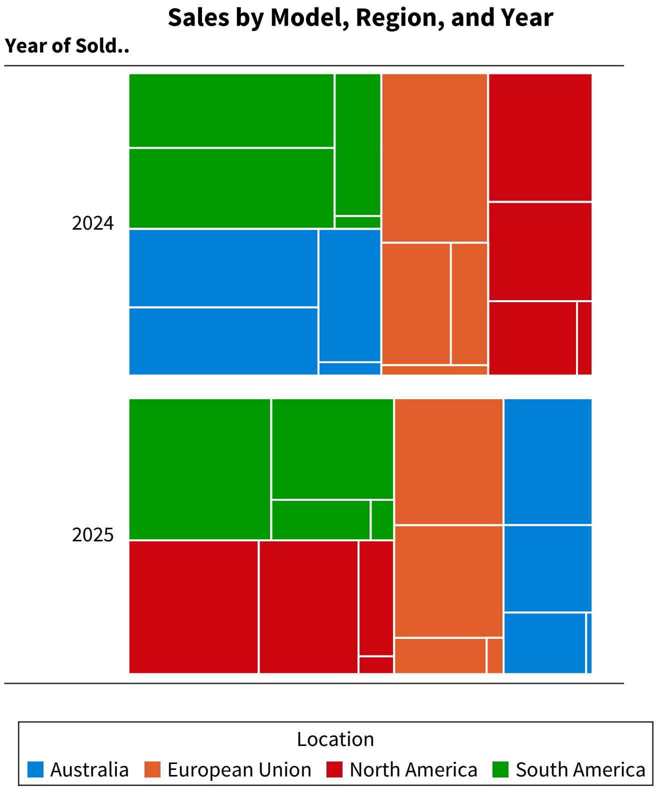
	Sold Date	
Location	**2024**	**2025**
Australia	$ 1,417,584	$ 1,134,785
European Union	$ 1,252,712	$ 1,376,503
North America	$ 1,239,689	$ 1,624,719
South America	$ 1,515,744	$ 1,742,311
Grand Total	**$ 5,425,729**	**$ 5,878,318**



With the visualization and the table, use Freytag's pyramid to tell the story of your company's sales by country from the perspective of an executive discussing performance on an investor call. Then, tell the story from the perspective of an executive making managerial decisions about sales performance.

EX 9.11 (LO 2, 3, 4) Data Auditing Review Data Analytics Workpapers Assume you are an audit senior staffed to the SWI, Inc. engagement. SWI is a public company that manufactures and distributes microchips and microprocessors internationally. Your engagement team staff has prepared a series of visualizations to identify the extent of testing that should be performed in each region by model. Specifically, the engagement team needs to understand the regions and models with the largest changes over the prior year.





1. Provide review comments to improve the dashboard. Identify misleading visualizations and provide feedback to improve the overall effectiveness of the visualization.
2. Prepare a dashboard visualization that communicates differences in sales by region by model.

EX 9.12 (LO 3) Data Financial Accounting Create Visualizations to Analyze Payroll Data Perform an analysis of salaries paid by department and month for the city of Chicago. Use visualization software to create visualizations showing the following:

1. Departments with the three highest total salaries for full time employees.
2. The five departments with the highest number of employees.
3. The 10 highest total salaries by job title.

EX 9.13 (LO 5) Data Managerial Accounting Create Interactive Visualizations U.S. Outdoor Adventures would like to use an interactive dashboard to evaluate product and shipping costs. Review the static visualizations using one of the Excel, PowerBI, or Tableau files prepared by the managerial accounting department. Convert the static visualizations into interactive visualizations so management can monitor the costs of products by sub-category and location and the shipping costs by sub-category, location, and shipping mode.

EX 9.14 (LO 2, 3) Data Auditing Use Visualizations to Communicate Unusual Journal Entries HEH, Inc. is a private company with a calendar year-end. You are a staff auditor assigned to the engagement, and your audit senior has asked you to analyze the general ledger transactions in the current year and identify any unusual items that may have been recorded. Create visualizations that illustrate unusual journal entries. For example, consider the following:

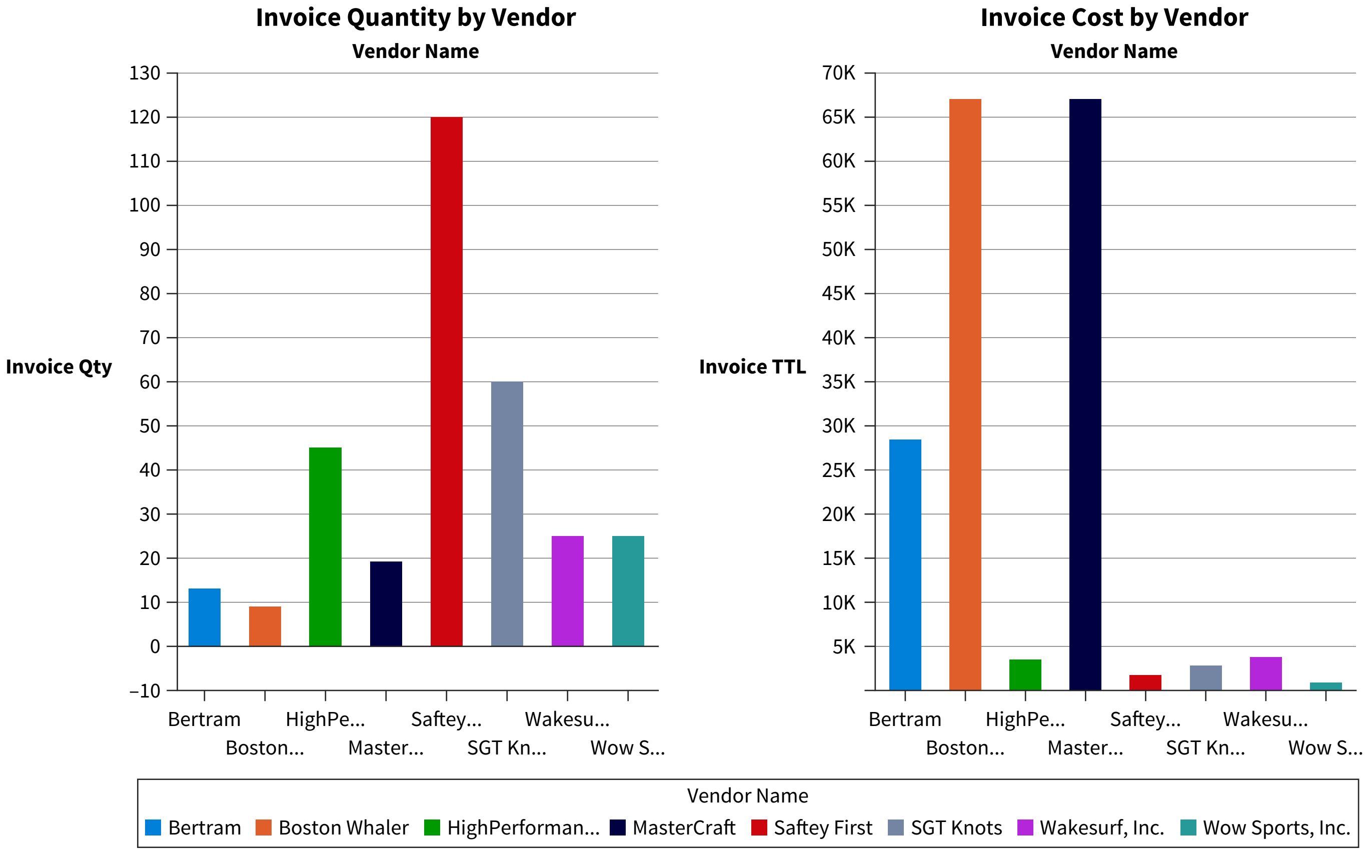
1. Journal entries that were recorded on a Saturday.
2. Journal entries where the memo item has the word adjustment.
3. Journal entries where the memo item has the word “plug.”

EX 9.15 (LO 2, 3) **Data** **Accounting Information Systems** **Use Visualizations to Communicate Log Analyses Results** You are an information systems accountant at your company, TBARk, which is a small retail company that sells both online and in brick and mortar stores. Quarterly, your team examines the log analysis of employees to ensure compliance with company policies. Key policies at the company include:

1. Employees should only login to one POS system at a time.
2. Employees should logout when they are not using the POS system to conduct a sale to a customer.
3. Corporate and back office employees should not login to the POS system.

You will communicate your results to the manager of your group, who is an expert in accounting information systems, controls, and log analyses. Prepare visualizations to test each of the polices using best practices to communicate the results of your analyses of each policy.

EX 9.16 (LO 3, 4) **Data** **Financial Accounting** **Identify Misleading Data Visualizations** You are a financial analyst for Adventure Sports and Outdoors, which is a retail company that sells boats, boating accessories, and safety gear. Your purchasing group is reviewing key vendor contracts. The purchasing group has performed analyses related to the dollar amount of spending with each vendor and has formed conclusions regarding purchasing behavior. The following purchasing group's analysis and conclusions were presented to you.

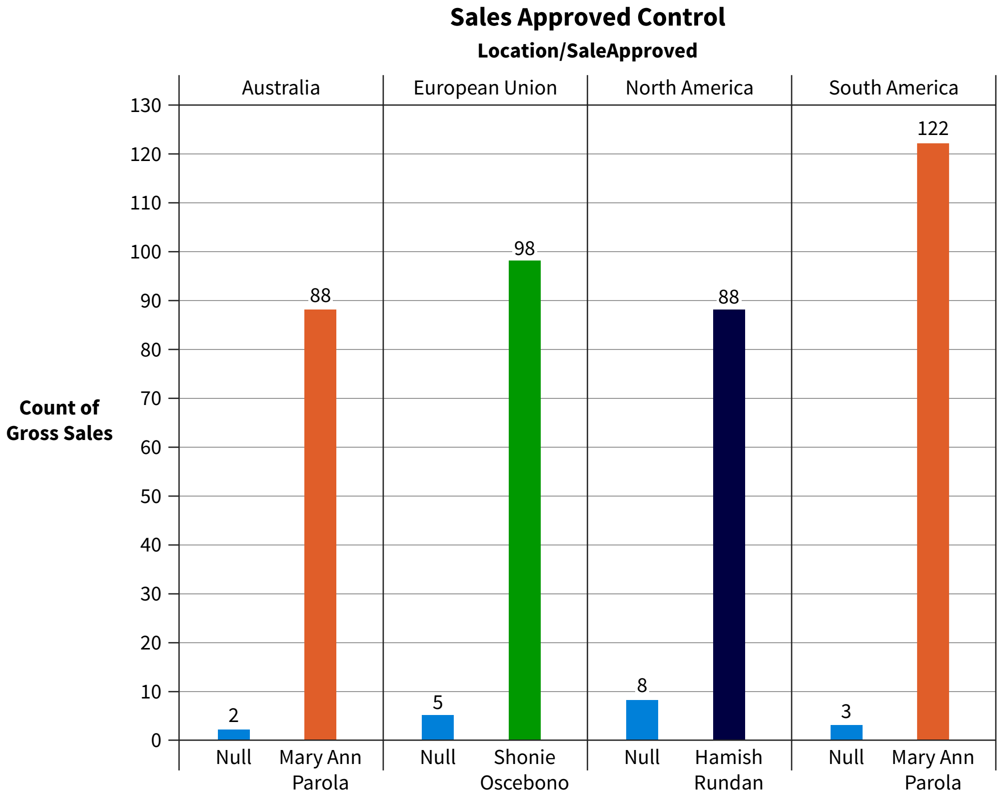


1. Identify misleading or ineffective elements of these visualizations.
2. Prepare visualizations that more accurately represent the purchasing activity for each vendor.

EX 9.17 (LO 4) **Data** **Accounting Information Systems** **Identify Misleading Visualizations in Control Testing** You are an accounting information systems accountant at SWI Inc. Your team is evaluating information technology controls associated with the sales approval process for the year 2025. SWI's sales approval process control is designed as follows:

- All sales above $10,000 and sales that have modifications to the general sales terms must be approved by the specified sales manager.
- There are three sales managers at SWI: Mary Ann Parola manages the Australia and South American regions, Hamish Rundan manages the North American regions, and Shonie Oscenbono manages the European Union region.

Your AIS accounting staff prepared the following visualizations to communicate the operating effectiveness of controls.

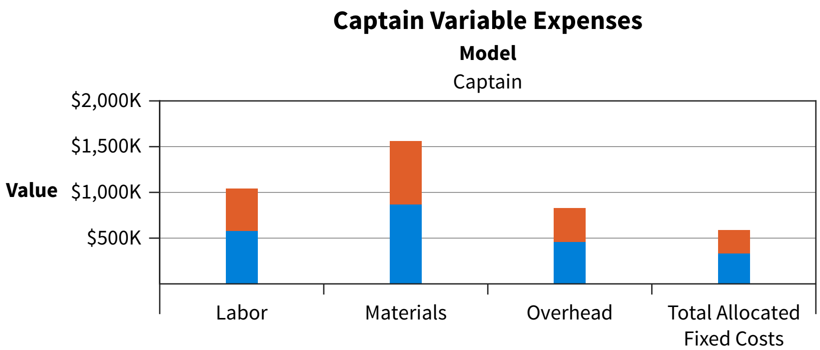


Review the visualization along with the data:

1. Identify instances where the visualization does not appropriately communicate conclusions.
2. Prepare a corrected visualization

EX 9.18 (LO 4) **Managerial Accounting** **Identify Misleading Visualizations** As a senior financial analyst at Super Scooters, your team is preparing for a presentation with a student group to educate them about variable expenses in the manufacturing process. You have received approval from the executive team to share company data, but you are aware that this is a novice audience.

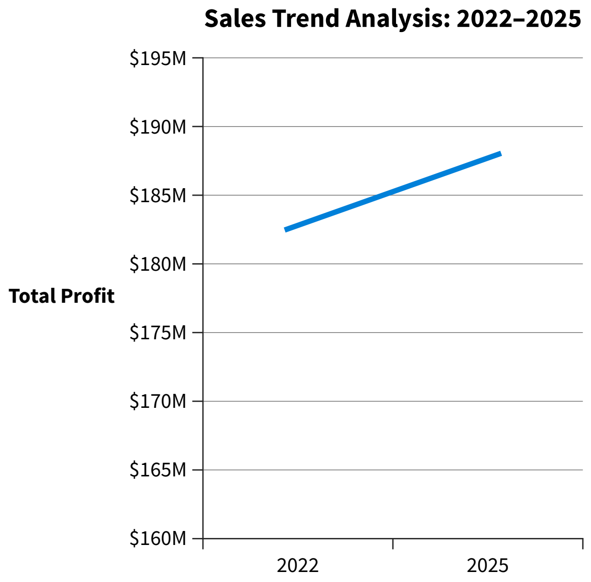
Your staff has prepared the following visualization to communicate the trend in labor, material, overhead, and total allocated costs associated with the production and sale of the Celeritas model.

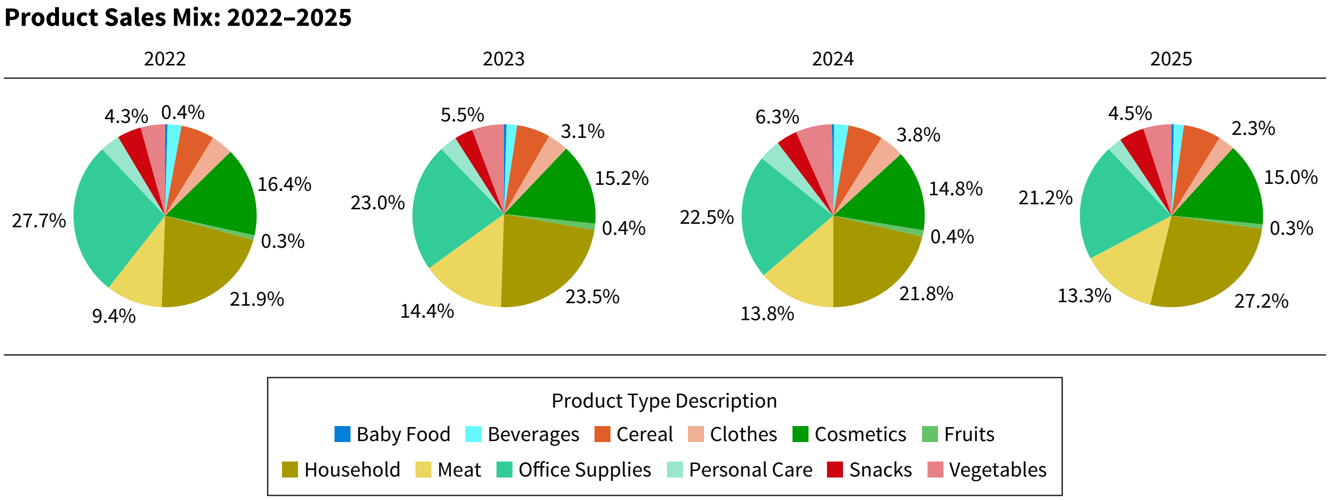


Review the visualization prepared by your staff and explain how it could be improved for presentation to a novice audience.

EX 9.19 (LO 5) Data Financial Accounting Build an Interactive Data Visualization Presentation for Sales HEH, Inc. sells bicycle parts B2B, which means that they sell primarily to companies that manufacture and assemble bicycles. Create two sales visualizations and an interactive dashboard that would allow you to present your analysis to the sales manager of the company. Remember to use best practices in creating your visualizations and creating the interactive dashboard.

EX 9.20 (LO 2, 3, 4) Data Auditing Evaluate a Client-Prepared Analysis of Sales Trends This is the first year your firm is auditing One Stop Shop. The team is performing analyses to gain an understanding of the client and the industry. One Stop Shop provided your team with an analysis of their sales trends and product mix from 2022 to 2025.





1. Evaluate the visualizations provided by the client and identify potential issues or problems with the visualizations.
2. Create more effective visualizations.

Problems

Pueblo Hospitality, Inc. (PHI) operates a chain of 48 hotels located in several states. Stephanie Putnam is the president and CEO of PHI. PHI's hotels are in the economy lodging segment. A typical economy lodging hotel has an average of 84 rooms, although PHI's hotels average 117 rooms. Properties are staffed by a general manager, front desk staff of 6 persons, a head housekeeper, 7 housekeepers, and a maintenance worker. Except for the general manager, employees are paid hourly and their assigned hours vary based on demand.

PHI uses the following performance benchmarks.

Measure	Target
Revenue per available room (RevPAR)	2 % increase from prior year
Customer satisfaction	7.5
Housekeeping productivity	30 minutes per room
Audit score	7.0

PR 9.1 **(LO 1, 3, 5)** **Data** **Financial Accounting** **Create a Data Story for Profit Analysis** Prepare an analysis of revenue per available room (RevPAR). Your analysis should cover revenue, profit, average revenue per available room and average competitor revenue per available room. Prepare interactive data visualizations to present to the CEO.

PR 9.2 **(LO 1, 2, 3, 5)** **Data** **Managerial Accounting** **Create a Dashboard for a Management Audience** Prepare an analysis of PHI's performance for each of the performance targets. Create a dashboard that would be useful for management to monitor performance. Ensure that the dashboard allows managers to see how individual hotels are performing, as well as the total company performance.

PR 9.3 **(LO 1, 2, 3, 5)** **Data** **Auditing** **Create Visualizations for Revenue Risk Assessment** You have been assigned to the audit team for the audit of Pueblo Hospitality. Your manager has asked you to use data visualization to gain a better understanding of revenue. Specifically, you have been asked to evaluate the relationship between the number of rooms rented and revenue and identify any unusual patterns or observations.

1. Prepare an analysis that evaluates revenue in the current year compared to the prior year.
2. Prepare an analysis that shows potential outliers by property ID.
3. Prepare a data story with your visualizations to give to your manager, and discuss your results.

Professional Application Case: Madison Public Library

The Madison Public Library (MPL) is an agency of the City of Madison, Wisconsin. The mission of the library is to "provide free and equitable access to cultural and educational experiences." Its vision is to be a "place to learn, share, and create." The library is governed by a nine-member board of directors appointed for three-year terms by the mayor of Madison. The library board works with the mayor, library staff, and the Madison Common Council to plan, fund, and implement public library service in Madison.

The library is supported financially by the city of Madison and the Madison Public Library Foundation. The foundation promotes and supports Madison's public library facilities, services, and programs. Its board is comprised of 30 members. Board initiatives include growing restricted gifts, improving fundraising, building strategic collaboration to fund innovation and urgent needs, and supporting racial equity and inclusion in the library and foundation activities through funding, staff, and board makeup.

The library is comprised of eight locations.

Library Locations

Library Name	Street Address	City	State	Zip Code
Alicia Ashman	733 N High Point Rd	Madison	WI	53717
Central	201 W Mifflin St	Madison	WI	53703
Hawthorne	2707 E Washington Ave	Madison	WI	53704
Lakeview	2845 N. Sherman Ave	Madison	WI	53704
Meadowridge	5726 Raymond Rd	Madison	WI	53711
Pinney	516 Cottage Grove Rd	Madison	WI	53716
Sequoya	4340 Tokay Blvd	Madison	WI	53711
South Madison	222 S Park St	Madison	WI	53713

MPL operates as a not-for-profit entity. The financial information for the previous four years follows.

Library Financial Information

	Year Ended December 31			
Library Revenues	**2025**	**2024**	**2023**	**2022**
City of Madison Library Appropriation	$ 17,703,566	$ 17,779,030	$ 16,915,564	$ 16,288,835
South Central Library System Contractual Services	$ 395,478	$ 337,246	$ 337,361	$ 356,336
Dane County Library System Contractual Services	$ 1,144,935	–	–	–
LINK Contractual Services	$ 404,255	$ 454,290	$ 454,255	$ 454,255
Fines and Fees	$ 335,984	$ 383,403	$ 395,421	$ 404,399
Grants	$ 602,994	$ 149,459	$ 1,010,390	$ 370,254
Other	$ 74,538	$ 121,886	$ 104,631	$ 124,395
Endowment	$ 20,000	$ 20,000	$ 20,000	$ 20,000
	$ 20,681,750	$ 19,245,314	$ 19,237,622	$ 18,018,474
Library Expenses				
Salaries and Benefits	$ 13,026,440	$ 12,659,647	$ 12,352,852	$ 11,474,221
Library Books, Media, and Databases:	$ 1,040,746	$ 1,039,586	$ 1,000,816	$ 1,046,644
Dane County Library System Contractual Services	$ 1,537,180			
Facilities	$ 1,227,112	$ 1,515,114	$ 1,456,628	$ 1,357,358
LINKcat Online Computer Operations	$ 623,845	$ 609,444	$ 611,337	$ 592,158
Debt Retirement	$ 2,826,376	$ 2,648,112	$ 2,745,463	$ 2,720,545
Supplies and Capital Assets	$ 482,606	$ 497,976	$ 390,440	$ 330,283
Purchased Services, Miscellaneous	$ 541,895	$ 745,755	$ 578,811	$ 604,312
	$ 21,306,200	$ 19,715,634	$ 19,136,347	$ 18,125,521
Net Inflow/(Outflow)	$ (624,450)	$ (470,320)	$ 101,275	$ (107,047)

This financial information shows that the library has incurred an increase in net outflows in the last two years. The MPL Board is concerned that if the net outflows continue, the libraries will have to reduce their community services. The following is a list of metrics that can be used to evaluate MPL performance.

Metric:	2025	2024	2023	2022
Checkouts	3,454,156	3,575,215	3,698,903	3,800,000
Digital Checkouts	462,416	382,068		289,309
Internet Uses	227,370	247,129	564,787	635,363
Library Card Registrations	15,544	12,154	11,775	13,245
Library Employees	137	135	131	128
Meeting Room Uses	22,714	22,278	23,010	20,782
Program Attendance (all ages)	107,447	136,303	134,666	110,744
Visits	1,779,552	1,911,287	1,965,014	2,170,000

PAC 9.1 Accounting Information Systems: Visualize Computer Systems Usage

Data **Accounting Information Systems** MPL has seen an increase in computer usage over the last two years. They want to ensure they are providing computers in the branches with the most usage and reducing the number of computers in the branches with lower usage. Use the MPL computer usage data to prepare an interactive visualization to help the accounting information systems department evaluate computer and technology usage by branch.

PAC 9.2 Auditing: Present Payroll Expense Analyses with a Data Story

Data **Auditing** As a member of the audit team examining MPL's payroll expense, you have gathered a list of current employees names and salary amounts. Prepare a descriptive analysis of payroll expense using data visualization. Summarize your findings in a data story.

PAC 9.3 Financial Accounting: Visualize Revenue and Expense Analysis

Data **Financial Accounting** You have been asked to prepare an analysis of revenue and expenses from 2016 to 2025. Use the MPL financial data to prepare visualizations that can be shown to your manager.

PAC 9.4 Managerial Accounting: Build an Interactive Performance Dashboard

Data **Managerial Accounting** You have been asked to prepare a dashboard that would allow management to view both financial and non-financial metrics for the library system. Use the managerial data to prepare a dashboard.

Le Grind Continuing Case: Communicate Results and Recommendations for the Gross Profit Analysis

Data Visit Wiley's online learning platform for the case background, additional questions, data, and more details about the continuing case.